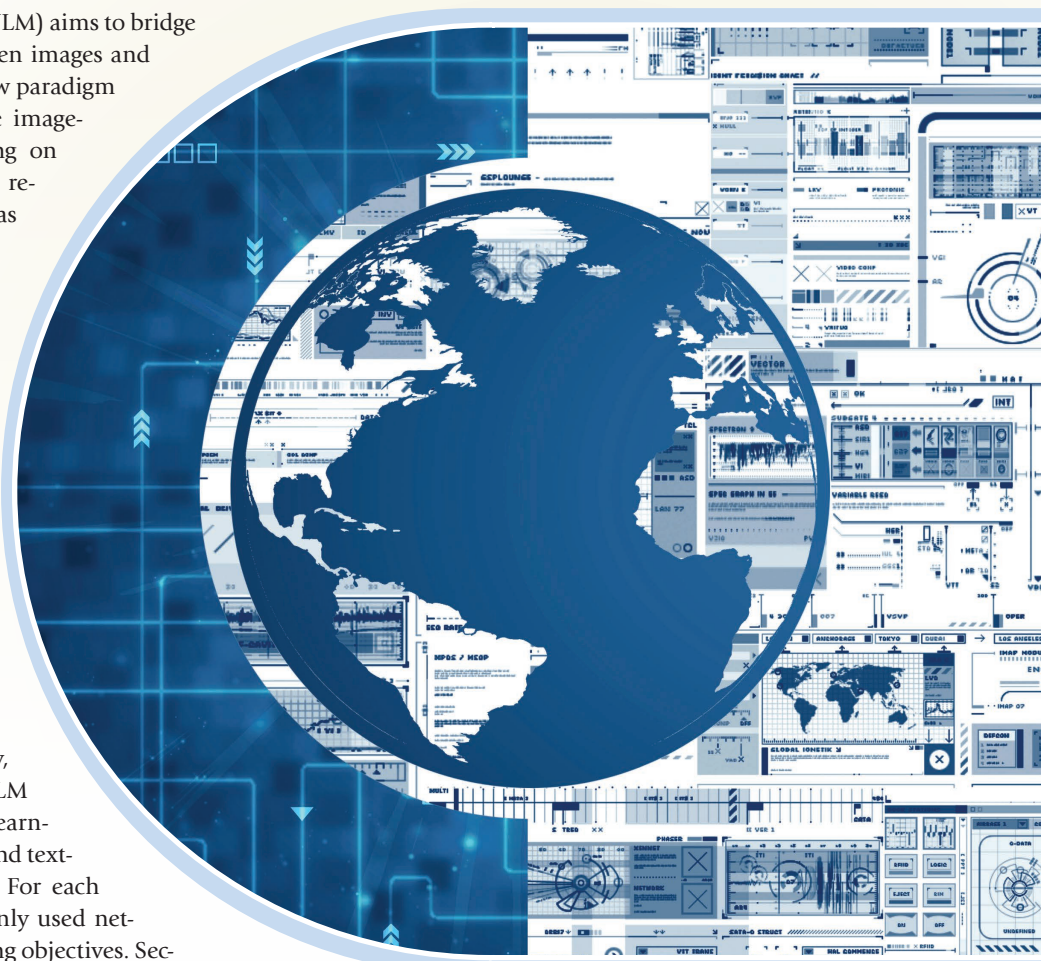


Vision-Language Modeling Meets Remote Sensing

Models, datasets, and perspectives

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Vision-language modeling (VLM) aims to bridge the information gap between images and natural language. Under the new paradigm of first pretraining on massive image-text pairs and then fine-tuning on task-specific data, VLM in the remote sensing (RS) domain has made significant progress. The resulting models benefit from the absorption of extensive general knowledge and demonstrate strong performance across a variety of remote sensing data analysis tasks. Moreover, they are capable of interacting with users in a conversational manner. In this article, we aim to provide the remote sensing community with a timely and comprehensive review of the developments in VLM using the two-stage paradigm. Specifically, we first cover a taxonomy of VLM in remote sensing: contrastive learning, visual instruction tuning, and text-conditioned image generation. For each category, we detail the commonly used network architecture and pretraining objectives. Second, we conduct a thorough review of existing works, examining foundation models and task-specific adaptation methods in contrastive-based VLM, architectural upgrades, training strategies, and model capabilities in instruction-based VLM as well as generative foundation models with their representative downstream applications. Third, we summarize datasets used for VLM pretraining, fine-tuning, and evaluation, with an analysis of their construction methodologies (including image sources and caption generation) and key properties, such as scale and task adapt-



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ability. Finally, we conclude this survey with insights and discussions on future research directions: cross-modal representation alignment, vague requirement comprehension, explanation-driven model reliability, continually scalable model capabilities, and large-scale datasets featuring richer modalities and greater challenges.

INTRODUCTION

VLM in remote sensing, aiming to bridge the information gap between remote sensing images and natural language, facilitates a deeper understanding of remote sensing scene

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semantics, like the attributes of ground objects and their relationship, and enables more natural human interaction with intelligent remote sensing data analysis models or methods [1], [2]. Since the introduction of remote sensing tasks, such as image captioning [3], visual question answering (VQA) [4], text-image (or image-text) retrieval [5], and text-based image generation [6], VLM in remote sensing has achieved significant success, driven by advancements in deep learning.

Early works on VLM primarily emphasize the careful design of model architectures, followed by supervised training from scratch on small-scale datasets. For example, in image captioning research, many efforts [7], [8], [9], [10] have been made to effectively combine convolutional neural networks (e.g., VGG [11] and ResNet [12]) with sequential models (e.g., LSTM [13] and Transformer [14]) before training on UCM-captions [3] and Sydney-captions [3] datasets. Under this classical construction paradigm, deep models often excel on test datasets but struggle to perform satisfactorily in large-scale deployments. Moreover, although these models are capable of describing image content, they fall short when tasked with answering questions about the images. In other words, they struggle to accomplish related tasks, such as VQA. The task-specific nature of these models seriously limits their applicability across diverse scenarios.

Recently, a new paradigm of pretraining followed by fine-tuning provides a promising solution to address the challenges mentioned previously. The core idea is to first pretrain a model on massive image-text data, enabling it to capture general knowledge that covers a wide range of visual and textual concepts, along with their underlying correspondence. The pretrained model is then fine-tuned on task-specific training data. The integration of general knowledge has been shown to enhance the model's generalization ability in a single task [15], [16] while also making it more versatile and capable of handling a variety of downstream tasks [17], [18]. Consequently, VLM with this new paradigm has emerged as a prominent research focus in the field of remote sensing.

To date, significant progress has been achieved, as illustrated in Figure 1. This includes works based on the following:

- 1) contrastive learning [19], such as GeoRSClip [15], SkyCLIP [16], and RemoteCLIP [20], which have driven substantial advancements in various cross-modal tasks and zero-shot image understanding tasks
- 2) learning an implicit joint distribution between text and images, like RS-SD [15], DiffusionSat [21], and CRS-Diff [22], which allow for image generation from text prompts
- 3) visual instruction tuning [23], such as GeoChat [18], LHRS-Bot [24], and SkySenseGPT [25], which have demonstrated improved performance, diverse capabilities, and conversational interactions in remote sensing data analysis.

Despite these remarkable achievements, it is widely acknowledged that VLM remains an open challenge. Indeed, existing works have not yet achieved the level of remote sensing experts in processing remote sensing data. To provide clarity and motivation for further advances in the research community, several surveys have reviewed VLM in remote sensing. For instance, Li et al. [2] summarize vision-language models from an application perspective and suggest potential research opportunities. However, due to time constraints, they primarily concentrate on vision-only foundation models and early works. Zhou et al. [26] review recent developments but lack an in-depth analysis of key designs, which is significant for inspiring future research. Moreover, datasets, as a prerequisite of visual-language modeling research, have not been given adequate attention in existing surveys.

In this work, we aim to provide a timely and comprehensive review of the literature, with a focus on VLM based on the *pretraining and fine-tuning* paradigm in the field of remote sensing. Specifically, we cover the following:

- 1) a taxonomy of VLM in remote sensing, detailing commonly used network architectures and pretraining objectives for each category
- 2) the latest advancements in contrastive-based, instruction-based, and generation-based VLM in remote sensing, highlighting key designs and downstream applications
- 3) progress in datasets for VLM pretraining, fine-tuning, and evaluation
- 4) several challenges and potential research directions.

Figure 2 presents an overview of this article.

THE TAXONOMY OF VISUAL-LANGUAGE MODELING IN REMOTE SENSING

Under the *pretraining and fine-tuning* paradigm, VLM in remote sensing can be divided into three distinct groups based on their strategies for bridging the two modalities in the pretraining phase: contrastive learning, visual instruction tuning, and text-conditioned image generation. In this

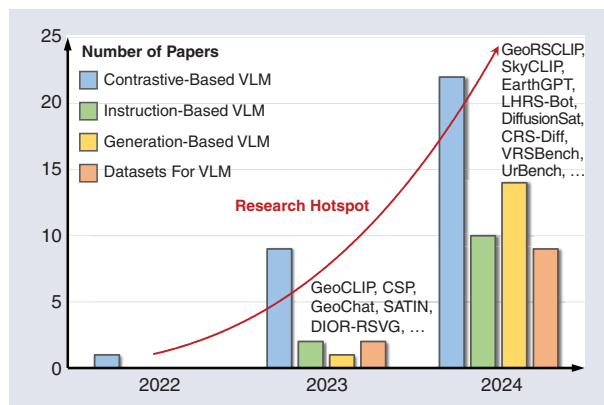


FIGURE 1. The number of publications on visual-language modeling in remote sensing using the pretraining and fine-tuning paradigm.

section, we present commonly used network architectures and pretraining objectives within each group.

CONTRASTIVE LEARNING

The motivation behind applying contrastive learning to VLM is training a model to map vision and language into a shared representation space, where an image and its corresponding text share similar representations while differing from other texts, which was first implemented by the pioneering work CLIP [27] in the field of computer vision. As illustrated in Figure 3(a), CLIP utilizes two independent encoders responsible for encoding visual and textual information. Given a batch of N image-text pairs $\{(x_i, y_i)\}_{i=1}^N$, the embeddings extracted by the two encoders followed by normalization are $\{e_i^I, e_i^T\}_{i=1}^N$. To achieve the similarity between visual and textual representations, CLIP is trained to maximize the cosine similarity of the embeddings of the i th image and the corresponding i th text ($i \in [N]$ with $[N] = \{1, 2, \dots, N\}$) while minimizing the cosine similarity of the embeddings of the i th image and j th text ($i, j \in [N]$, $i \neq j$). The loss is computed using InfoNCE [29] as follows:

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{2N} \sum_{i=1}^N (\log \sigma_{\tau}(s_{ii}^I, \{s_{ij}^I\}_{j=1}^N) + \log \sigma_{\tau}(s_{ii}^T, \{s_{ij}^T\}_{j=1}^N)) \quad (1)$$

where $s_{ij}^I = \mathcal{S}(e_i^I, e_j^I)$ and $s_{ij}^T = \mathcal{S}(e_i^T, e_j^T)$ are the similarity scores between the image and text embeddings, as computed by the function $\mathcal{S}(e, e') = e \cdot e' / \|e\| \|e'\|$. $\sigma_{\tau}(s_i, \{s_j\}_{j=1}^N) = \exp(s_i/\tau) / (\sum_{j=1}^N \exp(s_j/\tau))$ is the softmax function, which normalizes the similarity score s_{ii}^I or s_{ii}^T over the corresponding sets $\{s_{ij}^I\}_{j=1}^N$ or $\{s_{ij}^T\}_{j=1}^N$. τ is the temperature.

The original CLIP model was trained on 400 million image-text pairs collected from the Internet and has demonstrated impressive results across various computer vision tasks [30], [31], [32]. These advancements spark interest in extending its capability to advance VLM in remote sensing.

Two primary lines of research have been actively explored. The first, following the CLIP learning way, focuses on pre-training foundation models that are task agnostic but specifically adapted for the remote sensing domain. This includes efforts such as constructing large-scale image-text datasets [15], [16] and developing novel pretraining objectives [33], [34]. The second line explores the effective adaptation of pretrained CLIP models toward diverse downstream tasks, including image captioning [35], [36], zero-shot scene classification [37], [38], image-text retrieval [39], [40], etc.

VISUAL INSTRUCTION TUNING

Optimizing a model from scratch for image-text alignment is extremely resource intensive due to the need for vast amounts of data and computational power. Fortunately, many pretrained vision encoders and language models have been released. Pretrained vision encoders can provide high-quality visual representations, while pretrained language models, particularly large language models (LLMs), demonstrate advanced language understanding capabilities. As a result, recent works increasingly leverage these models to achieve image-text alignment through visual instruction tuning, as introduced by LLaVA [23] and MiniGPT-4 [41]. Figure 3(b) illustrates a network architecture for this type of work, consisting of three key components: a pretrained vision encoder, a connector, and an LLM. Specifically, the vision encoder compresses remote sensing images into compact visual representations, while the connector maps image embeddings into the word-embedding space of the LLM. The LLM then receives both visual information and language instructions to perform reasoning tasks. Different from CLIP, which directly takes images and corresponding texts as input, this type of work preprocesses image-text pairs to instruction-following data. In this setup, each image is accompanied by a question or a language instruction, requiring the LLM to describe the image, while the corresponding text serves as the ground truth for the LLM's

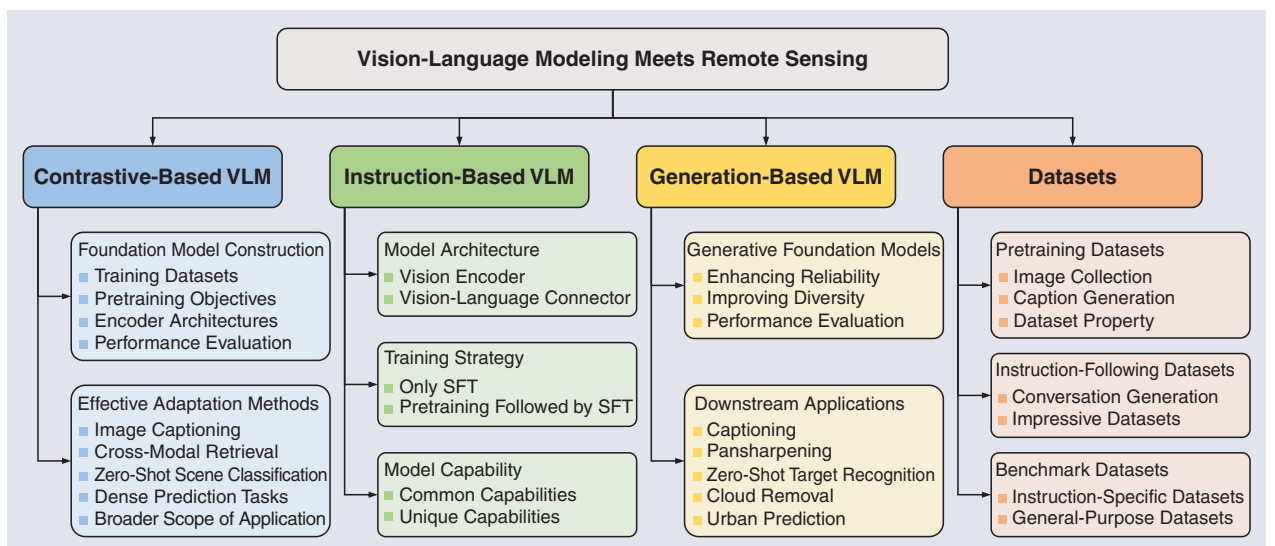


FIGURE 2. An overview of this survey.

predictions. Denote a batch of instruction-following data as $\{(x_i, q_i, y_i)\}_{i=1}^N$, where q_i is the question associated with the i th image. The pretraining objective is defined as follows:

$$\mathcal{L}_{\text{VIT}} = -\frac{1}{N} \sum_{i=1}^N \frac{1}{L_i} \sum_{j=1}^{L_i} \log P(w_j | x_i, q_i, y_{i,<j}) \quad (2)$$

where L_i is the length of the caption $y_i = \{w_1, w_2, \dots, w_{L_i}\}$, $y_{i,<j}$ denotes the masked sequence that contains only the first $j-1$ words, and $P(w_j | x_i, q_i, y_{i,<j})$ is the conditional probability of generating the caption word w_j given the image x_i , question q_i , and the previous words $y_{i,<j}$ in the caption. During pretraining, the vision encoder and LLM are typically kept frozen, with only the parameters of the connector being trainable.

In [23] and [41], the authors demonstrated that pretraining with visual instruction tuning can align vision and language representations while preserving extensive knowledge. Since then, advances have been made by modifying network architectures and creating high-quality pretraining datasets [24], [42]. In addition to improving alignment, another line of research focuses on supervised fine-tuning (SFT), aiming to enable the model to perform a variety of remote sensing image analysis tasks and interact with users in a conversational manner. This includes efforts to generate task-specific instruction-following data [18], [25], [43], design novel training strategies [42], [44], and incorporate cutting-edge vision encoders [44].

TEXT-CONDITIONED IMAGE GENERATION

Taking advantage of the advances in conditional image generation, a group of works [15], [21], [22] uses off-the-shelf generative models, primarily Stable Diffusion [28], to generate remote sensing images given text prompts, which essentially learn an implicit joint distribution between images and texts. As illustrated in Figure 3(c), their network architecture comprises three main components: a text encoder, a variational

autoencoder (VAE) [45], and a denoising U-Net. During training, the VAE encoder first transforms an image into a latent representation z_0 . Gaussian noise ϵ is then added to this latent representation at different time steps t , resulting in $z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$, where $\bar{\alpha}_t$ is a time-dependent scaling factor. Next, the conditioning representation c ,

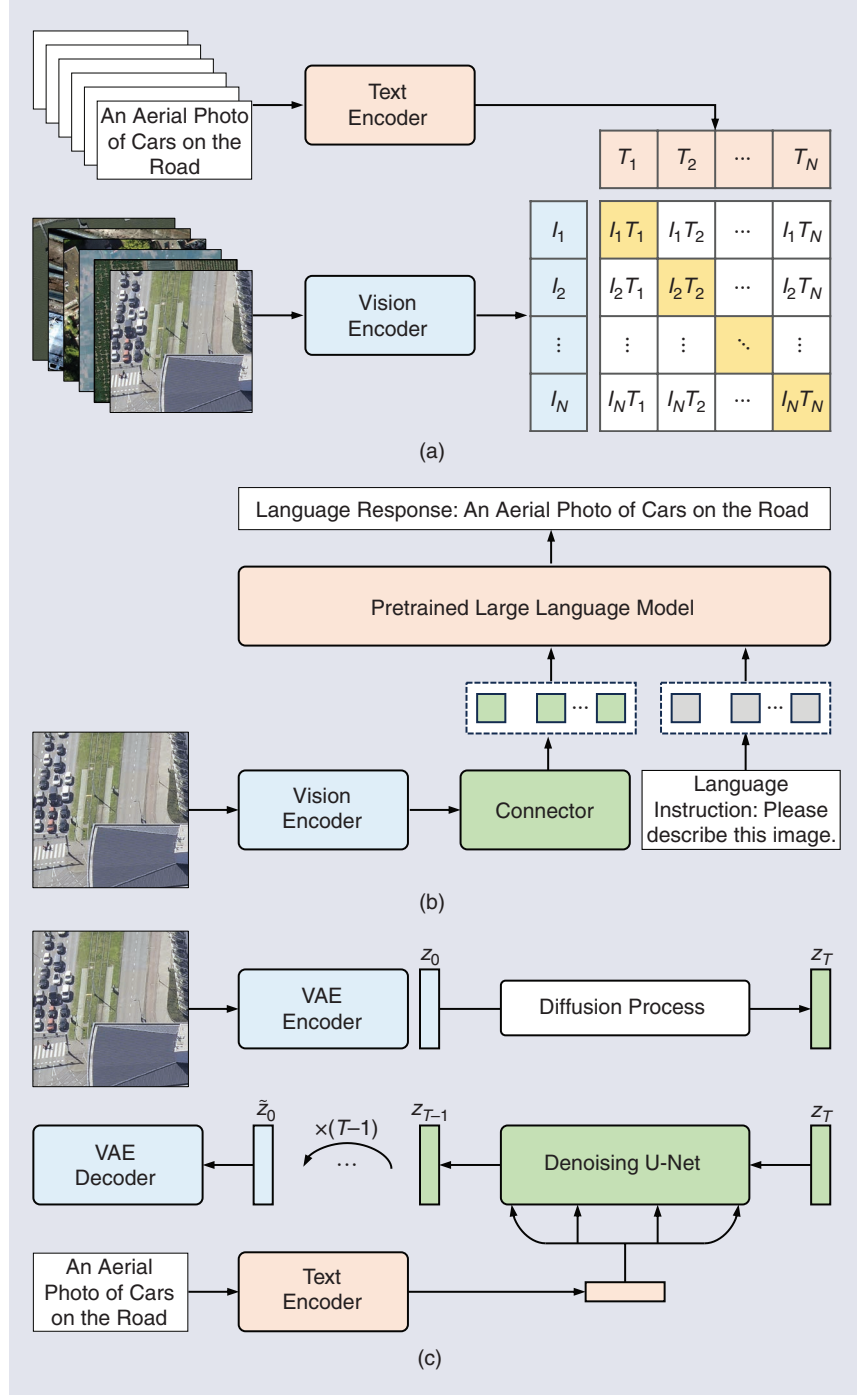


FIGURE 3. Based on the strategies employed to bridge remote sensing images and natural language during the pretraining phase, VLM in remote sensing can be categorized into three groups: (a) contrastive learning, (b) visual instruction tuning, and (c) text-conditioned image generation. This image is recreated by us based on [23], [27], and [28]. (a). Contrastive-based VLM. (b) Instruction-based VLM. (c) Generation-based VLM.

extracted from the text encoder, is provided as input alongside z_t to the denoising U-Net, which predicts the noise added at time step t . Finally, the VAE decoder upsamples the denoised latent representation to reconstruct the input image. Based on image-text pairs, the training objective is defined as follows:

$$\mathcal{L}_{\text{LDM}} = \mathbb{E}_{z_0, c, \epsilon, t} [\|\epsilon - \epsilon_\theta(z_t, t, c)\|_2^2] \quad (3)$$

where $\epsilon_\theta(z_t, t, c)$ is the model's predicted noise, parameterized by θ . The application of diffusion models in remote sensing has shown rapid development, encompassing areas such as remote sensing image generation, enhancement, and interpretation [46]. As this article focuses on VLM using diffusion models, we do not attempt to cover every instance of their application in remote sensing tasks. Instead, we specifically highlight works that integrate text conditions with diffusion models. For a more comprehensive overview of the advancements in diffusion models for remote sensing, please refer to [46].

Based on the fundamental principles outlined previously, two major research groups have emerged. The first group aims to develop generative foundation models for various remote sensing images, including satellite [21], aerial [47], hyperspectral [48], and multiresolution images [49]. The second group, in contrast, extends text-conditioned diffusion models to specific remote sensing tasks, such as image or change captioning [50], [51], pansharpening [52], and zero-shot target recognition [53].

CONTRASTIVE-BASED VISION-LANGUAGE MODELING

Most existing works on VLM fall into the group of employing contrastive learning. As mentioned previously, there are two main research directions being actively investigated, namely *foundation model construction* and *effective adaptation*. Specifically, foundation model construction concerns the large domain gap between natural and remote sensing images, aiming to learn visual representations with rich remote sensing scene semantics and well aligned with textual representations. On the other hand, effective adaptation answers the question of how to leverage pretrained CLIP models for specific remote sensing tasks. In the following sections, we analyze the existing works from these two directions.

FOUNDATION MODEL CONSTRUCTION

Table 1 summarizes contrastive-based vision-language models in remote sensing. To build foundation models, three key components need to be carefully designed: training datasets, pretraining objectives, and encoder architectures.

TRAINING DATASETS

Large-scale image-text datasets form the basis for constructing foundation models. Ready-made image-text datasets in remote sensing, e.g., UCM-captions and Sydney-captions,

suffer from limited data volume and insufficient image diversity, rendering them inadequate for pretraining models to capture general knowledge of the domain. Recognizing the availability of numerous remote sensing image datasets, [15] and [20] use open source datasets as the image source and develop image captioning methods to generate corresponding textual descriptions. Notably, [15] filters 11 commonly used image-text datasets using RS-related keywords and caption three large-scale RS image datasets (Million-AID [67], fMoW [60], and BigEarthNet [68]) with the aid of the tuned BLIP2 model [69], resulting in the RS5M dataset, which contains more than 5 million image-text pairs. Off-the-shelf vision-language models are indeed powerful tools for building large-scale image-text datasets due to their availability and ease of use, but ensuring captioning accuracy remains a significant challenge. To address this, [20] proposes a rule-based method called *mask-to-box* (M2B) and *box-to-caption* (B2C), which converts pixel-wise or bounding box annotations into natural language captions.

Another concern is that the semantic diversity of the generated captions is constrained by the limited number of predefined classes in open source remote sensing image datasets. Given this, Wang et al. [16] attempt to leverage rich semantic information contained in OpenStreetMap (OSM), allowing the textual descriptions to encompass not only a wide variety of object categories but also fine-grained subcategories and object attributes. Similar to [20], captions are assembled from object tags following predefined rules.

PRETRAINING OBJECTIVES

Instead of creating large-scale datasets, several works [33], [34], [55] explore new training objectives to facilitate model pretraining with a few remote sensing image-text pairs. For instance, S-CLIP [55] introduces caption-level and keyword-level pseudolabels to fine-tune the original CLIP model on massive unpaired remote sensing images alongside a few image-text pairs. Denote a large number of unpaired images as $\{u_i\}_{i=1}^M$ ($M \gg N$). The caption-level pseudolabel $q_i^c \in \mathbb{R}^N$ is based on the assumption that the semantics of an unpaired image u_i can be represented as a combination of those of paired images. Thus, q_i^c represents a probability distribution over the captions of N paired images, derived from the relationships between unpaired and paired images, which are formulated as an optimal transport problem. The keyword-level pseudolabel $q_i^k \in \mathbb{R}^K$ relies on the assumption that u_i shares keywords with visually similar images, representing the similarity between the embeddings of u_i and the keywords (drawn from the nearest paired image) $\{k_j\}_{j=1}^K$. The loss functions for two pseudolabels are defined as follows:

$$\mathcal{L}_{\text{caption}} = -\frac{1}{M} \sum_{i=1}^M \sum_{j=1}^N q_i^c \log \sigma_\tau(s_{ij}^U, \{s_{il}^U\}_{l=1}^N) \quad (4)$$

$$\mathcal{L}_{\text{keyword}} = -\frac{1}{M} \sum_{i=1}^M \sum_{j=1}^K q_{ij}^k \log \sigma_{\tau}(s_{ij}^U, \{s_{il}^U\}_{l=1}^K). \quad (5)$$

In (4), $s_{il}^U = \mathcal{S}(\mathbf{e}_i^U, \mathbf{e}_l^T)$ refers to the similarity score of the unpaired image embedding $\mathbf{e}_i^U$ and the caption embedding $\mathbf{e}_l^T$ of the paired image. Meanwhile, in (5), $s_{il}^U = \mathcal{S}(\mathbf{e}_i^U, \mathbf{e}_l^k)$ denotes the similarity score between $\mathbf{e}_i^U$ and the keyword-embedding $\mathbf{e}_l^k$. Since few image-text pairs are involved in the fine-tuning process, the overall training objective also includes the InfoNCE loss, as shown in the following equation:

$$\mathcal{L}_{\text{SCLIP}} = \mathcal{L}_{\text{InfoNCE}} + \frac{1}{2}(\mathcal{L}_{\text{caption}} + \mathcal{L}_{\text{keyword}}). \quad (6)$$

With a similar training data setup consisting of massive unpaired images and texts $\{u_i, v_i\}_{i=1}^M$ and limited

image-text pairs $\{(x_i, y_i)\}_{i=1}^N$, Set-CLIP [34] transforms the representation alignment between images and texts into a manifold matching problem, developing a multi-kernel maximum mean discrepancy loss $\mathcal{L}_{\text{MK-MMD}}$ and a semantic density distribution loss $\mathcal{L}_{\text{SDD}}$. The loss $\mathcal{L}_{\text{MK-MMD}}$ constrains the consistency of whole representation distributions of images and texts, thereby achieving macrolevel alignment, as formulated in the following equation:

$$\mathcal{L}_{\text{MK-MMD}} = \left\| \frac{1}{M+N} \sum_{i=1}^{M+N} \phi(\mathbf{e}_i^I) - \frac{1}{M+N} \sum_{i=1}^{M+N} \phi(\mathbf{e}_i^T) \right\|_{\mathcal{H}_{\text{RKHS}}}^2 \quad (7)$$

where $\phi(\cdot)$ is a linear combination of Gaussian and Polynomial kernel functions, used to map image and text

TABLE 1. A SUMMARY OF CONTRASTIVE-BASED VISION-LANGUAGE FOUNDATION MODELS IN REMOTE SENSING.

MODEL	VISION ENCODER	TEXT ENCODER	TRAINING DATASET	PRETRAINING OBJECTIVE	EVALUATION TASK	PUBLIC
GeoRSCLIP [15] ^a	ViT-B/32, ViT-H/14 [54]	Transformer [14]	RS5M [15]	InfoNCE loss [29]	Zero-shot scene classification Image-text retrieval Semantic localization	✓
RemoteCLIP [20] ^b	ResNet-50 [12], ViT-B/32, ViT-L/14 [54]	Transformer [14]	RET-3+SEG-4+DET-10 [20]	InfoNCE loss [29]	Zero/few-shot scene classification Image-text retrieval Object counting Linear/k-NN classification	✓
SkyCLIP [16] ^c	ViT-B/32, ViT-L/14 [54]	Transformer [14]	SkyScript [16]	InfoNCE loss [29]	Zero-shot scene classification Image-text retrieval Zero-shot fine-grained classification	✓
S-CLIP [55] ^d	ResNet-50 [12], ViT-B/32, ViT-B/16 [54]	Transformer [14]	RS-ALL [55] NWPU-RESISC45 [56]	InfoNCE loss [29] Pseudolabel losses [55]	Zero-Shot scene classification Image-text retrieval	✓
Set-CLIP [34]	ResNet [12]	Transformer [14]	RS-ALL [55]	InfoNCE loss [29] SSL loss [57] MK-MMD, SDD losses [34]	Zero-shot scene classification Image-text retrieval	✗
GRAFT [33] ^e	ViT-B/16 [54]	—	Internet-NAIP Dataset [33] Internet-Sentinel-2 Dataset [33]	Image/Pixel-level contrastive losses [33]	Zero-shot scene classification Zero-shot text-to-image retrieval Zero-Shot semantic segmentation Zero-shot VQA	✓
CSP [58] ^f	ResNet-50 [12]	Sinusoidal transform+FcNet [59]	fMoW [60]	InfoNCE loss [29]	Few-shot geo-aware scene classification	✓
GeoCLIP [61] ^g	ViT-L/14 [54]	Random Fourier features+MLP [62]	MP-16 [63]	InfoNCE loss [29] SSL loss [57]	Image geo-localization Text-query geo-localization Geo-aware image classification	✓
SatCLIP [64] ^h	ResNet-18, ResNet-50 [12], ViT-S/16 [54]	Spherical harmonics+Siren [65]	S2-100K [64]	InfoNCE loss [29]	Air temperature prediction Elevation prediction Median income estimation California housing price estimation Population density estimation Biome classification Ecoregion classification Country code classification Species recognition	✓
PIR-CLIP [66] ⁱ	ViT-B/32 [54]+ ResNet-50 [12]	Transformer [14]	RS5M [15]	InfoNCE loss [29] Affiliation loss [66]	Image-text retrieval	✓

^a<https://huggingface.com/Zilun/GeoRSCLIP>; ^b<https://github.com/ChenDelong1999/RemoteCLIP>; ^c<https://github.com/wangzhecheng/SkyScript>; ^d<https://github.com/alinelab/s-clip>; ^e<https://graft.cs.cornell.edu/>; ^f<https://gengchenmai.github.io/csp-website/>; ^g<https://github.com/VicenteVivan/geo-clip>; ^h<https://github.com/microsoft/satclip>; ⁱ<https://github.com/jaychempan/PIR-CLIP>.

Image-text retrieval involves image-to-text and text-to-image retrieval tasks. In the "Vision Encoder" column, we separate different architectures of vision encoders with commas, whereas in [66], the vision encoder is a combination of two architectures. "Public" refers to the availability of both code and model weights. S-CLIP has only open-sourced its code.

embeddings $\mathbf{e}_i^I, \mathbf{e}_i^T$ into Reproducing Kernel Hilbert Space (RKHS) $\mathcal{H}_{\text{RKHS}}$. Meanwhile, the loss $\mathcal{L}_{\text{SDD}}$ refines the alignment between the two modalities by ensuring that their probability density distributions remain similar in the representation space. It can be formulated as follows:

$$\mathcal{L}_{\text{SDD}} = \frac{1}{2} (\Gamma(I, T) + \Gamma(T, I)) \quad (8)$$

where $I = \{\mathbf{e}_i^I\}_{i=1}^{M+N}$ and $T = \{\mathbf{e}_i^T\}_{i=1}^{M+N}$ refer to embedding distributions of images and texts, respectively. $\Gamma(\cdot, \cdot)$ represents the Kullback-Leibler divergence, which measures the dissimilarity between two distributions. The format of $\Gamma(I, T)$ is given by the following:

$$\Gamma(I, T) = \sum_{i=1}^{M+N} \frac{\chi(\mathbf{e}_i^I, I)}{\sum_{j=1}^{M+N} \chi(\mathbf{e}_j^I, I)} \log \frac{\chi(\mathbf{e}_i^I, I) / \sum_{j=1}^{M+N} \chi(\mathbf{e}_j^I, I)}{\chi(\mathbf{e}_i^I, T) / \sum_{j=1}^{M+N} \chi(\mathbf{e}_j^I, T)} \quad (9)$$

with $\Gamma(T, I)$ defined similarly by swapping the roles of I and T . Here, $\chi(\cdot, \cdot)$ is an exponential probability density function that estimates the density value for each embedding in the corresponding distribution. Additionally, the self-supervised contrastive loss $\mathcal{L}_{\text{SSL}}$ is introduced to obtain robust feature representations for each modality independently, defined as follows:

$$\mathcal{L}_{\text{SSL}} = -\frac{1}{M+N} \sum_{i=1}^{M+N} \log \frac{\exp(\mathcal{S}(\mathbf{e}_i, \mathbf{e}_i^+)/\tau)}{\sum_{j=1}^{M+N} \exp(\mathcal{S}(\mathbf{e}_i, \mathbf{e}_j)/\tau)} \quad (10)$$

where $\mathbf{e}_i$ is an embedding, and $\mathbf{e}_i^+$ is the representation of the positive sample generated by augmentation techniques. Ultimately, the overall training objective is formulated as follows:

$$\mathcal{L}_{\text{Set-CLIP}} = \alpha \mathcal{L}_{\text{InfoNCE}} + \mu \mathcal{L}_{\text{SSL}} + \delta \mathcal{L}_{\text{MK-MMD}} + \eta \mathcal{L}_{\text{SDD}} \quad (11)$$

where α, μ, δ , and η are tunable hyperparameters to balance the loss terms.

Even a small set of image-text pairs requires expert knowledge to craft textual descriptions. To avoid textual annotations entirely, GRAFT [33] proposes utilizing collocated ground images as the bridge between satellite images and language. In doing so, a dataset of 2 million ground-satellite image pairs, denoted as $\{x_i, \{g_{ij}\}_{j=1}^{N_i}\}_{i=1}^N$, is collected to support model training. Building on this, a feature extractor is designed to map satellite images to the representation space of the CLIP model, which was trained on Internet image-text pairs. Since a satellite image x_i can cover a large ground area and thus be associated with multiple ground images $\{g_{ij}\}_{j=1}^{N_i}$, the extractor is optimized using the following loss:

$$\mathcal{L}_{\text{GRAFT}} = -\frac{1}{N} \sum_{i=1}^N \frac{1}{N_i} \sum_{j=1}^{N_i} \log \frac{\exp(\mathcal{S}(\mathbf{e}_i^I, \mathbf{e}_{ij}^G)/\tau)}{\sum_{a=1}^N \sum_{b=1}^{N_a} \exp(\mathcal{S}(\mathbf{e}_i^I, \mathbf{e}_{ab}^G)/\tau)} \quad (12)$$

where $\mathbf{e}_{ij}^G$ is the embedding of the j th ground image corresponding to the i th satellite image. This loss focuses solely on aligning image-level representations between the two types of images while ignoring the fact that a ground image g_{ij} can be mapped to a specific pixel location p_{ij} in the corresponding satellite image x_i . Thus, GRAFT trains the feature extractor with an additional pixel-level contrastive loss as follows:

$$\mathcal{L}_{\text{GRAFT}} = -\frac{1}{N} \sum_{i=1}^N \frac{1}{N_i} \sum_{j=1}^{N_i} \log \frac{\exp(\mathcal{S}(\mathbf{e}_{ij}^P, \mathbf{e}_{ij}^G)/\tau)}{\sum_{a=1}^N \sum_{b=1}^{N_a} \exp(\mathcal{S}(\mathbf{e}_{ij}^P, \mathbf{e}_{ab}^G)/\tau)} \quad (13)$$

where $\mathbf{e}_{ij}^P$ denotes the feature vector for pixel p_{ij} .

ENCODER ARCHITECTURES

Unlike natural images, where corresponding textual information typically describes the image content, the textual information for remote sensing images can be represented by their geographic coordinates (longitude and latitude), which are beneficial for tasks such as scene classification and object recognition [60]. Therefore, some works propose aligning representations from remote sensing images and their geographic coordinates, with particular attention given to the choice of location encoders [58], [61], [64]. A location encoder generally consists of a nonparametric functional positional encoding combined with a small neural network. In CSP [58], an existing 2D location encoder, namely Space2Vec's grid proposed in [59], is utilized. This encoder maps image coordinates into high-dimensional representations using sinusoid transforms for position encoding, followed by fully connected rectified linear unit layers. GeoCLIP [61], on the other hand, first applies equal Earth projection to the image coordinates to reduce distortions inherent in standard geographic coordinate systems. It then adopts random Fourier features [62] to capture high-frequency details, varying the frequency to construct hierarchical representations. These hierarchical representations are processed through separate multilayer perceptrons (MLPs), followed by element-wise addition, resulting in a joint representation. This design enables the location encoder to effectively capture the features of a specific location across multiple scales. In SatCLIP [64], the authors use a location encoder that combines spherical harmonics basis functions with sinusoidal representation networks to map geographic coordinates into latent representations [65].

In addition to adapting encoders for different types of textual information, another purpose of adjusting encoders is to improve the representations of visual concepts. Remote sensing images typically cover a large field of view and include a variety of objects. However, the corresponding textual descriptions often center on specific objects of interest and their relationships. Semantic noise, such as irrelevant objects and background, can interfere with the representation of key content in images, thereby obstructing

the alignment between image and text representations. In [66], this problem is addressed by using prior knowledge of remote sensing scenes to instruct the pretrained model in filtering out semantic noise before calculating the similarity between image and text embeddings. This is practically achieved by adding an instruction encoder and a transformer encoder layer on top of the vision encoder. The instruction encoder, pretrained on the scene classification dataset AID [70], generates instruction embeddings, which are used to filter image embeddings via a soft belief strategy. The filtered embeddings are then activated using instruction information through the transformer encoder layer, producing relevant image embeddings for subsequent alignment.

PERFORMANCE EVALUATION

Zero-shot scene classification and image-text retrieval are commonly used tasks to assess foundation models' capabilities to capture a wide range of visual and textual concepts, along with their correspondence. Table 2 lists the performance of contrastive-based vision-language foundation models on these two tasks. Note that it shows the best model performance in the original articles. For zero-shot scene classification, three conclusions can be drawn:

- 1) AID [70] is the most widely used dataset for evaluation, with RemoteCLIP achieving the best accuracy of 87.9%.
- 2) Using the same vision encoder (ViT-L/14), SkyCLIP, which is pretrained on images collected from Google Earth, outperforms RemoteCLIP only on PatternNet [74], a dataset sourced from Google Earth.
- 3) Comparing models pretrained on limited image-text pairs (i.e., S-CLIP, SetCLIP, and GRAFT) with RemoteCLIP, their performance is somewhat comparable, showing the importance of exploring data-effective pre-training. For cross-modal retrieval, PIR-CLIP achieves state-of-the-art performance on RSICD [78] and RSITMD [82]. In comparison, S-CLIP and SetCLIP perform significantly worse, with a performance gap exceeding 20%. This suggests that existing foundation models pretrained on limited image-text pairs are capable of acquiring general visual and textual concepts but may face challenges in learning the relationships between them.

EFFECTIVE ADAPTATION METHODS

Inspired by the outstanding performance of foundation models, numerous methods have been developed to effectively adapt pretrained models for specific remote sensing tasks, including image captioning, cross-modal retrieval, zero-shot classification, dense prediction, etc. This section presents the development of adaptation methods within the context of different downstream tasks as summarized in Table 3.

IMAGE CAPTIONING

Image captioning aims to describe the content of a given image using natural language. Vision-language foundation

models are well suited for this task as they align image and text representations. One can use the vision encoder of these models to extract representations of image content, which prepares the input for language models to generate captions [36], [83], [85]. Commonly used vision encoders for remote sensing image captioning are from CLIP, while language models for this task include GPT-2 [105], BERT [106], and OPT [107]. To improve caption accuracy, a few works are devoted to enhancing models' visual representation capabilities or further aligning images and texts.

For example, regarding representation enhancement, VLCA [83] introduces external attention [84] to capture potential correlations between different images, improving visual representations. MVP [36] fuses visual representations from pretrained vision-language models and pretrained vision models through stacked transformer encoders. It also leverages CLIP's vision encoder, followed by adaptive average pooling layers, to generate visual prefixes, which are concatenated with token embeddings. A BERT-based caption generator is subsequently developed to combine the fused visual representations and concatenated embeddings, enabling the generation of accurate captions. In contrast, BITA [85] focuses on improving the alignment of images and texts in the remote sensing domain by introducing an interactive Fourier transformer. In this design, learnable visual prompts are fed to the Fourier layer and interact with image embeddings from a pretrained vision encoder, capturing the most relevant visual representations. Through contrastive learning, these visual representations are aligned with textual representations, also extracted by the Fourier-based transformer. The interactive Fourier transformer then connects the frozen vision encoder with the frozen language model, leveraging the language model's generation and reasoning capabilities.

Beyond improving caption accuracy, generating detailed captions has also been explored [86], [87]. In [86], a two-stage instruction fine-tuning is proposed for the vision-to-language mapping layer to generate geographically detailed captions. The first stage, guided by the instruction "[region] Based on the provided region of the remote sensing image, describe the basic attributes of the main objects in that region," aligns geographic object regions with their attribute descriptions. The second stage, guided by the instruction "[image] Please provide a detailed description of this image," focuses on understanding the spatial distribution of geographic objects within images. In [87], image captioning is defined as the aggregation of information from multi-turn dialogues, where each turn serves to query the image content. In each turn, CLIP's vision encoder extracts image features, which are then input into an auto-regressive language model [105] along with previous questions, answers, and the current question to generate the response. After several dialogue turns, GPT-3 [108] summarizes the dialogue information to produce an enriched textual image description.

TABLE 2. THE PERFORMANCE OF CONTRASTIVE-BASED VISION-LANGUAGE FOUNDATION MODELS.

ZERO-SHOT SCENE CLASSIFICATION: TOP-1 ACCURACY

MODEL	VISION ENCODER	AID [70]	EuroSAT [71]	fMoW [60]	MILLION-AID [67]	MLRSNet [72]	RESISC [56]	OPTIMAL-31 [73]	PatternNet [74]	RSSCN7 [75]	RSC11 [76]	RSI-CB128 [77]	RSI-CB256 [77]	RSICD [78]	SIRI-WHU [79]	UCM [80]	WHU-RS19 [81]
SkyCLIP [16]	ViT-L/14 [54]	71.7	51.33	27.12	67.45	—	70.94	—	80.88	—	—	—	50.09	—	—	—	—
Set-CLIP [34]	ResNet-50 [12]	76.2	—	—	—	—	—	—	—	66.2	—	—	—	69.2	—	67.5	89
GRAFT [33]	ViT-B/16 [54]	—	63.76	—	—	—	—	—	—	—	—	—	—	—	—	—	—
GeoRSCLIP [15]	ViT-H/14 [54]	76.33	67.47	—	—	—	73.83	—	—	—	—	—	—	—	—	—	—
S-CLIP [55]	ViT-B/16 [54]	85.2	—	—	—	—	—	—	—	76.3	—	—	—	79.5	—	82.3	93.9
RemoteCLIP [20]	ViT-L/14 [54]	87.9	59.94	—	—	66.32	79.84	90.05	68.75	72.32	74.9	37.22	52.82	—	70.83	—	94.66

IMAGE-TEXT RETRIEVAL: RECALL@1, RECALL@5, AND RECALL@10

MODEL	VISION ENCODER	RSICD [78]	R@1	R@5	Mean	R@10	RSITMD [82]	Mean	R@1	R@5	R@10	UCM-captions [3]	Mean	R@1	R@5	R@10	Sydney-captions [3]	Mean
S-CLIP [55]	ResNet-50 [12]	4.2	18.4	—	—	—	—	—	11.6	—	—	45.7	—	14.9	50	—	—	—
Set-CLIP [34]	ResNet-50 [12]	—	19.6	—	—	—	—	—	—	—	—	46.3	—	—	51.1	—	—	—
SkyCLIP [16] [†]	ViT-L/14 [54]	—	—	—	23.7	—	—	30.75	—	—	—	—	72.22	—	—	—	—	—
RemoteCLIP [20]	ViT-L/14 [54]	18.39	37.42	51.05	35.62	28.76	52.43	63.94	48.38	19.05	54.29	80.95	51.43	—	—	—	—	—
GeoRSCLIP [15]	ViT-B/32 [54]	21.13	41.72	55.63	39.49	32.3	53.32	67.92	51.18	—	—	—	—	—	—	—	—	—
PIR-CLIP [66]	ViT-B/32 [54]	27.63	45.38	55.26	42.76	45.58	65.49	75	62.02	—	—	—	—	—	—	—	—	—
	+ ResNet-50 [12]																	

TEXT-TO-IMAGE RETRIEVAL

MODEL	VISION ENCODER	R@1	R@5	Mean	R@10	R@1	R@5	R@10	Mean	R@1	R@5	R@10	Mean	R@1	R@5	R@10	Mean
S-CLIP [55]	ResNet-50 [12]	4.2	16.8	—	—	—	—	—	—	11.1	43.5	—	—	17.8	55.1	—	—
Set-CLIP [34]	ResNet-50 [12]	—	17.4	—	—	—	—	—	—	—	44.1	—	—	—	55.2	—	—
SkyCLIP [16] [†]	ViT-L/14 [54]	—	—	19.97	—	—	—	30.58	—	—	—	—	59.33	—	—	—	—
RemoteCLIP [20]	ViT-L/14 [54]	14.73	39.93	56.58	37.08	23.76	59.51	74.73	52.67	17.71	62.19	93.9	57.93	—	—	—	—
GeoRSCLIP [15]	ViT-B/32 [54]	15.59	41.19	57.99	38.26	25.04	57.88	74.38	52.43	—	—	—	—	—	—	—	—
PIR-CLIP [66]	ViT-B/32 [54]	21.1	44.87	56.12	40.7	30.13	55.44	68.54	51.37	—	—	—	—	—	—	—	—
	+ ResNet-50 [12]																

RESISC, NWPU-RESISC45 [56]. Mean refers to the mean recall average of recall@1 (R@1), recall@5 (R@5), and recall@10 (R@10). [†]Indicates that the training data corresponding to the test dataset were not used to pretrain the model. Best results are highlighted in bold.

TABLE 3. SUMMARY OF CONTRASTIVE-BASED VISION-LANGUAGE FOUNDATION MODELS APPLIED TO REMOTE SENSING TASKS.

WORK	TASK	VISION-LANGUAGE FOUNDATION MODEL	ADAPTATION
VLCA [83]	Image captioning	CLIP ResNet-50, ViT [27]	Utilize external attention [84] to enhance visual representations extracted by CLIP's vision encoder.
MVP [36]	Image captioning	CLIP ViT-B/16 [27]	Fuse visual representations from CLIP and pretrained vision models.
BITA [85]	Image captioning	CLIP ViT-L/14 [27]	Introduce the Fourier-based transformer to align images and texts in the remote sensing domain.
MGIMM [86]	Image captioning	CLIP ViT-L/14 [27]	Use region-level and image-level instruction tuning for generating captions with detailed geographical information.
CFD-C [87]	Image captioning	CLIP ViT-B/16 [27]	Define image captioning as the aggregation of information from multimodal dialogues.
Prompt-CC [88]	Change captioning	CLIP ViT-B/32 [27]	Develop an image-level classifier, a feature-level encoder on top of the pretrained vision encoder for detecting the presence of changes and extracting discriminative features.
CISEN [40]	Image-text retrieval	CLIP ResNet-50, ViT-B/32+Transformer [27] GeoRSCLIP ViT-B/32+Transformer [15]	Use two-stage training to progressively fuse fine-grained semantic features to enrich visual representation.
KTIR [89]	Image-text retrieval	BLIP ViT-B/16+BERT [90]	Incorporate knowledge into text features for better image-text alignment.
Zavras et al. [39]	Zero-shot RGB-multispectral retrieval Zero-shot scene classification	CLIP ViT-B/32, ViT-B/16, ViT-L/14, ViT-L/14@336 [27]	Use two-stage fine-tuning to achieve alignment between distinct image modalities in the CLIP representation space.
WEICOM [91]	Composed-to-Image retrieval	CLIP ViT-L/14+Transformer [27] RemoteCLIP ViT-L/14+Transformer [20]	Use a weighting parameter for combining image-query and text-query retrieval results, achieving a composed query.
Rahhal et al. [37]	Zero-shot scene classification	Thirteen CLIP/Open-CLIP Models [27]	Use tuning prompts by adding task-relevant context words, including "remote sensing image" and "scene."
Lan et al. [92]	Few-shot fine-grained ship classification	CLIP ResNet-50, ViT-B/16+Transformer [27]	Inject domain priors to adapt frozen vision encoder to remote sensing scenes. Design hierarchical learnable prompts to capture rich task-specific knowledge.
RS-CLIP [35]	Zero-shot scene classification	CLIP ViT-L/14+Transformer [93]	Use pseudolabeling and curriculum learning to fine-tune the pretrained model in multiple rounds.
DSVA [38]	Zero-shot scene classification	CLIP ViT-B/32+Transformer [27]	Employ the pretrained model to annotate attributes for each class.
Text2Seg [94]	Zero-shot semantic segmentation	CLIP ViT-B/16+Transformer [27]	Employ CLIP to endow segmentation masks with semantic categories.
Lin et al. [95]	Semantic segmentation	CLIP ViT-B/16+Transformer [27]	Modify CLIPSeg decoder [96] to receive CLIP joint image-text embeddings as input and generate segmentation masks.
SegEarth-OV [97]	Open-vocabulary semantic segmentation	CLIP ViT-B/16+Transformer [27]	Develop SimFeatUp to robustly upsample low-resolution CLIP embeddings. Execute subtraction operations between patch embeddings and the class token embedding to alleviate global bias.
SCM [98]	Unsupervised change detection	CLIP [27]	Employ CLIP to identify objects of interest in bitemporal images, helping filter out pseudo-changes.
ChangeCLIP [99]	Change detection	CLIP ResNet-50, ViT-B/16+Transformer [27]	Employ CLIP to construct and encode multimodal input data for change detection. Design a transformer-based decoder to combine vision-language features with image features, predicting change maps.
BAN [100]	Change detection	CLIP ViT-B/16, ViT-L/14 [27] RemoteCLIP ViT-B/32, ViT-L/14 [20] GeoRSCLIP ViT-B/32, ViT-L/14 [15]	Design bridging modules to inject general knowledge from foundation models into existing change detection models.
Bazi et al. [101]	VQA	CLIP ViT+Transformer [27]	Employ CLIP to encode images and questions.
Czerkawski et al. [102]	Cloud presence detection	CLIP [27]	Apply CLIP to this task via tuning prompts or adding linear classifier.
TGN [103]	Text-based image generation	CLIP [27]	Utilize CLIP to classify generated images, ensuring semantic class consistency.
AddressCLIP [104]	Image address localization	CLIP ViT-B/16+Transformer [27]	Align images with scene captions and address texts by contrastive learning. Introduce image-geography matching to constrain image features with the spatial distance.

Unlike previous works that caption a single image, Prompt-CC [88] utilizes pretrained models to describe differences between bitemporal images, a task known as *change captioning*. Based on bitemporal visual representations from CLIP, Prompt-CC introduces an image-level classifier to detect the presence of changes and a feature-level encoder to extract discriminative features that identify the specific changes.

CROSS-MODAL RETRIEVAL

Cross-modal retrieval is the task of retrieving data from one modality by using queries from another modality. Based on vision-language foundation models, there is ongoing research that explores retrieval between images and text [40], [89], [91] as well as between images captured with different imaging parameters [39]. Image-text retrieval involves the challenges of finding textual descriptions that correspond to given image queries and vice versa. Taking advantage of advances in vision-language foundation models, existing works use pretrained encoders to encode images and text separately, mapping them into a shared representation space for similarity measurement. Typically, CISEN [40] adopts encoders from CLIP and GeoRSCLIP as the backbone for its retrieval network. Given the abundance of objects in remote sensing images, CISEN is trained in two stages to enhance visual representation. In the first stage, an image adapter [109] is trained to map global visual features into textual-like features. In the second stage, a feature pyramid network is used to integrate the textual-like features into local visual features, which are then utilized to enrich global visual features. Rather than focusing on effective representation, KTIR [89] aims to improve alignment by explicitly incorporating knowledge into text features. The knowledge is derived from textual information using off-the-shelf knowledge sources like RSKG [110] and ConceptNet [111]. Once converted to knowledge sentences, the knowledge is processed by the text encoder to extract knowledge features, which are then fused with text features via a single cross-attention layer.

Image-image retrieval involves searching for relevant images across different imaging parameters, such as matching red, green, blue (RGB) images with multispectral images, as explored in [39]. The authors use a pretrained CLIP text encoder as the classification head and fine-tune the CLIP vision encoder alongside a newly added multispectral-specific encoder. The training is conducted in stages; first, the CLIP vision encoder is fine-tuned on RGB images composed from multispectral images, followed by fine-tuning the new encoder on multispectral images. This encoder is guided to produce discriminative representations that can achieve accurate image classification while ensuring that its representations are similar to those of RGB images extracted by the CLIP vision encoder.

Considering the complexity of Earth's surface, single-modality queries, such as text queries, require users to fully articulate their needs to pinpoint relevant images. This raises the barrier to using retrieval models. To address this issue, Psomas et al. [91] introduce the composed-to-image retrieval task, which searches for remote sensing images based on a composed query of image and text. The retrieved images share the same scene or object category as the image query and reflect the attribute defined by the text query. To achieve this, the authors in [91] propose calculating similarity scores for the image and text query separately and then normalizing and combining these scores using a convex combination controlled by a weighting parameter λ , which adjusts the contribution of each modality. As shown in Figure 4, when $\lambda = 0$, composed-to-image retrieval simplifies to image-to-image retrieval, while at $\lambda = 1$, it simplifies to text-to-image retrieval.

ZERO-SHOT SCENE CLASSIFICATION

Zero-shot scene classification challenges models to identify remote sensing images from scene categories that were not seen during training. The idea of applying vision-language foundation models to this task is straightforward; since these models align images and texts, zero-shot scene classification can be achieved by comparing image embeddings with text embeddings extracted by the text encoder,

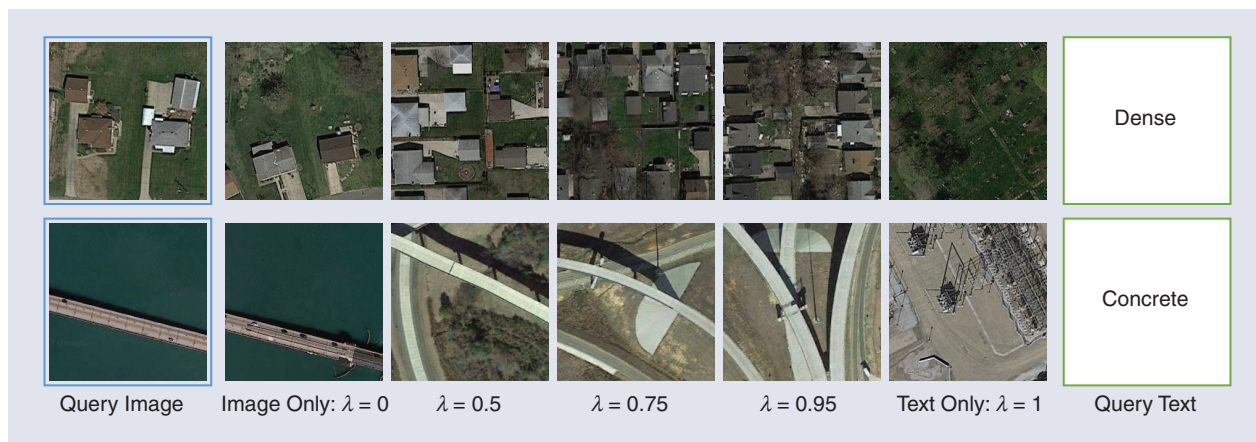


FIGURE 4. An illustration of composed image retrieval for remote sensing [91].

which takes as input textual descriptions specifying the unseen classes. By default, these descriptions follow the format “a photo of a [CLASS],” where the class token is replaced by the specific class name, such as “farmland,” “forest,” or “playground.” The input text, also known as the *prompt*, plays a crucial role in the performance of foundation models on downstream tasks. As a result, several works [35], [37] pay attention to prompt tuning and suggest adding task-relevant context words to improve performance. An example is provided by [37], where the authors replace “photo” with terms such as “remote sensing image,” “top view image,” “satellite image,” and “scene.” This slight modification results in more than a 5% increase in the accuracy of the CLIP model with the ViT-B/32 vision encoder on the UCM dataset [80]. However, as noted in other experiments from [37], prompts subjected to extensive tuning are not guaranteed to be optimal for the task and may not be suitable for other test datasets.

To avoid manual prompt tuning, Lan et al. [92] model context words in a prompt as learnable vectors, which are combined with the class token embeddings before being input to the text encoder. Moreover, the class token embeddings for each category are organized across multiple levels of granularity, taking the form: “a photo of a ship, the primary type is [CLASS1], secondary type is [CLASS2], final type is [CLASS3],” aimed at the task of few-shot fine-grained ship classification. Compared to hand-crafted prompts, these hierarchical learnable prompts incorporate richer task-specific knowledge.

Most works employ and freeze pretrained CLIP models. However, due to the significant domain gap between web images and remote sensing images, their performance tends to be limited. To mitigate this, one can inject remote sensing domain priors into the vision encoder [92] or fine-tune the entire model using pseudolabeling techniques [35]. Typically, Lan et al. [92] introduce a lightweight network that is trained on data from seen classes to capture the domain prior. This prior is then combined with image embeddings output by the vision encoder, allowing the pretrained model to adapt better. On the other hand, RS-CLIP [35] leverages the strong transferability of the pretrained model to automatically generate pseudolabels from unlabeled images, which are used to fine-tune the model. Additionally, a curriculum learning strategy is developed to gradually select more pseudolabeled images for model training in multiple rounds, further boosting the model’s performance in zero-shot scene classification.

In addition to comparing the similarity between embeddings of images and unseen-class texts, DSVa [38] introduces a solution that uses pretrained models to annotate attributes for each scene class and predict scene categories by evaluating the similarity between attribute values derived from image embeddings and those associated with scene classes. For automatic attribute annotation, textual descriptions have the form of “This photo contains [ATTRIBUTE],” where the attribute token is replaced by specific

attribute names, such as “red,” “cement,” or “rectangle.” Meanwhile, the attribute value for each class is calculated by measuring the similarity between embeddings of attribute text and images belonging to that class.

DENSE PREDICTION TASKS

Dense prediction tasks, such as semantic segmentation and change detection, which produce pixel-level predictions for input images, have recently benefited from the application of vision-language foundation models. Typically, Text2Seg [94] uses a CLIP model to classify category-agnostic segmentation masks generated by the Segment Anything Model (SAM) [112], enabling zero-shot semantic segmentation of remote sensing images. Lin et al. [95] instead modify the CLIPSeg decoder [96] to receive joint image-text embeddings from the CLIP model as input, producing a binary segmentation mask. The decoder is composed of transformer blocks, convolution layers, and linear interpolation layers.

Without the need for additional decoders or segmentation models, one can perform an upsampling operation on image embeddings from CLIP and compare the similarity between image patch embeddings and text embeddings to produce segmentation results. However, empirical findings in [97] suggest that for CLIP with a vision encoder based on ViT-B/16, the image embeddings are downsampled to 1/16 of the input image size. This downsampling leads to distorted object shapes and poorly fitting boundaries in segmentation masks. Furthermore, CLIP’s self-attention causes global information from the class token embedding to be attached to the patch embeddings, which significantly degrades performance in semantic segmentation. To handle these issues, SegEarth-OV [97] incorporates SimFeatUp on top of the CLIP vision encoder to restore lost spatial information in image embeddings. Subsequently, subtraction operations are executed between patch embeddings and the class token embeddings before similarity measurement, alleviating global bias in patch embeddings. In particular, SimFeatUp is a learnable upsampler consisting of a single parameterized Joint Bilateral Upsampling (JBU) operator. It is trained with a frozen CLIP model, a learnable downsampler, and a lightweight content retention network. The training objective is to ensure that the image embeddings, after the up-down-sampling process, remain similar to those from CLIP. Meanwhile, the image generated by the content retention network, which takes the upsampled image embeddings as input, should closely resemble the input image to CLIP.

When it comes to change detection, similar adaptation strategies used in segmentation tasks, such as combining SAM with CLIP and designing decoders, are applied. For instance, in SCM [98], CLIP is applied after SAM to identify objects of interest in bitemporal images, helping to filter out pseudo-changes. In ChangeCLIP [99], CLIP is used to construct and encode multimodal input data for change detection tasks, while a transformer-based decoder combines

bitemporal vision-language features with image features to produce change maps. To be specific, the authors predefine 56 common land cover categories in remote sensing images and use the CLIP model to identify bitemporal images from these categories. Each image is paired with textual descriptions of the foreground and background, formatted as “remote sensing image foreground objects, [Predict Classes]” and “remote sensing image background objects,” respectively. This way allows bitemporal texts to highlight the changing object, providing additional information for change detection.

Alternatively, only the vision encoders of foundation models are adopted as most existing datasets provide only bitemporal images for identifying changes. In BAN [100], the authors introduce bridging modules to inject general knowledge extracted from the vision encoders of foundation models into existing change detection models like BiT [113] and ChangeFormer [114]. Since the input image sizes for foundation models and change detection models may differ and not all general knowledge contributes to predicting changes, the bridging modules are responsible for selecting, aligning, and injecting this knowledge. As shown in Figure 5, these modules consist of three main components: layer normalization and a linear layer to mitigate the distribution inconsistency between the image features from the two models, cross-attention to obtain valuable information from general knowledge, and bilinear interpolation to solve the misalignment problem. The bridging modules are placed between the two encoders, executing multi-level knowledge injection.

BROADER SCOPE OF APPLICATION

In addition to the remote sensing tasks previously discussed, there are several emerging applications of contrastive-based vision-language foundation models, including the following:

- 1) *VQA*: VQA attempts to provide answers to questions related to the content of images. In [101], CLIP is used to extract both visual and textual representations from images and questions. Two decoders, followed by two classifiers, are then developed to capture the intradependencies and interdependencies within and between these representations. The final answer is determined by averaging the predictions from both classifiers.
- 2) *Cloud presence detection*: Cloud presence detection involves identifying satellite images that are affected by clouds. The authors in [102] explore several strategies to adapt CLIP for this task. One approach, similar to zero-shot scene classification [37], involves selecting prompts such as “This is a satellite image with clouds” and “This is a satellite image with clear sky” and then classifying satellite images via similarity measurement. Another strategy follows CoOp [115], combining learnable context with the class token embeddings. Alternatively, the vision encoder can be employed by itself, with a linear classifier added on top. Unlike the first strategy, the other approaches necessitate training for prompts or classifiers.
- 3) *Text-based image generation*: Text-based image generation refers to the task of creating images from textual descriptions, which can help mitigate the issue of class imbalance commonly found in remote sensing data. Typically, TGN [103] utilizes CLIP to classify generated images,

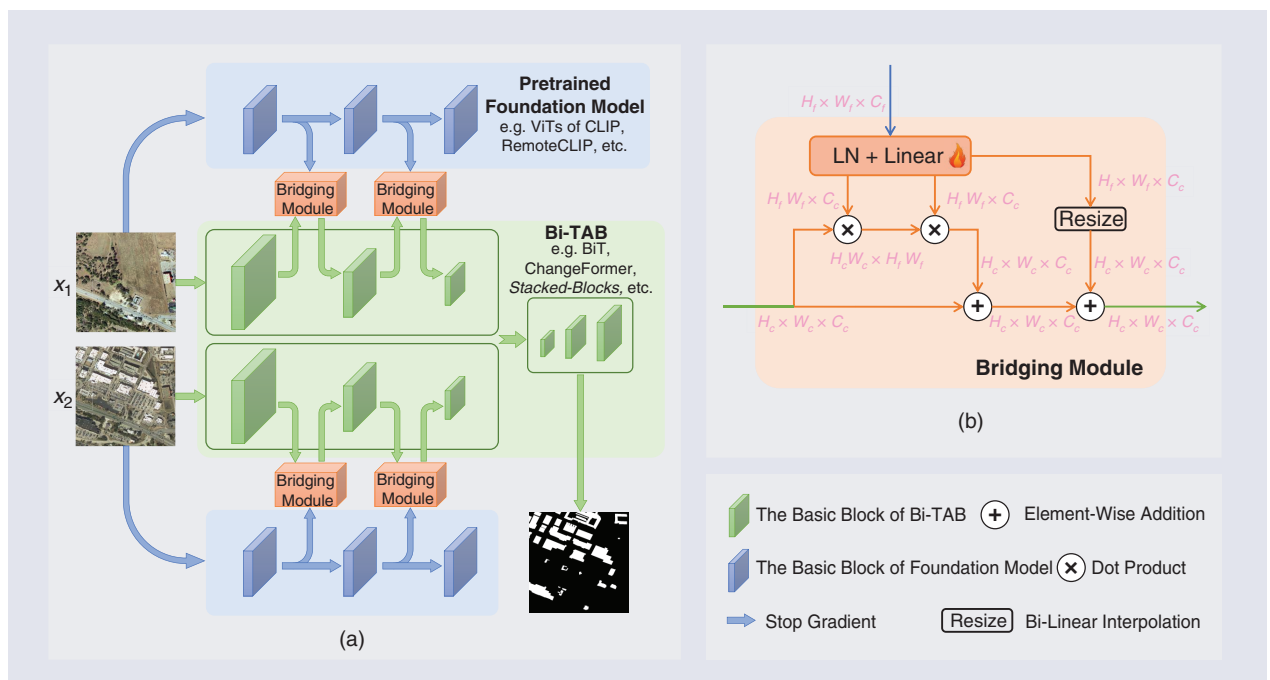


FIGURE 5. Illustration of BAN architecture [100]. (a) The detailed structure of BAN. (b) The detailed structure of the bridging module. LN: layer normalization.

ensuring semantic class consistency between input images and generated ones.

- 4) **Image address localization:** Image address localization aims to predict the readable textual address where an image was taken [104]. Different from image geo-localization in [61], which predicts GPS coordinates (i.e., latitude and longitude) from images, this task outputs semantic address text such as “Grant Street, Downtown.” Utilizing existing foundation models like CSP [58] and GeoCLIP [61], a viable solution is predicting GPS coordinates with a foundation model and then converting them into readable addresses. However, this mapping from coordinates to addresses often presents ambiguities. A recent work, AddressCLIP [104], introduces a novel idea for performing image address localization in an end-to-end manner. In particular, AddressCLIP applies contrastive learning to align images with scene captions and address texts. Furthermore, image-geography matching is developed to bring features of geographically proximate images closer together while distancing features of images that are far apart geographically. Figure 6 shows the framework of AddressCLIP.

INSTRUCTION-BASED VISION-LANGUAGE MODELING

Today, instruction-based VLM is advancing rapidly. Since 2023, more than 10 impressive vision-language models have emerged, as shown in Table 4. They are not only versatile, capable of performing a range of remote sensing image analysis tasks, but also able to interact with users in a conversational manner. This broadens the accessibility of intelligent models beyond experts in remote sensing, facilitating their widespread deployment and application. This

section presents critical developments in terms of *model architecture*, and *training strategy*, *model capability*.

MODEL ARCHITECTURE

As CLIP has been trained to align image and text representations, most works [18], [24], [25], [42], [132] directly employ its vision encoder. The commonly used LLMs include the LLaMA family [119], [130] and its derivative, the Vicuna family [118]. Developed by Meta, the first-generation LLaMA [130] adopts transformer architectures ranging from 7 to 65 billion parameters, pretrained on approximately 1.4 trillion tokens of publicly available text corpora. Despite its smaller scale, LLaMA outperforms proprietary models like GPT-3 [108]. Its successor, LLaMA 2 [119], introduces significant improvements, including expanded pretraining data (2 trillion tokens), architectural upgrades such as grouped-query attention [136], and a scaled-up 70-billion-parameter variant. Through SFT and reinforcement learning with human feedback, LLaMA 2 was further optimized for dialogue, yielding the LLaMA 2-Chat series [119]. Parallel to this development, the research community developed Vicuna [118] by fine-tuning LLaMA models on 70,000 conversations from ShareGPT. These models achieve more than 90% of ChatGPT’s quality. Regarding model architecture, existing works are devoted to improving visual encoders to enhance visual perception and designing connectors to promote the alignment between the two modalities. The following encoders were developed.

VISION ENCODER

Through a mask image modeling pretext task, EVA [116] incorporates geometry and structure information into CLIP’s visual representations, leading to improved performances

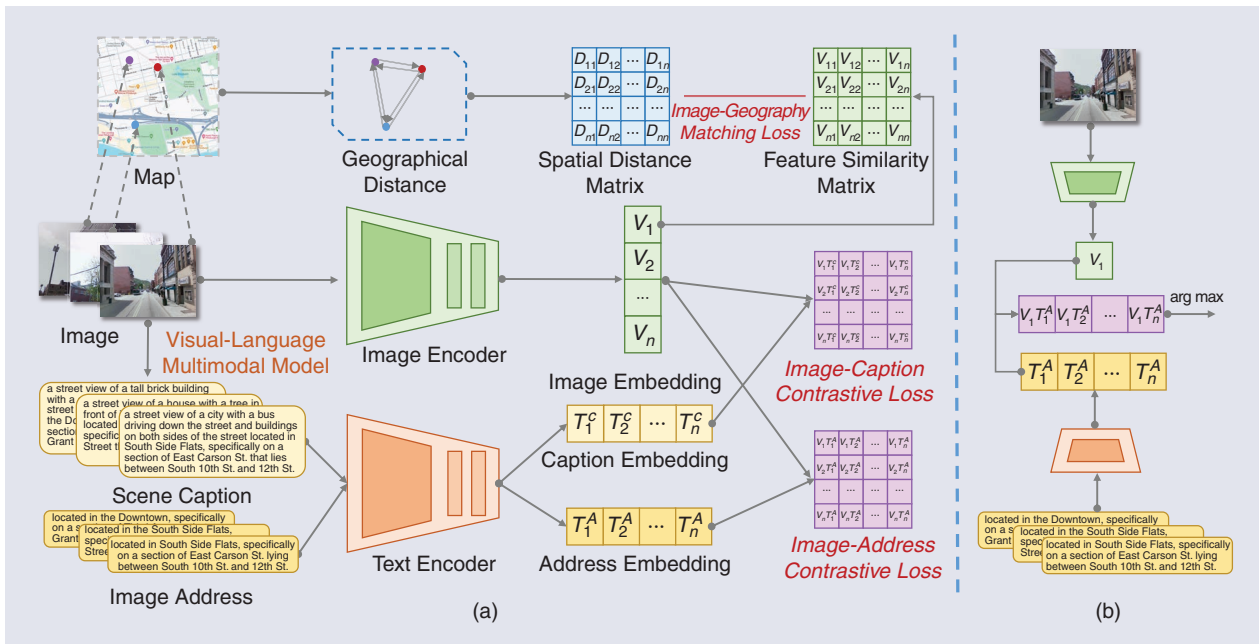


FIGURE 6. An illustration of AddressCLIP framework [104].

TABLE 4. SUMMARY OF INSTRUCTION-BASED VISION-LANGUAGE MODELS IN REMOTE SENSING.

MODEL	VISION ENCODER	CONNECTOR	LLM	TRAINING DATASET	TRAINING STRATEGY	PUBLIC
RSGPT [17]	EVA CLIP-G [116]	Q-Former [117] +Linear layer	Vicuna-7B/13B [118]	RSICap [17]	- Load pretrained weights of InstructBLIP [117]. - Fine-tune Q-Former and linear layer.	X
SkyEyeGPT [43]	EVA CLIP-G [116]	Linear layer	LLaMA 2-Chat [119]	SkyEye-968k [43]	- Load pretrained weights of [116] and [120]. - Employ LoRA [121] for linear layer and LLM in two stages.	X
EarthGPT [122]	DINOv2 ViT-L/14 [123] + CLIP ConvNeXt-L [27]	Linear layer	LLaMA 2-13B [119]	LAION-400M [124] COCO Caption [125] MMRS-1M [122]	- Load pretrained weights of [27], [119], and [123]. - Train linear layer and LLM on LAION-400M and COCO Caption. - Employ bias tuning [126] for LLM on MMRS-1M.	X
EarthMarker [44]	DINOv2 ViT-L/14 [123] +CLIP ConvNeXt [27]	Projection layer	LLaMA 2-13B [119]	COCO Caption [125] RSVP-3M [44] RefCOCO [127] RefCOCO+ [128]	- Load pretrained weights of [27], [119], and [123]. - Train projection layer on COCO Caption and RSVP subset. - Fine-tune attention layers of LLM on RefCOCO and RefCOCO+. - Employ LoRA [121] for LLM on RSVP-3M.	X
Popeye [129]	DINOv2 ViT-L/14 [123] +CLIP ViT-L/14 [27]	Projection layer	LLaMA-7B [130]	COCO Caption [125] MMShip [129]	- Load pretrained weights of [27], [123], and [130]. - Train projection layer and employ LoRA [121] for LLM on COCO Caption. - Fine-tune the projection layer and employ LoRA and instruction adapters for LLM on MMShip.	X
LHRS-Bot [24] ^a	CLIP ViT-L/14 [27]	Vision perceiver	LLaMA 2-7B [119]	LHRS-Align [24] LHRS-Instruct [24] Multitask dataset [24] LLaVA-Instruct-158K [23]	- Load pretrained weights of [27], and [119]. - Train vision perceiver on LHRS-Align. - Fine-tune vision perceiver and employ LoRA [121] for LLM on LHRS-Instruct, multitask dataset. - Fine-tune LLM on LHRS-Instruct, the multitask dataset, and the LLaVA-Instruct-158K subset.	✓
VHM [42] ^b	CLIP ViT-L/14 [27]	MLP	Vicuna-v1.5-7B [118]	VersaD [42] VersaD-Instruct [42] HnstD [42] VariousRS-Instruct [42]	- Load pretrained weights of [27], and [118]. - Train vision encoder, MLP, and LLM on VersaD. - Fine-tune MLP and LLM on VersaD-Instruct, HnstD, and VariousRS-Instruct.	✓
GeoChat [18] ^c	CLIP ViT-L/14 [27]	MLP	Vicuna-v1.5-7B [118]	Multimodal Instruction Dataset [18]	- Load pretrained weights of [93], [118], and [131]. - Employ LoRA [121] for LLM.	✓
RS-LLaVA [132] ^d	CLIP ViT-L@336 [27]	MLP	Vicuna-v1.5-7B/13B [118]	RS-Instructions [132]	- Load pretrained weights of [27] and [118]. - Train MLP on a general image-language dataset. - Employ LoRA [121] for LLM.	✓
SkySenseGPT [25] ^e	CLIP ViT-L/14 [27]	MLP	Vicuna-v1.5 [118]	FIT-RS [25] Additional instruction dataset [25]	- Load pretrained weights of [27], [118], and [131]. - Fine-tune MLP and employ LoRA [121] for LLM.	✓
IFShip [133]	CLIP ViT-L/14 [27]	MLP	Vicuna-13B [118]	TITANIC-FGS [133]	- Load pretrained weights of [27] and [118]. - Employ LoRA [121] for MLP and LLM.	X
TEOChat [134] ^f	CLIP ViT-L/14 [27]	MLP	LLaMA 2 [119]	TEOChatlas [134]	- Load pretrained weights of [27] and [135]. - Employ LoRA [121] for LLM.	X

^a<https://github.com/NJU-LHRS/LHRS-Bot>; ^b<https://github.com/opendatalab/VHM>; ^c<https://github.com/mbzuai-oryx/geochat>; ^d<https://github.com/BigData-KSU/RS-LLaVA>; ^e<https://github.com/Luo-Z13/SkySenseGPT?tab=readme-ov-file>; ^f<https://github.com/ermongroup/TEOChat?tab=readme-ov-file>.

LoRA: Low-Rank Adaptation. Detailed information about training datasets can be found in Tables 12 and 13. "Public" refers to the availability of both code and model weights.

across a wide range of visual perception tasks. Consequently, RSGPT [17] and SkyEyeGPT [43] adopt EVA as their vision encoder. An alternative to complementing CLIP is utilizing the strengths of diverse vision encoders. CLIP

learns visual representations through language supervision, which limits the information retained about the image. Captions only approximate the main content of images, often failing to present complex pixel-level details.

Targeting this limitation, some works [44], [122], [129] propose combining vision encoders from CLIP and DINOv2 [123]. DINOv2 learns visual representation from images alone via self-supervised learning, enabling it to capture both image-level and pixel-level information. Moreover, given the varied object sizes in remote sensing images, these works refine visual representations by incorporating multiscale information. In [122], a vision encoder with the convolutional neural network architecture is used to extract multiscale visual features. In [44] and [129], the input image is downsampled to different resolutions and then respectively fed into two vision encoders. The encoded visual features are transformed to the same dimension and concatenated channel-wise.

VISION-LANGUAGE CONNECTOR

The linear layer and MLP are widely used as vision-language connectors, serving as key components in most models [18], [25], [42], [43], [122], [132], [133], [134]. In contrast, RSGPT [17] and LHRS-Bot [24] explore alternative connector architectures. Following InstructBLIP [117], RSGPT includes an instruction-aware query transformer (Q-Former) as an intermediate module between the vision encoder and LLM. The Q-Former is designed to extract task-relevant visual representations by interacting additional query embeddings with instruction and image embeddings. It achieves this through the attention mechanism; first, self-attention is applied to implement interaction between instruction and query embeddings, and then cross-attention is employed between the query and image embeddings. The resulting output from the Q-Former, after passing through a linear layer, is then input into the LLM along with the instruction embeddings to generate responses. LHRS-Bot proposes incorporating multilevel image embeddings to sufficiently capture the semantic content of images. To mitigate the computational burden and the risk of overwhelming language information with excessive visual information, it introduces a set of learnable queries for each level of visual representation. These queries are used to summarize the semantic information of each level through stacked cross-attention and MLP layers. As a result, a dedicated visual perceiver is developed, with experimental results demonstrating that, compared to single-level visual representations and a two-layer MLP, multilevel representations paired with the visual perceiver improve the model's performance in scene classification, VQA, and visual grounding (VG) tasks.

TRAINING STRATEGY

Training instruction-based vision-language models typically involves two stages: pretraining for modality alignment and SFT for following task-specific instructions.

ONLY SUPERVISED FINE-TUNING

Due to the lack of large-scale image-text datasets specifically designed for the remote sensing domain, most works

target the SFT stage using carefully crafted instruction-following datasets [17, [18], [25], [43], [133], [134]. To preserve the general knowledge embedded in pretrained vision encoders, the vision encoder is typically kept frozen during training, with the connector or the LLM undergoing fine-tuning. For instance, RSGPT [17] fine-tunes the connector, while GeoChat [18] and TEOChat [134] fine-tune the LLM. SkyEyeGPT [43], SkySenseGPT [25], and IFShip [133] fine-tune both the connector and LLM. To avoid the expense of full-parameter tuning, Low-Rank Adaptation (LoRA) [121] is often adopted, which introduces low-rank learnable matrices into the layers of the connector [43], [133] or LLM [18], [25], [43], [133], [134].

PRETRAINING FOLLOWED BY SUPERVISED FINE-TUNING

A couple of recent works have investigated how to implement the pretraining stage to boost model performance. Based on the choice of training data, they can be categorized into two groups: those that combine available image-text pairs from multiple domains for pretraining and those that direct attention toward creating large-scale RS image-text datasets. For combining available data, EarthGPT [122], Popeye [129], and RS-LLaVA [132] utilize natural image-text datasets, while EarthMarker [44] integrates data from both computer vision and remote sensing domains. COCO Caption [125] is a commonly used pretraining dataset. Unlike RS-LLaVA, which limits pretraining to the connector, the other three models perform pretraining for the connector and the LLM.

To address the domain gap between remote sensing images and natural images, researchers have developed large-scale RS image-text datasets, such as LHRS-Align [24] and VersaD [42], both of which contain more than 1 million training samples. Leveraging the LHRS-Align dataset, Muhtar et al. [24] design a three-stage curriculum learning strategy for LHRS-Bot. First, the vision perceiver is pretrained on LHRS-Align. Next, the vision perceiver and LLM are fine-tuned on the LHRS-Instruct subset and multitask dataset, equipping the model to handle multimodal tasks. Finally, the LLM undergoes additional fine-tuning on all instruction data from LHRS-Instruct, the multitask dataset, and the LLaVA-Instruct-158K subset to fully unlock LHRS-Bot's capabilities. Note that these models still keep the vision encoder frozen. In contrast, Pang et al. [42] unfreeze the vision encoder, MLP, and LLM for pretraining on VersaD. They subsequently fine-tune the MLP and LLM on customized datasets, including VersaD-Instruct, Hnstd, and VariousRS-Instruct. Their experimental results show that the model pretrained with RS data significantly outperforms the one only fine-tuned with RS data across multiple tasks, confirming the importance of incorporating extensive RS visual knowledge by pretraining.

MODEL CAPABILITY

Table 5 lists the capabilities of existing instruction-based vision-language models in remote sensing. Key observations include the following:

- 1) Most models are developed for general-purpose remote sensing data analysis [18], [24], [25], [42], [43], [44], [122], [134], with only Popeye [129] and IFShip [133] specifically tailored for remote sensing images of ships. These models primarily process optical images but have expanded to include synthetic aperture radar (SAR) [122], [129] and infrared (IR) [122] images.
- 2) Most models support conversational interaction with users and can perform a variety of tasks [18], [24], [25], [42], [43], [44], [122], [134], ranging from single-image analysis to temporal series analysis [134].
- 3) The granularity of analyzed information has progressed from the image level to the region level [25], [44] and even to the point level [44].

In this section, we examine the models' performance on common capabilities and offer an in-depth exploration of unique capabilities.

COMMON CAPABILITIES

As shown in Table 5, most models are capable of performing remote sensing tasks such as image captioning, scene classification, VQA, and VG. Accordingly, we present model performance on these tasks in Tables 6–8. To ensure a fair comparison, we report only the models' performance on public datasets. Although multiturn conversation (MTC) is also a common capability among existing models [18],

[24], [25], [42], [43], [44], [122], [129], [134], it is challenging to evaluate quantitatively. Figure 7 provides examples of conversations between various models and users.

- 1) *Image captioning*: Based on the input remote sensing image and the language instruction, vision-language models generate a description of image content. Examples of instructions include "Describe this image in detail" [17], [43] or "Please provide a one-sentence caption for the provided remote sensing image in detail" [122]. From Table 6, we can observe that SkyEyeGPT [43] outperforms RSGPT [17] on the RSICD [78], UCM-captions [3], and Sydney-captions [3] datasets in terms of BLEU [137], METEOR [138], and ROUGE_L [139], though it falls short of RSGPT in CIDEr [140] scores. On the UCM-captions dataset, RS-LLaVA achieves BLEU scores comparable to SkyEyeGPT while surpassing it in all other metrics. On the NWPU-Captions [141] dataset, EarthGPT [122] achieves state-of-the-art results, with a remarkable improvement in CIDEr (nearly 30%) over EarthMarker [44].
- 2) *Scene classification*: Table 7 presents evaluations on 10 remote sensing scene classification datasets under both fully supervised and zero-shot settings. The absorption of extensive remote sensing visual knowledge improves model accuracy and generalization in scene classification tasks. Specifically, in the supervised setting, VHM [42] and EarthGPT [122] boost classification

TABLE 5. CAPABILITY COMPARISONS OF INSTRUCTION-BASED VISION-LANGUAGE MODELS IN REMOTE SENSING.

MODEL	SIT	CAP.	CLS	VQA	VG	OD	ORR	MTC	OTHERS
	Opt./SAR/IR	Img./Reg./Pt.	Img./Reg./Pt.						
RSGPT [17]	✓/x/x	✓/x/x	x/x/x	✓	x	x	x	x	x
SkyEyeGPT [43]	✓/x/x	✓/x/x	✓/x/x	✓	✓	✓	x	✓	Video captioning Referring expression generation
EarthGPT [122]	✓/✓/✓	✓/✓/✓	✓/x/x	✓	✓	✓	x	✓	x
EarthMarker [44]	✓/x/x	✓/✓/✓	✓/✓/✓	x	x	✓	✓	✓	x
Popeye [129] [†]	✓/✓/✓	✓/x/x	x/x/x	x	x	✓	x	✓	x
LHRS-Bot [24]	✓/x/x	✓/x/x	✓/x/x	✓	✓	x	✓	✓	Object counting Object attribute recognition Image property recognition
VHM [42]	✓/x/x	✓/x/x	✓/x/x	✓	✓	x	✓	✓	Object counting Geometric measurement Building vectorizing Image property recognition Multilabel classification Honest question answering
GeoChat [18]	✓/x/x	✓/✓/✓	✓/x/x	✓	✓	x	x	✓	x
RS-LLaVA [132]	✓/x/x	✓/x/x	x/x/x	✓	x	x	x	x	x
SkySenseGPT [25]	✓/x/x	✓/✓/✓	✓/x/x	✓	✓	✓	✓	✓	Multilabel classification Image/region scene graph generation
IFShip [133] [†]	✓/x/x	✓/x/x	✓/x/x	✓	x	x	x	✓	x
TEOChat [134]	✓/x/x	✓/✓/✓	✓/x/x	✓	✓	x	x	✓	Change detection Temporal scene classification Temporal referring expression Spatial change referring expression Image/region change question answering

[†]Indicates the model's capability tailored for remote sensing images of ships. SIT: supported image type; Cap: captioning; CLS: classification; OD: object detection; ORR: object relationship reasoning; MTC: multiturn conversation. We present the Opt./SAR/IR column to indicate whether the model supports input of optical (Opt.), SAR, or infrared (IR) images, and we present the Img./Reg./Pt. column to denote the granularity of the analyzed information, which includes image level (Img.), region level (Reg.), or point level (Pt.).

performance to more than 93% on the NWPU-RESISC45 [56] dataset, which includes 45 scene categories with image spatial resolutions ranging from 0.2 m to more than 30 m. For zero-shot scene classification, LHRS-Bot [24], VHM, and SkySenseGPT [25] demonstrate impressive generalization, achieving accuracy above 91% on the AID [70] dataset and more than 93% on the WHU-RS19 [142] dataset.

SkySenseGPT notably achieves the highest accuracy, reaching 92.25% and 97.02% on these datasets, respectively. Despite these remarkable results, low-resolution and fine-grained scene classification remain significant challenges. The EuroSAT [71] dataset, used for land use and land cover classification, has an image spatial resolution of 10 m, while fMoW [60] contains more than 1 million images across 63 scene categories. On these datasets, LHRS-Bot [24] achieves an accuracy below 57%.

- 3) VQA: The RSVQA-HR [4] and RSVQA-LR datasets are widely adopted to assess model performance in VQA tasks, as shown in Table 7. Under a supervised setting, RS-

GPT [17] establishes strong performance baselines on both datasets. Subsequent models, including EarthGPT [122], GeoChat [18], VHM [42], and SkySenseGPT [25], have progressively approached these benchmarks in both supervised and zero-shot settings. The latest model, SkySenseGPT, narrows the zero-shot performance gap to approximately 13%, coming closest to the baseline on

TABLE 6. THE PERFORMANCE OF INSTRUCTION-BASED VISION-LANGUAGE MODELS ON IMAGE CAPTIONING.

MODEL	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr
RSICD [78]							
RSGPT [17]	70.32	54.23	44.04	36.83	30.1	53.34	102.94
SkyEyeGPT [43]	86.71	76.66	67.31	59.99	35.35	62.63	83.65
UCM-captions [3]							
RSGPT [17]	86.12	79.14	72.31	65.74	42.21	78.34	333.23
RS-LLaVA [132]	90	84.88	80.3	76.03	49.21	85.78	355.61
SkyEyeGPT [43]	90.71	85.69	81.56	78.41	46.24	79.49	236.75
Sydney-captions [3]							
RSGPT [17]	82.26	75.28	68.57	62.23	41.37	74.77	273.08
SkyEyeGPT [43]	91.85	85.64	80.88	77.4	46.62	77.74	181.06
NWPU-Captions [141]							
EarthMarker [44]	84.4	73.1	62.9	54.3	37.5	70	162.9
EarthGPT [122]	87.1	78.7	71.6	65.5	44.5	78.20	192.6

BLEU, METEOR, ROUGE_L, and CIDEr are short for Bilingual Evaluation Understudy [137], Metric for Evaluation of Translation with Explicit Ordering [138], Recall-Oriented Understudy for Gisting Evaluation [139], and Consensus-based Image Description Evaluation [140]. Best results are highlighted in bold.

TABLE 7. THE PERFORMANCE OF INSTRUCTION-BASED VISION-LANGUAGE MODELS ON SCENE CLASSIFICATION AND VQA.

SCENE CLASSIFICATION: TOP-1 ACCURACY										
MODEL	RESISC [56]	CLRS [145]	NaSC-TG2 [146]	UCM [80]	AID [70]	WHU-RS19 [142]	SIRI-WHU [79]	EuroSAT [71]	METER-ML [147]	fMoW [60]
GeoChat [18]	—	—	—	84.43 [†]	72.03 [†]	—	—	—	—	—
EarthMarker [44]	—	—	—	86.52[†]	77.97 [†]	—	—	—	—	—
EarthGPT [122]	93.84	77.37[†]	74.72[†]	—	—	—	—	—	—	—
LHRS-Bot [24]	83.94	—	—	—	91.26 [†]	93.17 [†]	62.66 [†]	51.4[†]	69.81	56.56
VHM [42]	94.54	—	—	—	91.7 [†]	95.8 [†]	70.88 [†]	—	72.74	—
SkySenseGPT [25]	—	—	—	—	92.25[†]	97.02[†]	74.75[†]	—	—	—
VQA										
RSVQA-HR [4] TEST SET 1			RSVQA-HR [4] TEST SET 2				RSVQA-LR [4]			
MODEL	PRESENCE	COMPARISON	AVG. ACC.	PRESENCE	COMPARISON	AVG. ACC.	PRESENCE	COMPARISON	RURAL/URBAN	AVG. ACC.
SkyEyeGPT [43]	84.95	85.63	85.29	83.5	80.28	81.89	88.93	88.63	75	84.19
RSGPT [17]	91.86	92.15	92	89.87	89.68	89.78	91.17	91.7	94	92.29
LHRS-Bot [24]	—	—	—	—	—	—	88.51	90	89.07	89.19
RS-LLaVA [132]	—	—	—	—	—	—	92.27	91.37	95	88.1
GeoChat [18]	—	—	—	58.45 [†]	83.19 [†]	72.3 [†]	91.09	90.33	94	90.7
EarthGPT [122]	—	—	—	62.77 [†]	79.53 [†]	72.06 [†]	—	—	—	—
VHM [42]	—	—	—	64 [†]	83.5 [†]	73.75 [†]	90.11	89.89	88	89.33
SkySenseGPT [25]	—	—	—	69.14 [†]	84.14 [†]	76.64 [†]	91.07	92	95	92.69

RESISC: NWPU-RESISC45 [56]; Avg. Acc.: average accuracy. [†]indicates zero-shot performance. Best results are highlighted in bold.

TABLE 8. PERFORMANCE OF INSTRUCTION-BASED VISION-LANGUAGE MODELS ON VG.

MODEL	RSVG [143]		DIOR-RSVG [144]
	Val	Test	
SkyEyeGPT [43]	69.19	70.5	88.59
EarthGPT [122]	—	—	76.65
LHRS-Bot [24]	—	73.45	88.1
VHM [42]	—	—	56.17

The evaluation metric adopts Acc@0.5, which means that the intersection over union (IoU) of the predicted box is at least 0.5 with the ground truth bounding box.

test set 2 of the RSVQA-HR dataset and surpassing RS-GPT on the RSVQA-LR dataset in the supervised setting.

4) VG: VG aims to locate specific objects within an image based on a natural language expression. Vision-language models accomplish this task by providing the coordinates of the target object. Table 8 presents the performance of several models on RSVG [143] and DIOR-RSVG [144] datasets, using an intersection over union (IoU) threshold of 0.5 adopted as the evaluation metric. LHRS-Bot [24] achieves the highest score on the RSVG test set, while SkyEyeGPT [43] leads on the DIOR-RSVG test set. Due to differences in metric calculations, VHM [42] achieves a score of 57.17% on DIOR-RSVG.

New chat (First round)

User: 

User: Describe this image in detail

SkyEyeGPT: This is a high-resolution aerial image showing a baseball field on the left side of the image with two baseball fields, and two tennis courts on the right side of the image, there's a large grassy area surrounded by trees.

User: [refer] where can I locate the baseball field at the top?

SkyEyeGPT: 

User: [detection] the two brown baseball fields

SkyEyeGPT: 

User: [vqa] How many tennis courts are there?

SkyEyeGPT: 2

User: [caption] Write a short description for the image.

SkyEyeGPT: two tennis courts with a baseball field in the background

User: [identify] This {<62><50><81><69>} is

SkyEyeGPT:  two tennis courts

User: [vqa] Classify the image within one of the given classes: sports field, forest, river, parking lot, overpass, baseball diamond, freeway, agricultural, chaparral, beach, buildings, intersection, dense residential, sparse residential, medium residential, airplane, tennis court, mobile home park, golf course, storage tanks, runway, harbor. Answer it using the above category word or phrase.

SkyEyeGPT: sports field

New chat (Second round)

User: 

User: Describe the video frame image in detail.

SkyEyeGPT: The scene features a large body of water, with several boats filled with people rowing in unison and forming lines. Some of the boats are carrying flags, and the participants can be seen wearing uniforms. There is a parking lot and several buildings near the shore, possibly serving as a viewing area for the race. In the top-left corner, there is a truck and a car parked next to each other, suggesting that they might have been used to transport equipment or spectators to the event.

User: [vqa] what is the event of this video frame image?

SkyEyeGPT: boating

User: [caption] Provide a concise depiction of this video frame image.

SkyEyeGPT: people are rowing in boats on a river surrounded by buildings

(a)

User: What can you see from this image?

SkyEyeGPT: From this image, I can see a beautiful and well-organized city with many water bodies, including lakes. The city appears to be a mix of residential and commercial areas, with several buildings of different heights and architectural styles. Additionally, there is a golf course, which suggests that the city provides leisure activities for its residents.

User: How many water bodies can you see?

SkyEyeGPT: Based on the image, I can see at least five water bodies.

User: [VG] The coordinates of the largest water bodies.

SkyEyeGPT: [0.02, 0.25, 0.24, 0.42]

User: [CLS] Residential or Commercial?

SkyEyeGPT: Residential

User: Is there any playground? If have, where is it?

SkyEyeGPT: Yes, there is a playground in the image. It can be found on the right side of the road.

User: [VG] Playground

SkyEyeGPT: [0.82, 0.50, 0.99, 0.83]

User: Describe the primary type of infrastructure visible in the center of the image.

SkyEyeGPT: The image prominently features a highway interchange with multiple lanes.

User: How many major roadways can you identify in the image?

SkyEyeGPT: There appear to be at least two major roadways intersecting. One is a multi-lane highway running vertically, and the other is a horizontal roadway crossing it.

User: Can you spot any residential areas, and how are they arranged in relation to the main roads?

SkyEyeGPT: There is a residential area visible to the left (west) side of the vertical highway. The homes are arranged in a typical suburban pattern, with streets laid out in a block formation.

User: What can you infer from this image?

SkyEyeGPT: From this image, I can infer that the area depicted is a densely populated residential neighborhood with many houses and trees. The neat arrangement of the houses and the presence of roads suggest a well-planned and organized community. The image also suggests that the area is relatively affluent, as there are many white buildings and a dense concentration of houses.

User: How many buildings?

SkyEyeGPT: There are many buildings in the dense residential area. The exact number is not visible, but there appear to be at least a dozen or more buildings in the area.

User: What type of facility is depicted in the image?

SkyEyeGPT: The image appears to show an airport terminal with a large, open area and multiple baggage claim areas.

User: Describe the layout of the terminal buildings.

SkyEyeGPT: The terminal buildings have a large, open layout with several rows of gates and baggage claim areas. The layout suggests that the airport is well-organized and efficiently designed to accommodate a high volume of passengers and aircraft.

(b)

FIGURE 7. Examples of conversations between vision-language models and users. (a) SkyEyeGPT [43]. (b) LHRS-Bot [24].

UNIQUE CAPABILITIES

Current research seeks to develop versatile vision-language models capable of handling various remote sensing image analysis tasks in a conversational manner. For instance, EarthMarker [44] supports not only language instructions but also visual prompts (e.g., boxes and points), enabling the model to perform fine-grained image understanding, such as region/point-level captioning and referring object classification. TEOChat [134] excels in analyzing time-series remote sensing images, detecting changes of interest (in the form of the bounding box), and answering questions related to changes. Coincidentally, LHRS-Bot [24] and VHM [42] both feature the ability to qualitatively recognize object attributes (e.g., color) and image properties (e.g., resolution and modality) while also quantitatively counting objects. Additionally, VHM offers insights into model honesty, which is vital for applications such as national defense security. Beyond single-object analysis, relationship analysis between objects is increasingly recognized as essential for understanding complex remote sensing scenes, drawing attention in recent models [24], [25], [42], [44]. This section delves into these unique capabilities, with a focus on their implementations as detailed.

- 1) *Fine-grained image understanding*: Region-level image understanding is challenging but achievable. One can include the coordinates of the target region in language instructions to direct the model's attention to a specific local region, as demonstrated in EarthGPT [122] and GeoChat [18]. While using precise coordinates is effective, it lacks flexibility and is challenging to extend to the point level. In contrast, EarthMarker [44] leverages visual prompting marks to guide the model to interpret specific regions or points. To ensure that the model understands the relationship between visual prompts and the whole image, the visual prompts share the same visual encoder and projection layer with the input image. Their embeddings are combined with instruction embeddings before being fed into the LLM. Visual prompts allow EarthMarker to perform multigranularity RS image interpretation at the image, region, and point levels.
- 2) *Time-series image analysis*: Change detection, the process of identifying changes on Earth's surface from sequences of remote sensing images captured over the same area, plays an important role in many applications, such as urban planning [148] and war damage assessment [149]. The work TEOChat [134] pioneers change detection in a conversational manner. Given a sequence of remote sensing images, TEOChat employs a shared vision encoder to generate embeddings for each image. These embeddings are then projected to the input space of the LLM via a two-layer MLP. The LLM processes these projected image embeddings alongside instruction embeddings to produce change detection results in the form of bounding boxes. To train TEOChat, the authors of [134] develop TEOChatlas, the first instruction-following dataset tailored for time-series image analysis tasks.

Leveraging this dataset, TEOChat extends its capabilities beyond change detection to include tasks such as change question answering, temporal scene classification, temporal referring expression, and spatial change referring expression. Figure 8 provides specific examples illustrating each of these tasks.

- 3) *From qualitative recognition to quantitative analysis*: Existing models mostly concentrate on qualitative recognition tasks, excelling at describing the attributes or categories of images and objects in a nonnumerical manner. In contrast, LHRS-Bot [24] and VHM [42] broaden their capabilities to include quantitative image analysis, answering "how many" questions. LHRS-Bot is equipped to count objects in an image and estimate image resolution, using single-choice questions with two to four candidate answers. This question format requires a certain level of user expertise as the user needs to predefine several candidate answers, one of which must be correct. VHM goes a step further by not only handling these tasks but also measuring object size. Unlike LHRS-Bot, all quantitative tasks in VHM adopt open-ended question formats, which may be more suitable for practical application. On the test set of the VariousRS-Instruct [42], VHM achieves mean absolute errors of 0.24 for image resolution estimation, 6.75 for object counting, and 12.82 for geometric measurement. These results highlight the potential of vision-language models to advance quantitative analysis in remote sensing images. The successful implementation of such analysis relies heavily on the development of specialized instruction-following datasets, which are discussed in detail in the "Instruction-Following Datasets" section.
- 4) *Endowing models with honesty*: Most instruction-following datasets designed for remote sensing tasks exclusively contain factual questions that query real objects within the images and are accompanied by affirmative answers. This makes vision-language models susceptible to lying, such as providing an affirmative answer regarding a category of a nonexistent object. A dishonest vision-language model is particularly problematic in remote sensing applications, especially in critical areas like national defense security, as it may lead to serious negative consequences, including inaccurate intelligence gathering. Given this, Pang et al. [42] are the first to offer insights into the honesty of models. They create HnstD, an RS-specific honest dataset comprising questions with factual and deceptive categories. By utilizing it as an additional instruction-following dataset, VHM is endowed with honesty. As shown in Figure 9, VHM refuses to answer questions regarding the color or position of nonexistent objects.
- 5) *Object relationship understanding*: Remote sensing images often encompass a vast number of objects owing to their long-distance imaging. Comprehensively understanding the relationships between objects within an image is essential for interpreting complex remote

sensing scenes [150], [151]. Consequently, the latest models, i.e., LHRs-Bot [24], VHM [42], EarthMarker [44], and SkySenseGPT [25], extend their capabilities to understand relationships between objects in an image. Specifically, LHRs-Bot and VHM target spatial relationships, using simple rules like “top” or “top right corner” to indicate object relationships. EarthMarker, on the other hand, explores functional relationships, such as the “overpass” and “toll station” potentially contributing to the transportation infrastructure around the “stadium.” Meanwhile, SkySenseGPT describes semantic relationships, for example, the “crane” is over the “boat.” This promising understanding of relationships benefits from the design of specialized instruction-following datasets, as discussed in the “Instruction-Following Datasets” section.

GENERATION-BASED VISION-LANGUAGE MODELING

Similar to research on contrastive-based VLM, generation-based VLM follows two major research directions: the construction of foundation models concerning the characteristics of remote sensing images and their application to promote various remote sensing data analysis tasks. This section first presents the development of *generative foundation models* and then discusses representative *downstream applications*.

GENERATIVE FOUNDATION MODELS

Building an effective generative foundation model is a formidable task because one needs to consider how to ensure the reliability and diversity of the generated images. Table 9 provides a summary of existing generative foundation models in remote sensing.

ENHANCING RELIABILITY

Text, often in the form of descriptions of ground objects within images [15], [22], [47], [155], [158], [165] or semantic categories of images [21], [48], has been a common condition for conditional image generation. For instance, the text “a satellite view of San Francisco showing the bay, a street, and a bridge” is used to guide RS-SD [15], while the text “a fMoW satellite image of a car dealership in the United States of America” constrains DiffusionSat [21]. However, it is evident that these textual descriptions alone struggle to fully encapsulate the variety of objects and intricate relationships present within a satellite image. The lack of sufficient constraint information poses a challenge to generating reliable images. To address this challenge, additional conditions, such as metadata or images, are increasingly utilized to constrain the generation process.

- 1) *Metadata*: Metadata, such as latitude, longitude, ground sampling distance, cloud cover, and imaging time (year, month, and day), are adopted in both DiffusionSat [21]



FIGURE 8. Examples showcasing the capabilities of TEOChat [134] in time-series image analysis.

and CRS-Diff [22]. In contrast, MetaEarth [49] and Text2Earth [158] focus on spatial resolution, and GeoSynth [155] focuses on geographic location (latitude and longitude). Compared to text conditions, metadata are more easily available as they are inherently embedded within remote sensing images. Furthermore, they allow generative foundation models to be trained on large-scale image datasets, benefiting from the diverse geographic distribution of these datasets. Consequently, the key problem lies in addressing how to inject metadata conditions into diffusion models. Identical to the encoding of diffusion time steps, DiffusionSat and MetaEarth process metadata values through sinusoidal encoding, followed by MLPs. The resulting metadata embeddings are added with time step embeddings before being fed into the diffusion model. CRS-Diff maps metadata values to a fixed range and encodes them using different MLPs. The metadata embeddings are concatenated with text and content embeddings, providing global control information. For latitude and longitude, an alternative approach is to utilize a pretrained location encoder. An example is [155] where the authors employ SatCLIP [64] to extract location embeddings and use CoordNet, which takes location and time step embeddings as input, to integrate these conditions into the diffusion model. CoordNet consists of 13 multihead cross-attention blocks, each including a zero-initialized feed-forward layer.

- 2) *Image*: Metadata, such as geographic location, determines the visual appearance of objects in generated images to some extent. For example, the architectural styles of Chinese and European buildings are notably distinct. How-

ever, metadata primarily impose constraints at a macro level, allowing considerable flexibility in the generated objects, which can result in uncontrollable object shapes. As a result, recent works have investigated the use of image-form conditions to enable more precise control over the image generation process. Based on the information conveyed by the image, image conditions are roughly split into low-level visual conditions and high-level semantic conditions. Low-level visual conditions pertain to the geometric information of images, such as edges, line segments, and sketches, which constrain the shape and structure of objects in generated images, as demonstrated in works [22] and [48]. Off-the-shelf algorithms or models, including holistically nested edge detection (HED) [168], LETR [169], and the model proposed in [170], are employed to obtain these image conditions.

High-level semantic conditions, as the name implies, refer to the semantic information of images, providing constraints on object categories and their relationships in generated images. These conditions can be further categorized into two subgroups. The first group [21], [47], [49], [158] conditions on associated remote sensing images, exemplified by works such as MetaEarth [49], which uses low-resolution images to prompt the diffusion model to produce high-resolution images, and GPG2A [47], which synthesizes aerial images guided by layout maps derived from corresponding ground images. The second group [22], [48], [155] leverages abstract representations of image content, as seen in works like CRS-Diff [22] and GeoSynth [155]. CRS-Diff utilizes road maps, segmentation masks, and deep features to constrain the layout and objects within generated

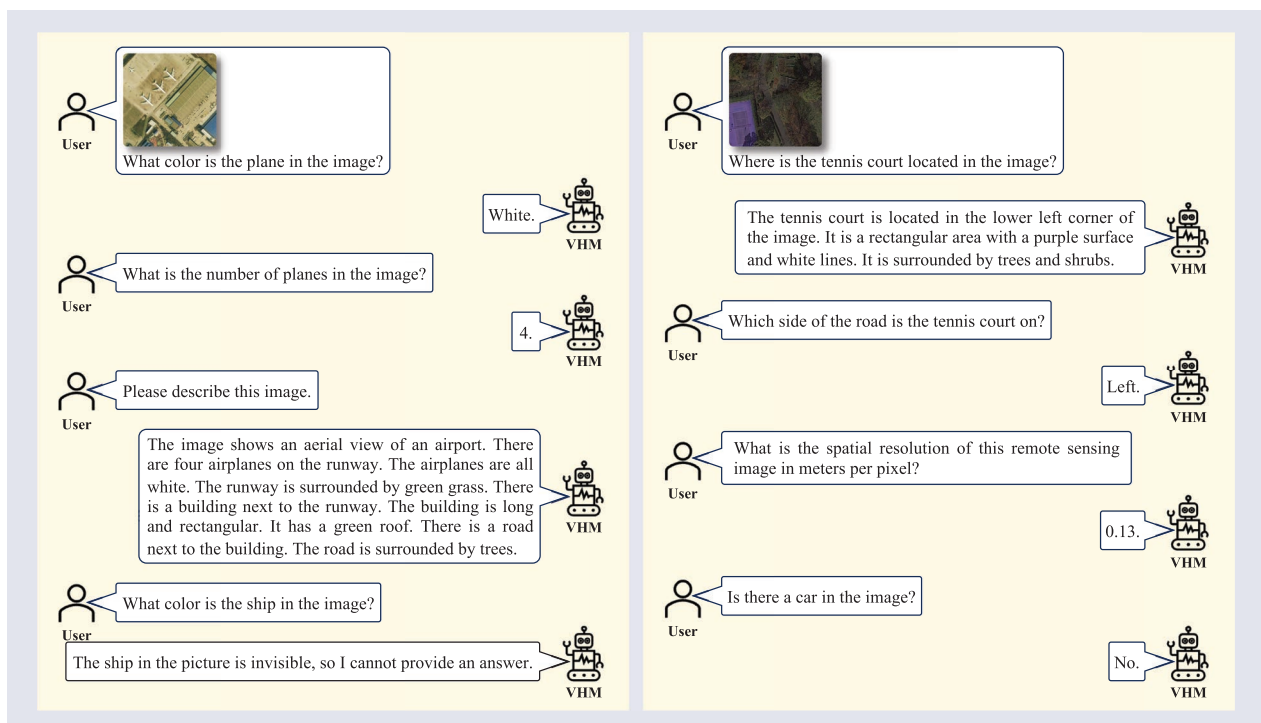


FIGURE 9. Conversations between users and VHM [42].

images, while GeoSynth employs OSM images to achieve similar constraints on layout and object placement.

Image-form conditions are typically injected into diffusion models using ControlNet [154]. ControlNet replicates the encoder blocks and the middle block of Stable Diffusion's U-Net and incorporates several zero convolutions. It processes both conditioning representations and noisy latent representations as inputs, with its outputs added to the decoder blocks and middle block of Stable Diffusion's U-Net. In GeoSynth [155] and GPG2A [47], ControlNet is used for condition injection with different inputs—GeoSynth provides layout maps derived from ground images, while GPG2A inputs OSM images, text, and diffusion time steps. To enhance multicondition injection, DiffusionSat [21] extends ControlNet into a 3D version capable of accepting a sequence of satellite images. This 3D ControlNet retains the replicated encoder and middle blocks, with each followed by a temporal layer consisting of a 3D zero-convolution and a temporal pixel-wise transformer. Following Uni-ControlNet [171], the

authors in [22] and 48] perform multiscale condition injection and use attentional feature fusion [172] to combine conditioning and latent representations. Unlike these works relying on ControlNet, MetaEarth and Text2Earth directly concatenate latent and image condition representations. MetaEarth encodes low-resolution images via RRDBNet [157] with upsampling and convolution layers, while Text2Earth processes masked images using a VAE encoder [45].

IMPROVING DIVERSITY

Herein, diversity refers to two aspects; first, the generated objects exhibit a variety of semantic categories, with each category featuring varied visual characteristics, and second, the generated images capture a broad range of variations in imaging conditions (e.g., season and illumination) and imaging sensors (e.g., spatial resolution and viewpoint). By conditioning on metadata, the diversity of the generated data is improved, as shown in Figure 10. DiffusionSat [21] modifies geographic coordinates, imaging time, and

TABLE 9. SUMMARY OF GENERATION-BASED VISION-LANGUAGE FOUNDATION MODELS IN REMOTE SENSING.

MODEL	DIFFUSION MODEL	CONDITION ENCODER	CONDITION			TRAINING DATASET	GenImgType	PUBLIC
			Text	Metadata	Image			
RS-SD [15] ^a	Stable Diffusion [28]	CLIP Transformer [27]	✓	×	×	1% RS5M [15]	Optical	✓
DiffusionSat [21] ^b	Stable Diffusion [28]	CLIP Transformer [27] Sinusoidal Projection+MLP 3D ControlNet [21]	✓	Latitude Longitude GSD Cloud cover Imaging time	Satellite image	fMoW [60] Satlas-small [152] SpaceNet [153]	Optical	✓
CRS-Diff [22] ^c	Stable Diffusion [28]	CLIP Transformer [27] MLP ControlNet [154] FFN	✓	Latitude Longitude GSD Cloud cover Imaging time	HED MLSD Depthmap Sketch Road map Seg. mask Deep feature	RSICD [78] fMoW [60] Million-AID [67]	Optical	✓
GeoSynth [155] ^d	Stable Diffusion [28]	CLIP Transformer [27] SatCLIP Location Encoder [64] CoordNet [155] ControlNet [154]	✓	Latitude Longitude	OSM Image	Satellite-OSM Dataset [155]	Optical	✓
MetaEarth [49]	DDPM [156]	Sinusoidal Projection+MLP RRDBNet [157]+Upsampling	×	Resolution	Low-Resolution Image	Multiresolution dataset [49]	Optical	×
Text2Earth [158]	Stable Diffusion [28]	CLIP Transformer [27] Projection Layer VAE Encoder [45]	✓	Resolution	Masked Image	Git-10M [158]	Optical	×
HSIGene [48] ^e	Stable Diffusion [28]	CLIP Transformer [27] ControlNet [154] FFN	✓	×	HED MLSD Sketch Seg. mask Deep feature	Xiongan [159] Chikusei [160] DFC2013 [161] DFC2018 [162] Heihe [163]	Hyperspectral	✓
GPG2A [47] ^f	Improved DDPM [164]	CLIP Transformer [27] ControlNet [154]	✓	×	Ground image	VIGORv2 [47]	Optical	✓
RSDiff [165]	Imagen [166]	T5 [167]	✓	×	×	RSICD [78]	Optical	×

^a<https://huggingface.co/Zilun/GeoRSSD>; ^b<https://github.com/samar-khanna/DiffusionSat>; ^c<https://github.com/Sonettoo/CRS-Diff>; ^d<https://github.com/mvrl/GeoSynth>; ^e<https://github.com/LiPang/HSIGene>; ^f<https://github.com/AhmadArrabi/GPG2A>.

GenImgType refers to the type of generated image, e.g., optical or hyperspectral images. GSD: ground sampling distance; HED: holistically nested edge detection; MLSD: multiscale line segment detection; Seg. Mask: Segmentation Mask. Public refers to the availability of both code and model weights.

ground sampling distance, resulting in varied stadiums, different scenes, and multiresolution images, respectively. In addition to considering the perspective of conditions, some works [49], [165] explore new frameworks for generating multiresolution images. In [165], cascaded diffusion models [166] are employed, where one model generates low-resolution images conditioned on text embeddings, and the other superresolves these low-resolution images. This framework achieves the gradual generation of high-resolution satellite images purely from textual descriptions. In [49], a resolution-guided self-cascading framework is proposed to generate images at various resolutions instead of being limited to the two resolutions described in [165]. As illustrated in Figure 11, the generation process unfolds over multiple stages, with each stage conditioned on the low-resolution image output from the preceding stage and its corresponding spatial resolution. The resolution increases at a fixed ratio with each stage. Considering the resolution gap

between adjacent stages, the image condition is injected through a sequential process involving an image encoder, followed by several upsampling and convolution layers. Building on this framework, the authors introduce sliding windows with overlaps and a noise sampling strategy to generate continuous unbounded scenes.

PERFORMANCE EVALUATION

With these efforts, existing generative foundation models have demonstrated the capability to generate optical remote sensing images [15], [21], [22], [47], [49], [155], [165] or hyperspectral remote sensing images [48]. Their performance is typically evaluated using metrics such as Fréchet Inception Distance (FID) [173] and Inception Score (IS) [174]. For text-conditioned models, CLIP [27] is employed to measure the similarity between generated images and their corresponding textual descriptions, as detailed in Table 10. In addition to the direct evaluations, some works

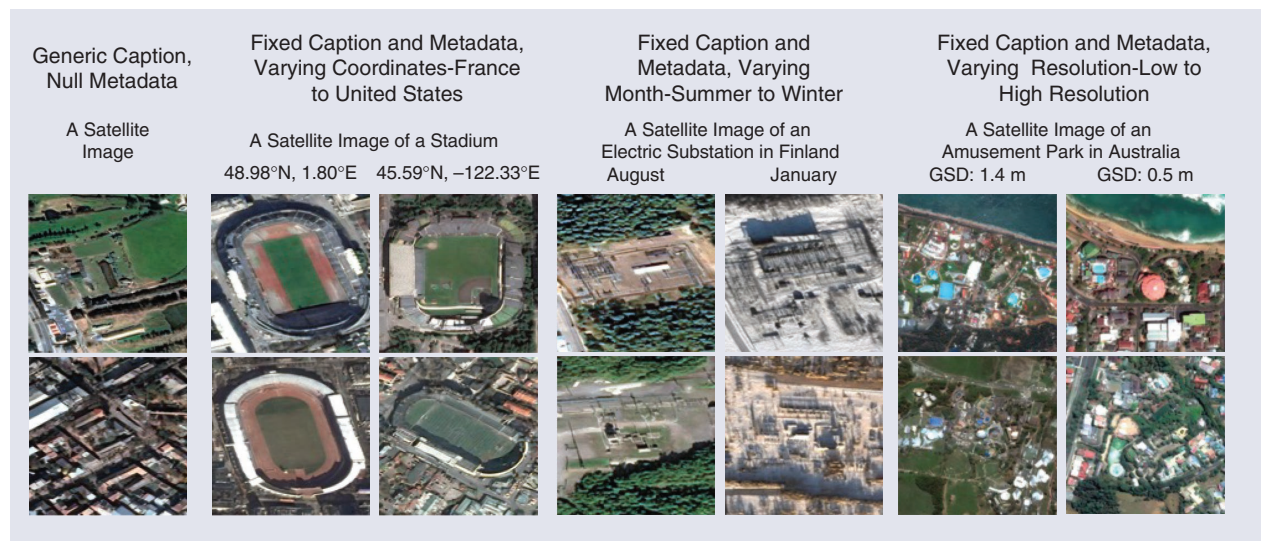


FIGURE 10. Examples of remote sensing images generated by DiffusionSat [21]. GSD: ground-sampling distance.

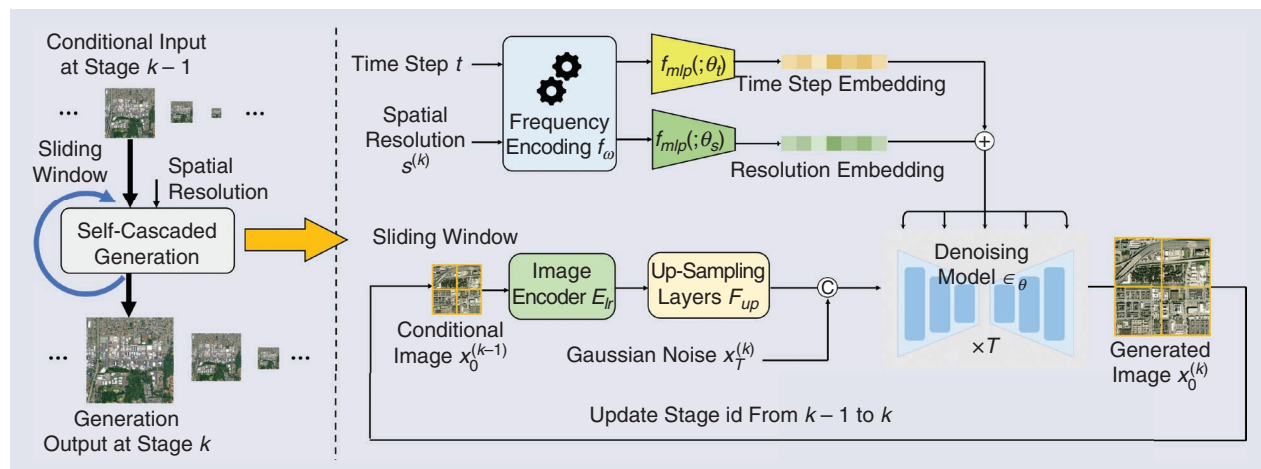


FIGURE 11. An illustration of MetaEarth architecture [49].

assess model performance by applying them or the generated images to specific downstream tasks and measuring their impact [21], [22], [47], [48], [49]. For example, DiffusionSat [21] is applied to tasks such as superresolution, temporal generation, and in-painting, while CRS-Diff [22] uses its generated images to augment training datasets of the road detection model.

DOWNSTREAM APPLICATIONS

Generative foundation models serve as powerful and versatile tools for advancing various remote sensing tasks. Their exceptional image generation capabilities have been utilized to tackle challenges such as limited image availability [175] and the high costs associated with annotation [176], [177]. Additionally, the conditionally controllable nature of these models makes them particularly suitable for remote sensing image enhancement tasks. Examples include tasks like superresolution [178], [179], where low-resolution images serve as conditions, and cloud removal [180], [181], where SAR images are used as conditions. Moreover, their advantages in learning diverse remote sensing image distribution equip them to handle the complexity of remote sensing scenes, improving the accuracy of remote sensing image interpretation tasks such as change detection [182], [183] and land cover classification [184]. These advancements predominantly center on the ability of generative foundation models to model image distribution, which has been comprehensively reviewed in [46]. In this discussion, however, we shift the focus to recently emerging applications that show how the joint distribution between images and texts in generative foundation models promotes advancements in remote sensing tasks. This includes tasks such as image or change captioning [50], [51], pansharpening [52], and zero-shot target recognition [53], as illustrated in Table 11. This section delves into representative implementations in conjunction with specific tasks.

CAPTIONING

Recent works [50], [51] propose generating textual descriptions conditioned on visual features rather than generating images from textual descriptions, thus applying generative models for captioning remote sensing images. In [50], the authors focus on extracting both global and local features of remote sensing images to enhance the visual conditions for the generative model, addressing challenges such as intra-class diversity, interclass similarity, and varying object sizes. They then improve the interaction between noisy representations and visual conditions in the decoder by sequentially interacting with global and local conditions, aiming for generated captions that better align with the image content. In [51], the difference features of bitemporal images condition the generative model to generate change captions from standard Gaussian noise. These difference features are integrated into the generative model using cross-attention, followed by stacking self-attention. Inspired by the success of generative models in change detection [188], the authors in [186] employ these models as a powerful feature extractor for multilevel multi-time step features of bitemporal images. A difference encoder, based on a time-channel-spatial attention mechanism, is then utilized to extract discriminative information from the features of bitemporal images, and a decoder generates change captions from this information.

PANSHARPENING

Pansharpening is the process of fusing high-resolution panchromatic images with low-resolution multispectral images to produce high-resolution multispectral images. The idea of applying generative foundation models to this task involves formulating it as an image generation problem conditioned on panchromatic and multispectral images. To enhance the generalizability of the pansharpening model and enable it to uniformly handle multispectral images from different satellites, TMDiff [52] incorporates textual descriptions, such as “Text Prompts of GaoFen-2 Satellite,” as identifiers to specify the satellites that capture multispectral images. These descriptions, combined with panchromatic-multispectral image pairs, jointly guide the generation process. Specifically, the textual descriptions are processed by a physics-informed HyperNet to get text embeddings, which are then used to modulate the denoising network. Meanwhile, the image pair is injected into the decoder of the denoising network via a condition encoder that shares the same architecture as the network’s encoder. To address the variability in spectral band configurations among different satellites, TMDiff, built on the U-Net architecture, organizes

TABLE 10. PERFORMANCE OF GENERATIVE FOUNDATION MODELS.

MODEL	CONDITION	DATASET	FID	IS	CLIP-SCORE
DiffusionSat [21]	Text+Metadata	fMoW [60]	15.8	6.69	17.2
CRS-Diff [22]	Text	RSICD [78]	50.72	18.39	20.33
	HED		30.18	—	—
	MLSD		55.75	—	—
	Depthmap		54.4	—	—
	Sketch		68.29	—	—
	Seg. mask		70.05	—	—
	Road map		94.16	—	—
	Deep feature		44.93	—	—
GeoSynth [155]	Coordinate+OSM Image	Satellite-OSM [155]	11.9	—	30.3
	Text+OSM Image		12.97	—	29.8
HSIGene [48]	MLSD	AID [70]	—	—	79.27
	HED+MLSD		—	—	81.52
	HED+MLSD+Sketch		—	—	81.4
	HED+MLSD+Sketch+Seg. Mask		—	—	81.83
RSDiff [165]	Text	RSICD [78]	66.49	7.22	—
Text2Earth [158]	Text	RSICD [78]	24.49	—	—

CLIP-Score [27] measures the similarity between generated images and corresponding textual descriptions.

each block with stacked modulated ResBlock, a swish activation layer, and frequency-aware downsampling (or upsampling). Compared to PanDiff [191], which relies solely on image conditions, TMDiff demonstrates impressive generalization capabilities.

ZERO-SHOT SYNTHETIC APERTURE
RADAR TARGET RECOGNITION

Due to the high costs associated with SAR imaging and the limited azimuth range for capturing images, SAR datasets are often small, making it essential to explore SAR target recognition in zero-shot settings. From the perspective of SAR image simulation, adopting generative foundation models for this task is a natural solution. Typically, Wang et al. [53] employ generative models to create 3D models of targets conditioned on semantic information, such as the category and structure of targets. Since most existing generative models are trained on RGB natural images or optical remote sensing images, the authors adopt a two-step process; they first use the generative model to generate optical images based on the target semantic information, and then they transform these optical images into 3D models using TripoSR [192] rather than directly generating 3D models or SAR images.

CLOUD REMOVAL

Passive remote sensing is susceptible to cloud interference. By conditioning on cloud-contaminated images and SAR images, generative models have been employed for cloud removal of remote sensing images [180], [181]. Such solutions require geographically aligned multimodal image pairs, which are challenging to obtain in practical applications. To overcome this limitation, Czerkawski and Tachtatzis [189]

propose a novel approach that uses historical edge information and textual descriptions (e.g., a cloud-free satellite image) to constrain the generation process. Experimental results suggest that general-purpose generative models may not be directly applicable for cloud removal of remote sensing images as they tend to generate undesirable artifacts in cloud-contaminated areas.

URBAN PREDICTION

Urban prediction aims to forecast future urban layouts based on current urban layouts and planned change maps, providing support for urban planning. The pioneering work, UP-Diff [190], incorporates ConvNeXt [193] to encode current urban layouts and planned change maps into embeddings, which are subsequently injected into the decoder of diffusion models through cross-attention layers. Additionally, the text condition, derived from encoding “a remote sensing photo,” guides the model to generate outputs in a satellite image style.

DATASETS

Large-scale datasets are an essential prerequisite for vision-language research under the two-stage paradigm. Thus, a considerable amount of current research [15], [16], [17], [24], [25], [42], [194], [195], [196] focuses on dataset construction. In this section, we summarize these efforts, detailing the properties of datasets and their creation methods, with the aim of providing both convenience and inspiration for further research. Based on their usage, existing datasets can be broadly categorized into three groups: *pretraining datasets*, *instruction-following datasets*, and *benchmark datasets*.

TABLE 11. SUMMARY OF GENERATION-BASED VISION-LANGUAGE FOUNDATION MODELS APPLIED TO REMOTE SENSING TASKS.

WORK	TASK	GENERATIVE FOUNDATION MODEL	ADAPTATION
VCC-DiffNet [50]	Image captioning	D3PM [185]	Extract local and global features to enhance visual conditions. Enhance interactions between noisy representations with visual conditions by sequentially interacting global and local conditions.
Diffusion-RSCC [51]	Change captioning	Self-design	Design cross-mode feature fusion module to inject the difference features of bitemporal images. Design stacking self-attention module to recreate change captions from Gaussian noise iteratively.
MADiffCC [186]	Change captioning	SR3 [187], [188]	Use a generative model to extract multilevel multi-time step features of bitemporal images. Design an encoder using the time-channel-spatial attention to obtain discriminative information. Design a decoder guided by gated multihead cross-attention for generating change captions.
TMDiff [52]	Pansharpening	SR3 [187]	Use text and panchromatic-multispectral image pairs together to condition the generation process, enhancing generalization.
Wang et al. [53]	Zero-shot SAR Target Recognition	Stable Diffusion [28]	Generate optical images conditioned on target semantic information, aiding in the creation of 3D SAR target models.
Czerkawski and Tachtatzis [189]	Cloud Removal	Stable Diffusion [28]	Use text and historical edge information to constrain cloud-free image generation.
UP-Diff [190]	Urban Prediction	Stable Diffusion [28]	Use text, current urban layouts, and planned change maps as conditions, guiding the model to generate future urban layouts.

PRETRAINING DATASETS

Pretraining datasets, which consist of remote sensing images and their corresponding texts, play a crucial role in infusing the model with a broad range of visual and language concepts. This requires the images in pretraining datasets to be sufficiently diverse and rich. Currently, there are two alternative sources for collecting images; one is combining various open source remote sensing image datasets [15], [17], [20], [42], [83], [158], [195], and the other is utilizing public geographic databases [16], [24], [40], [158], [197]. Once the images are collected, corresponding textual descriptions can be generated through manual annotation [17], [40]. Although manual annotation is easy to implement and highly accurate, the complexity and diversity of remote sensing images make it costly, which can significantly limit the dataset size. As a result, there is a growing shift toward automatic image captioning using rule-based methods [16], [20] or off-the-shelf models [15], [24], [42], [158], [195], [197]. This section begins by presenting the image collection strategies of existing pretraining datasets, followed by an introduction to their caption generation methods.

IMAGE COLLECTION

Object detection (OD) datasets in remote sensing typically feature diverse ground objects, such as DOTA [198], containing 1,793,658 instances across 18 categories, and DIOR [199], including 192,472 instances spanning 20 categories. These datasets have quickly garnered attention and have been instrumental in constructing pretraining datasets, leading to the development of RSICap [17] and DIOR-Captions [83]. Compared to relying on a single dataset, combining multiple datasets further enriches the diversity of images and objects while significantly increasing the dataset size. For instance, RET-3+SEG-4+DET-10 [20] integrates three datasets for image retrieval, four for image segmentation, and 10 for OD. VersaD [42] further emphasizes multiresolution and multidomain coverage, drawing from datasets such as CVUSA [215] (0.08-m resolution), Million-AID [67] (0.5–153-m resolution, covering diverse geographies), and LoveDA [212] (featuring urban and rural environments). RS5M [15] prioritizes large-scale image datasets, incorporating sources like BigEarthNet [68] (590,326 images), fMoW [60] (1,047,691 images), and Million-AID (more than 1 million images). Moreover, RS5M integrates PUB11, a dataset of more than 3 million remote sensing-related image-text pairs filtered from 11 computer vision datasets, including LAION-400M [124] and CC3M [221]. PUB11 creation involved sequential steps such as invalid image checking, deduplication, filtering using a CLIP model, and employing a remote sensing image detector. These pretraining datasets support the development of vision-language models beyond specific airborne or satellite sensors. An exception is ChatEarthNet [195], which is tailored for satellite image analysis and sources images from the *Sentinel-2* collected in SatlasPretrain [152].

In addition to open source image datasets, public geographic databases contribute to the extensive collection

of remote sensing images [16], [24], [40], [158], [197]. For example, LuoJiaHOG [40] derives its images from Google Maps, determining global sampling points through spatial analysis and the evaluation of landscape indices. This method ensures the inclusion of images representing diverse topographies and varying economic conditions across countries and regions. Leveraging a customized OSM database, LHRS-Align [24] collects remote sensing images from Google Earth, ensuring that their centers are aligned with OSM features. These images then undergo a series of processing steps, including resizing to a uniform size, deduplication to remove images dominated by vast ocean areas or obscured by clouds, and pruning using a trained network. The final dataset encompasses images from 9,259 cities across 129 countries. Similarly, Git-10M [158] consists mainly of remote sensing images from Google Earth, which are gathered from both randomly selected and manually curated regions worldwide. Manual screening is performed to discard redundant scenes, and an image enhancement model is applied to all collected images, thereby improving dataset quality. SkyScript [16] and RSTeller [197] are sourced from Google Earth Engine. Among these, the authors of SkyScript carefully choose 10 image collections from Google Earth Engine, such as SWISSIMAGE 10-cm RGB imagery and *Landsat 9 C2 T1 TOA Reflectance*, forming a multisource multiresolution image pool. In contrast, RSTeller relies solely on one image collection, the National Agriculture Imagery Program, which provides aerial images covering most of the continental United States and parts of Hawaii, with a ground sampling distance of 0.6 m.

CAPTION GENERATION

Texts associated with remote sensing images in pretraining datasets, also referred to as image captions, are typically human-understandable sentences that describe various aspects of the images. These descriptions often include information about image properties, overall scenes, and specific local objects. In existing pretraining datasets, only RSICap [17] and DIOR-Captions [83] rely entirely on manual annotation to generate image captions. In [17], five remote sensing experts carried out the annotation process following predefined principles. The principles consist of three points: first, describing image attributes; second, describing object attributes; and third, providing a description of the overall scene before detailing specific objects. In [83], the authors refer to bounding boxes from the original image dataset to describe the most prominent or abundant objects. Although manual annotation guarantees high-quality image-text pairs, it is a time-intensive and laborious process, particularly for large-scale datasets. To address this, later datasets have adopted automated methods, employing rules or models to generate captions. This approach significantly reduces the cost of dataset creation while greatly expanding its scale.

1) *Rule-based captioning*: Motivated by the fact that open source image datasets are accompanied by image semantic

information in the form of bounding boxes, segmentation masks, and class names, Liu et al. [20] propose the B2C and M2B methods to convert heterogeneous annotations into natural language captions. In detail, B2C, designed for bounding box annotations, generates five distinct captions for each image by considering the location (bounding box center) and the number of objects. Meanwhile, M2B, tailored for segmentation mask annotations, converts segmentation masks into bounding boxes by first extracting contours for each class and then sorting the contour points to define the bounding boxes. By applying M2B followed by B2C, it becomes convenient to create image-text datasets from image segmentation datasets. With a similar motivation, Wang et al. [16] leverage semantic information from OSM to caption remote sensing images from Google Earth Engine as the two can be linked through geo-coordinates. Due to its uncurated nature, semantic information from OSM is rich but often messy. Consequently, not all this information is suitable for describing images, especially if it is not visually discernible. To address this, the authors leverage CLIP embeddings of semantic information as input and apply a binary logistic regression model to filter out invisible information. After filtering, a description for each object is crafted by assembling OSM semantic data using connecting words such as “of,” “is,” and “and.” Each image includes two captions: one describing the object used to determine the image boundary and another describing multiple objects within the image by assembling captions of individual objects based on their geospatial relationships. To reduce noisy image-text pairs, they finally use a CLIP model to estimate the similarity between images and texts and perform image filtering.

- 2) *Model-based captioning*: Numerous LLMs are available for generating remote sensing image captions, including the ChatGPT series [200], Gemini [216], Vicuna [118], and Mixtral [220]. To effectively apply these models, the key research focus lies in prompt design as caption quality heavily depends on the prompt. The fundamental principle of prompt design is to align with the input requirements of the LLM being used. For multimodal LLMs such as ChatGPT-4V, Gemini-Vision, BLIP-2 [69], and MiniGPT-4 [41], which can process both text and image inputs, the prompt can simply instruct the model to generate a caption like “generate a description for the image’s visual content” [15]. However, to improve caption quality, prompts are optimized by adding constraints on the model’s response [40], [42], [195] or incorporating semantic information about the image [40], [158], [195]. For instance, VersaD [42] defines prompts to ensure that the captions generated by Gemini-Vision encompass information about image properties, object attributes, and scene context while preventing the model from describing uncertain objects. Furthermore, it imposes constraints on the format of the model’s response. LuoJiaHOG [40] prompts MiniGPT-4 to follow principles such as describing object attributes, reducing

vague words, using “next” instead of “up,” adding synonyms, and more. Incorporating image semantic information into the prompt helps guide the model’s focus toward the image content of interest while enriching the captions. For example, in ChatEarthNet [195], the distribution and proportion of various land cover types in the image, derived from the European Space Agency’s WorldCover project, are embedded into the prompt to guide ChatGPT-4V in describing the land cover types of interest. In LuoJiaHOG, manually corrected OSM semantics are provided to MiniGPT-4, allowing the model to describe objects of interest within the images.

For models like ChatGPT-3.5, Vicuna-v1.5, and Mixtral, which primarily handle text inputs, providing image semantic information in the prompt allows the models to simulate “seeing” the image and generate captions. LHRs-Align [24] and RSTeller [197] both utilize OSM semantics to assist the model in understanding the image. Since the quality of the semantic information directly impacts the quality of the captions, a significant portion of the effort in caption generation for these two datasets is devoted to carefully processing the semantic data. For example, similar to SkyScript [16], LHRs-Align filters out invisible semantic information using rules, followed by manual inspection. Additionally, LHRs-Align removes duplicate semantics for each image and applies a threshold to ensure semantic balance across the entire dataset. Given that an image is often associated with abundant OSM semantic data, providing all of it to the LLM can be overwhelming, making it difficult for the model to generate accurate captions. Therefore, RSTeller focuses on describing the primary object in each image, defined by the largest size or longest length, and refers to the OSM Wiki to interpret the OSM semantics, thereby reducing the ambiguity of the semantic information. Based on the processed OSM semantic data, LHRs-Align and RSTeller use custom templates to integrate semantic information into the prompt. To enhance the model’s understanding of the captioning task, a few examples are provided for the model.

Based on the captions generated by models, some techniques are proposed to enrich the captions further. In RS5M [15], the meta-information of images, e.g., longitude, latitude, and timestamp, is structured into readable sentences and combined with the generated captions. Additionally, RS5M includes rotation-invariant captions, which are those that exhibit the most stable similarity to the image features, regardless of their rotation, among the generated captions. In RSTeller, the LLM is guided to create multiple revisions from the generated captions with different tones, resulting in each image being accompanied by at least two captions and up to five in total.

DATASET PROPERTY

Table 12 provides an overview of existing pretraining datasets for VLM in remote sensing, with several key points worth noting as follows:

TABLE 12. SUMMARY OF PRETRAINING DATASETS FOR VLM IN REMOTE SENSING.

DATASET	NO. OF PAIRS	IMAGE SOURCE	IMAGE SIZE	IMAGE RESOLUTION (M)	CAPTION GENERATION	NO. OF CAPTIONS PER IMAGE	AVG. CAP. LENGTH	PUBLIC
RSICap [17] ^a	2,585	DOTA-v1.5 [198]	512 × 512	—	Manual annotation	1	60	✓
DIOR-Captions [83]	16,565	DIOR [199]	800 × 800	—	Manual annotation	2	11	×
ChatEarthNet [195] ^b	173,488	SatlasPretrain [152]	256 × 256	10	ChatGPT-3.5 [200] ChatGPT-4V [200]	1–2	155/90	✓
RET-3+SEG-4 +DET-10 [20] ^c	828,725	AUAI [201] DOTA [198] ISAID [202] Potsdam [203] RSOD [204] Valhingen [205] CrowdAI [213] fMoW [60] BigEarthNet [68] Million-AID [67]	224 × 224 ~1,920 × 1,080	—	M2C+B2C Generation [20]	5	—	✓
VersaD [42] ^d	1,390,405	CARP [206] HRRSD [207] LEVIR [208] RSICD [78] Standford [209] VisDrone [210] CVACT [214] LoveDA [212] fMoW [60]	512 × 512	0.08–153	Gemini-Vision [216]	1	369	✓
RS5M [15] ^e	5,062,377	PUB11 [15]	—	—	BLIP-2 (6.7B) [69]	1–5	49/87	✓
Git-10M [158] ^f	>10,000,000	Google Earth SkyScript [16]	—	0.5–128	ChatGPT-4o [200]	1	52	✓
LuojiaHOG [40]	94,856	Google Maps	1,280 × 1,280	—	Manual Annotation MiniGPT-4 [41]	1	124	×
LHRS-Align [24] ^g	1,150,000	Google Earth	768 × 768	1	Vicuna-v1.5 (13B) [118]	1	—	✓
RSTeller [197] ^h	2,539,256	Google Earth Engine	448 × 448	0.6	Mixtral-7B [220]	2–5	54	✓
SkyScript [16] ⁱ	2,600,000	Google Earth Engine	—	0.1–30	Rule-based Assembly [16]	2	—	✓

^a<https://github.com/Lavender105/RSCTTab-readme-ov-file>; ^b<https://zenodo.org/records/11003436>; ^c<https://huggingface.co/datasets/grqy1026/RemoteCLIP>; ^d<https://github.com/opendatadb/VHM>; ^e<https://huggingface.co/datasets/Zilun/RS5M>; ^f<https://modelscope.cn/datasets/cybuai111/Git-10M/>; ^g<https://github.com/NUJ-LHRS/LHRS-Bot>; ^h<https://huggingface.co/datasets/SkytherinGe/RSTeller>; ⁱ<https://github.com/wangzhecheng/SkyScript>. The average caption length (Avg. Cap. Length) in ChatEarthNet is calculated for captions generated by ChatGPT-3.5 and ChatGPT-4V separately and in RS5M for PUB11 and other remote sensing datasets separately.

- 1) *Geographic coverage*: The datasets RS5M, VersaD, and ChatEarthNet benefit from large-scale remote sensing image datasets, while SkyScript, LHRS-Align, LuojiaHOG, and Git-10M derive from public geographic databases. Each of these datasets provides relatively comprehensive global coverage, with Git-10M being the largest (more than 10 million image-text pairs). In contrast, RSTeller's coverage is primarily limited to the continental United States and parts of Hawaii, potentially introducing geographic bias in trained models.
- 2) *Scene diversity*: Large-scale datasets such as RS5M, LHRS-Align, and SkyScript have, to some extent, ensured diversity in remote sensing scenes. Taking this a step further, the creators of LuojiaHOG employ geospatial analysis to collect images from global regions with varied topography and different development levels. Meanwhile, the Git-10M team manually selects specific areas to ensure comprehensive coverage of representative scenes, including urban areas, forests, mountains, and deserts.
- 3) *Caption quality*: Manually annotated or rule-generated image captions typically achieve high accuracy. For instance, manual inspection of 1,000 randomly sampled image-text pairs from SkyScript demonstrates 96.1% precision. Conversely, model-generated captions inevitably contain errors despite considerable efforts to mitigate noise injection. For smaller-scale datasets like ChatEarthNet, manual correction can significantly improve caption quality. However, this approach becomes prohibitively expensive for million-scale datasets such as VersaD and RS5M, making caption errors unavoidable. Surprisingly, the authors of VersaD experimentally verified that models pretrained on the noisy dataset VersaD (82.3%) outperformed those trained on the more accurate dataset SkyScript, suggesting that rich-content and long captions may make models less sensitive to noise.

- 4) *Distinctive characteristics*: Current pretraining datasets predominantly contain English text, with DIOR-Captions being the only exception that provides bilingual captions in both Chinese and English. This enables the exploration of cross-lingual vision-language alignment. Moreover, while most datasets focus on optical images, ChatEarthNet stands out by facilitating a deeper understanding of multispectral images by offering satellite images with nine specific bands, each paired with textual descriptions.

INSTRUCTION-FOLLOWING DATASETS

Instruction-following datasets are specifically designed for the SFT of instruction-based vision-language models, allowing them to perform specific remote sensing tasks. For contrastive or generation-based vision-language models, the emphasis during task-specific applications lies more on designing network architectures than on developing new datasets, as discussed in the “Effective Adaptation Methods” and the “Downstream Applications” sections. Therefore, in the fine-tuning phase of VLM, we exclusively summarize instruction-following datasets. This type of dataset is composed of images paired with conversations, structured as instructions (or questions) and answers. Its construction involves image collection and conversation generation, with images primarily sourced from open source remote sensing image datasets, as shown in Table 13. The main challenge lies in generating conversations, which is addressed in this section, followed by an overview of notable instruction-following datasets.

CONVERSATION GENERATION

Since sourced image datasets come with task-specific annotations, the most intuitive idea is to convert these annotations into a dialogue format. Currently, three methods are commonly used for this conversion: template-based transformation, large model assistance, and manual annotation. Next, we demonstrate each method with examples from specific tasks.

- 1) *Template-based transformation*: To create task-specific conversations, open source image datasets for the given task are typically used. Answers are derived directly from dataset annotations, while instructions are either manually defined or randomly selected from a pool generated by LLMs. This pool contains instructions tailored to specific tasks, conveying the same meanings with varied phrasing. For example, in MMRS-1M [122], classification datasets such as NWPU-RESISC45 [56] and EuroSAT [71], along with OD datasets like DIOR [199] and DOTA [198], are selected to create instruction-following data for classification and detection tasks. For scene classification, the authors design instructions using the template “What is the category of this remote sensing image? Answer the question using a single word or phrase. Reference categories include category 1, ..., and category n.” For OD, the template is “Detect all objects shown in

the remote sensing image and describe using oriented bounding boxes.” In SkyEye-968k [43], corresponding datasets are selected for image captioning and VG tasks, including UCM-captions [3], Sydney-captions [3], RSVG [143], and DIOR-RSVG [144]. Separate instruction pools are constructed for each task. The image captioning pool includes instructions such as “Briefly describe this image” or “Provide a concise depiction of this image.” Meanwhile, the VG pool contains instructions like “Give me the location of referring expression” or “Where is referring expression?”. Based on task-specific instruction-following data, MTCs can be generated by mixing data from different tasks [43], [122].

- 2) *Large model assistance*: Simple template-based transformations are insufficient for generating instruction-following data for complex reasoning tasks, which equip vision-language models with higher-order cognitive abilities, such as making decisions and identifying relationships. Additionally, MTCs created through multitask mixing after template transformation often lack diversity. To address these issues, LLMs are utilized to generate instruction-following data. By providing a few manually defined in-context examples, the models learn to create high-quality instruction-answer pairs based on image captions and other related information. For instance, Kuckreja et al. [18] prompt Vicuna [118] to generate 30,000 detailed image descriptions, 65,000 MTCs, and 10,000 examples of complex reasoning based on short descriptions supplied. Similarly, Muhtar et al. [24] use Vicuna [118] to produce instruction-following data based on image captions from RSITMD [82] and NWPU-Captions [141]. To ensure that the generated conversations focus on the visual content of the images, their prompts are designed to generate questions that inquire about object types, actions, locations, relative positions between objects, and more. However, considering the limitations of RSITMD and NWPU-Captions, where captions are short and lack detailed content, the authors supplemented their data with additional image-caption pairs from LHRS-Align [24]. GPT-4 is prompted with image captions, object bounding boxes, and object attributes to generate detailed image descriptions, complex reasoning, and MTCs that incorporate questions about object locations and counts.
- 3) *Manual annotation*: In constructing instruction-following datasets, manual annotation primarily serves to provide accurate information that supports template-based or large model-assisted conversation generation methods rather than directly creating conversations. For example, in FIT-RS [25], an instruction-following dataset for understanding semantic relationships between objects, Luo et al. [25] manually annotate very high-resolution remote sensing images with detailed scene graph labels, laying the groundwork for generating conversations focused on relationship comprehension tasks. In TITANIC-FGS [133], the authors manually summarize the common and private features of

fine-grained objects, which are used to populate predefined templates for creating object descriptions. These descriptions are subsequently used to prompt GPT-4, enabling the generation of MTCs that simulate human-like logical decision-making.

The three methods can be combined to facilitate the construction of instruction-following datasets [25], [44], with the goal of creating rich and diverse samples across a variety of tasks. To help vision-language models better distinguish between tasks, a common practice is to incorporate task-specific identifiers into the instructions, as demonstrated in [18], [24], [42], and [43]. One example is [43] where the authors introduce task identifiers such as “[caption],” “[vqa],” and “[refer]” to specify captioning, VQA, and VG tasks, respectively.

IMPRESSIVE DATASETS

Table 13 provides an overview of existing instruction-following datasets. MMSHIP [129] and TITANIC-FGS [133] are specifically tailored for ship image analysis, whereas the others are designed for general-purpose remote sensing data analysis. Most datasets include instruction data for tasks such as captioning, VQA, VG, and MTCs [18], [24], [25], [43], [122], [133], [134], [228], [228]. Among these, MMRS-1M [122] stands out for featuring instruction data that involve not only optical remote sensing images but also SAR and IR images. To achieve this, three SAR image datasets, i.e., AIR-SARShip-2.0 [237], SSDD [224], and HRISD [223], along with six IR image datasets (HIT-UAV [245], Sea-shipping [242], IR-security [240], Aerial-mancar [238], Double-light-vehicle [241], and Oceanic ship [246]) are incorporated into the creation of instruction-following data. Beyond supporting common tasks, some datasets incorporate innovative instruction data that endow vision-language models with impressive capabilities. These include fine-grained image understanding, time-series image analysis, quantitative analysis, honest question answering, and object relationship comprehension. The remainder of this section offers a detailed discussion of these datasets.

- 1) *RSVP-3M*: RSVP-3M [44] is the first visual prompting instruction dataset for remote sensing, with samples consisting of images, visual prompts, and conversations. The visual prompts take the form of masks that match the size of the corresponding image. These masks include bounding boxes or points, highlighting regions or points of interest specified by the user within the image. For a specific task, visual prompts in the instructions are described as “each marked point” or “each marked region,” directing the model to focus on the indicated areas and perform the corresponding analysis. For example, in point-level classification, the instruction can be “Please identify the category of each marked point in the image,” with the model’s response formatted as “<Mark 1>: Label 1\n<Mark 2>: Label 2\n.” Similarly, for region-level classification, instructions like “Please identify the category of each marked region in the image” yield “<Region 1>: Label 1\n<Region 2>: Label 2\n.”

To preserve the model’s image-level capabilities, image-level visual prompts are represented as bounding boxes with dimensions [0, 0, width, height]. Based on these definitions, RSVP-3M source images from 28 existing datasets spanning tasks such as image classification, image caption, OD, and instance segmentation. Ground truth bounding boxes or masks from these datasets are used as visual prompts. Furthermore, GPT-4V is employed to automatically generate instruction data, resulting in more than 3 million samples. This large-scale dataset empowers EarthMarker [44] with a multigranularity understanding of remote sensing images across image, region, and point levels for tasks like classification, captioning, and relationship analysis.

- 2) *TEOChatlas*: TEOChatlas [134] seeks to unlock the potential of vision-language models for time-series image analysis. It provides instruction data for a variety of temporal tasks, including temporal scene classification, change detection (represented by bounding boxes), temporal referring expression, spatial change referring expression, and image/region change question answering. These tasks span two real-world applications: disaster response and urban development monitoring. Figure 8 illustrates examples of each task. To ensure diverse image sequence lengths and image sources, TEOChatlas integrates datasets such as xBD [256], S2Looking [255], QFabric [254], and fMoW [60], which cover bitemporal, pentatemporal, and multitemporal sequences. These datasets are sourced from six different sensors, namely *WorldView-2/3*, *Sentinel-2*, *GaoFen*, *SuperView*, and *Beijing-2*. Building on this diverse data foundation, conversation generation is assisted by GPT-4o [200], resulting in 245,210 samples tailored for temporal tasks. In addition, TEOChatlas incorporates 308,861 samples from the instruction-following dataset of GeoChat [18], which focuses on single-image analysis. To help the model differentiate between single-image tasks and temporal tasks, task-specific instructions are supplemented with prompts that explicitly specify the input consists of a sequence of images and, optionally, indicate the resolution and sensor name of the input images.
- 3) *LHRS-Instruct and VariousRS-Instruct*: LHRS-Instruct [24] and VariousRS-Instruct [42] initiate the exploration of applying vision-language models for quantitative image analysis. In LHRS-Instruct, GPT-4 is prompted with image captions and detailed information about the types and coordinates of objects within the image. This allows GPT-4 to generate conversations centered on the number of objects, thereby equipping LHRS-Bot [24] with the capability to perform object counting. VariousRS-Instruct, on the other hand, extends its functionality beyond object counting by incorporating instruction data designed for geometric measurement tasks, aimed at estimating the length and width of objects within images. These instruction data are derived from the DOTA [198] and FAIR1M [225] datasets through a process of

TABLE 13. SUMMARY OF INSTRUCTION-FOLLOWING DATASETS FOR FINE-TUNING VISION-LANGUAGE MODELS IN REMOTE SENSING.

DATASET	TASK	DATA SOURCE					NO. OF SAMPLE TRAIN/TEST	PUBLIC
MMSHIP [129]	OD	DOSR [222]	DOTA ship subset [198]	HRSID [223]	SSDD [224]		81,000/—	X
Hnstd [42] ^a	Honest question answering	DOTA v2 [198]	FAIR1M [225]				45,000/1,642	✓
VersaD-Instruct [42] ^b	Complex reasoning MTC	DIOR [199]	DOTA v2 [198]	FAIR1M [225]			30,000/—	✓
RS-Instructions [132]	Image captioning VQA	RSIVQA [226]	RSVQA-LR [4]	UAV [227]	UCM-captions [3]		5,506/1,552	✓
RS-GPT4V [228]	VG Complex reasoning Image/region captioning VQA	DIOR-RSVG [144] RSIVQA [226]	FloodNet [229] RSVQA-HR [4]	NWPU-Captions [141] RSVQA-LR [4]	RSICD [78] Sydney-captions [3]	RSITMD [82] UCM-captions [3]	991,206/258,419	X
SkyEye-968k [43] ^c	VG Image/video captioning VQA MTC	CapERA [230] NWPU-Captions [141] RSVQA-HR [4] UCM-captions [3]	DIOR-RSVG [144] RSICD [78] RSVQA-LR [4] UCM-Conversa [43]	DIOR-Conversa [43] RSITMD [82] RSVG [143]	DOTA-Conversa [43] RSIVQA [226] Sydney-captions [3]	ERA-VQA [43] RSPG [43] Sydney-Conversa [43]	968,000/—	✓
Multitask dataset [24] ^d	VG Scene classification Image captioning VQA	DIOR-RSVG [144] RSITMD [82]	fMoW [60] RSVG [143]	METER-ML [147] RSVQA-HR [4]	NWPU-RESISC45 [56] RSVQA-LR [4]	RSICD [78] UCM-captions [3]	42,322/—	✓
TITANIC-FGS [133]	Fine-grained ship classification Ship image captioning Ship Image VQA MTC	Google	Baidu				16,876/2,053	X
RSVP-3M [44]	Scene/region/point classification Image/region/point captioning ORR MTC	DIOR-RSVG [144] Hi-UCD [231] LEVIR [208] Optimal-31 [73] SOTA [232] VisDrone [210]	DOSR [222] HRRSD [207] MAR20 [233] Potsdam [203] SOTA [232] WHU [234]	DOTA v2 [198] HRSC2016 [211] NWPU-Captions [141] RS46-WHU [204] UAVid [235] WHU-RS19 [142]	FAIR1M [225] iSAID [202] NWPU-RE-SISC45 [56] RSITMD [82] UCAS-AOD [236]	FAST [232] Kuckreja et al. [18] NWPUVHR10 [237] RSOD [204] Vaihingen [205]	3,648,884/—	X
Kuckreja et al. [18] ^e	VG Complex reasoning Scene classification Image/region captioning VQA MTC	DIOR [199] RSVQA [4]	DOTA [198]	FAIR1M [225]	FloodNet [229]	NWPU-RE-SISC45 [56]	306,000/12,000	✓
LHRS-Instruct [24] ^f	Object counting Image captioning Complex reasoning Object attribute recognition Object relationship analysis MTC	LHRS-Align [24]	NWPU-Captions [141]	RSITMD [82]			39,800/—	✓
MMRS-1M [122] ^g	VG OD Scene classification Image/region captioning VQA MTC	Aerial-mancar [238] DOTA [198] FGSCR-42 [239] IR-security [240] RSICD [78] RSVQA-LR [4] UCM [80]	AIR-SARShip-2.0 [237] Double-light-vehicle [241] FloodNet [229] NWPU-Captions [141] RSITMD [82] Sea-shipping [242] UCM-captions [3]	CRSVQA [243] DSCR [244] HIT-UAV [245] NWPU-RE-SISC45 [56] RSIVQA [226] SSDD [224] WHU-RS19 [142]	DIOR [199] EuroSAT [71] HRRSD [207] NWPUVHR10 [237] RSOD [204] Sydney-captions [3] VisDrone [210]	DIOR-RSVG [144] FAIR1M [225] HRSID [223] Oceanic ship [246] RSSCN7 [75] UCAS-AOD [236]	1,005,842/—	✓

(Continued)

TABLE 13. SUMMARY OF INSTRUCTION-FOLLOWING DATASETS FOR FINE-TUNING VISION-LANGUAGE MODELS IN REMOTE SENSING (Continued).

DATASET	TASK	DATA SOURCE					NO. OF SAMPLE	PUBLIC
.....							TRAIN/TEST	
FIT-RS [25] ^h	VG	extended					1,440,681/360,170	✓
	OD	STAR [247]						
	Image/region captioning							
	VQA							
	Multilabel classification							
	ORR							
	Image/region scene graph generation							
	MTC							
VariousRS-Instruct [42] ⁱ	VG	BANDON	CrowdAI [213]	DeepGlobe	DIOR-RSVG	DOTA v2 [198]	76,445/13,020	✓
	Object counting	[248]	FBP [250]	[252]	[144]	METER-ML		
	Scene classification	FAIR1M [225]	MtS-WH [251]	fMoW [60]	GID [253]	[147]		
	Building vectorizing	MSAR [249]	UCM-captions	NWPU-RE-	Potsdam [203]	RSITMD [82]		
	Geometric measurement	RSVQA-LR [4]	[3]	SISC45 [56]				
	Image property recognition							
	VQA							
	Multilabel classification							
TEOChatlas [134] ^j	VG	fMoW [60]	Kuckreja et al. [18]	QFabric [254]	S2Looking [255]	xBD [256]	554,071/—	✓
	Change detection							
	Scene classification							
	Image/region captioning							
	VQA							
	Temporal scene classification							
	Temporal referring expression							
	Spatial change referring expression							
	Image/region change question answering							

^a<https://github.com/opendatalab/VHM>; ^b<https://github.com/opendatalab/VHM>; ^c<https://huggingface.co/datasets/ZhanYang-nwpu/SkyEye-968k>; ^d<https://github.com/NJU-LHRS/LHRS-Bot?tab=readme-ov-file>; ^e<https://github.com/mbzuai-oryx/geochat>; ^f<https://github.com/NJU-LHRS/LHRS-Bot?tab=readme-ov-file>; ^g<https://github.com/wivizhang/EarthGPT>; ^h<https://huggingface.co/datasets/II-13/FIT-RS>; ⁱ<https://github.com/opendatalab/VHM>; ^j<https://huggingface.co/datasets/jirvin16/TEOChatlas>.
UAV: unmanned aerial vehicle.

template-based conversion. Ground truth for object sizes is determined using image resolution and the sizes of objects' bounding boxes. Beyond quantitative tasks, VariousRS-Instruct also introduces new qualitative tasks such as building vectorizing and multilabel classification. These advancements contribute to the evolution of a versatile vision-language model for comprehensive remote sensing image analysis.

- 4) *HnstD*: HnstD [42] aims to enhance the honesty of vision-language models. It incorporates four recognition tasks: identifying the relative positions between objects and their presence, color, and absolute positions. Except for the presence task, all others include both factual and deceptive instructions. Specifically, deceptive instructions regarding object color arise from either the absence of objects or their presence in panchromatic images, while those for relative and absolute positions stem from the ab-

sence of objects. This combination of factual and deceptive instructions helps prevent models from producing affirmative answers to unreasonable user queries. To construct HnstD, template-based transformations are applied to images sourced from the DOTA [198] and FAIR1M [225] datasets. The model's response formats vary by task: "yes" or "no" for the presence task, free-form text for the color task, and option selection from candidates for relative and absolute position tasks. Figure 12 presents samples from the HnstD dataset.

- 5) *FIT-RS*: FIT-RS [25] is designed to advance vision-language models' ability to comprehend fine-grained semantic relationships, introducing tasks such as object relationship reasoning (ORR) and scene graph generation at both region and image levels. ORR is further divided into relation detection and relation reasoning. Relation detection requires the model to

predict relationships between targets based on the provided categories and locations of subjects and objects, while relation reasoning demands the model to additionally infer the categories of the subject or object along with their relationships. Region-level scene graph generation involves describing an object's relative size, position, and visibility, followed by generating all subject-relation-object triples. Image-level scene graph generation extends this by creating scene graphs for entire images, appending isolated objects (with no relationships) to the answer in an OD format. These tasks challenge vision-language models to accurately interpret relationships between targets and distinguish between subjects and objects within those relationships. To construct FIT-RS, the scene graph generation dataset STAR [247], containing more than 400,000 triplets across 48 object categories and 58 semantic relationship categories, serves as the data source. Template-based transformations are applied to generate task-specific instruction data. Similar to most instruction-following datasets, FIT-RS also supports tasks such as captioning and VQA. In total, it comprises 1,800,851 samples.

BENCHMARK DATASETS

Benchmark datasets are essential for evaluating and fairly comparing the performance of different models. To foster research in VLM for remote sensing, efforts have been made to develop benchmark datasets that integrate remote sensing images with text. Consequently, datasets like UCM-captions [3], RSICD [78], and RSVQA [4] have been established and are widely utilized to assess models' abilities in visual perception, text understanding, and image-text alignment [15], [20], [24], [42]. However, the limited scale and task-specific design of these existing datasets have proven inadequate for a comprehensive evaluation of modern vision-language models. These advanced models, benefiting from the two-stage training paradigm and large-scale training datasets, excel at handling diverse remote sensing image analysis tasks in a conversational manner. In response to this, researchers have begun crafting large-scale benchmark datasets that encompass diverse tasks and are carefully designed to align with the input-output requirements of models, aiming to challenge vision-language models and push the boundaries of their capabilities. This section highlights these new research advancements [194], [196], [257], [258], [259], categorizing benchmark datasets



Factual:
Q: Is there a large vehicle in the image? Just answer yes or no.
A: yes

(a)



Factual:
Q: Where is the baseball diamond located in the image?
A: Sorry, the object is invisible B: right C: center D: left E: bottom right
A: D

Deceptive:
Q: Whereabouts in the aerial image is the soccer ball field?
A: center B: top right C: bottom right D: Sorry, the object is invisible E: top left
A: D

(c)



Factual:
Q: Where is the large vehicle in relation to the small vehicle?
A: To the top right of the small vehicle B: To the inside of the small vehicle C: To the left of the small vehicle D: Sorry, at least one object is invisible E: To the bottom right of the small vehicle
A: C

(d)



Factual:
Q: What is the actual color of the plane in the remote sensing image? If this question is unanswerable, please provide the reason.
A: white

Deceptive:
Q: Please specify the actual color of the swimming pool in this aerial image. If this question is unanswerable, please provide the reason.
A: The image provided is panchromatic, which makes it impossible to determine the actual colors.

Deceptive:
Q: What is the actual color of the tennis court in the remote sensing image? If this question is unanswerable, please provide the reason.
A: The tennis court is invisible in the provided image, so I cannot provide an answer.

(b)



Deceptive:
Q: Where is the plane in relation to the small vehicle?
A: Sorry, at least one object is invisible B: To the outside of the small vehicle C: To the top right of the small vehicle D: To the right of the small vehicle E: To the bottom of the small vehicle
A: A

FIGURE 12. Samples in the HnstD dataset [42]. (a) Presence. (b) Color. (c) Absolute position. (d) Relative position.

into two types: instruction-specific datasets and general-purpose datasets, based on the organization of their samples. Note that remote sensing image datasets, such as AID [70] and NWPU-RESISC45 [56], which have been adapted for evaluating the performance of vision-language models through appropriate transformations, are comprehensively reviewed in [67] and [260].

INSTRUCTION-SPECIFIC DATASETS

This type of dataset is specifically designed for instruction-based vision-language models, with samples typically presented as image-instruction-answer pairs. Its creation follows a similar process to instruction-following datasets, involving image collection and conversation generation. As summarized in Table 14, most benchmark datasets source their images from open source image datasets and leverage methods such as template transformation, assistance from LLMs, and human annotation to produce high-quality conversations. Detailed methods for conversation generation are provided in the “Instruction-Following Datasets” section, while this section focuses on dataset properties, including task formulations, the number of samples, and question format. From Table 14, three key observations can be made. First, to align with the development trend of multifunctional capabilities in vision-language models, benchmark datasets now include an increasing variety of tasks, extending beyond common ones like image captioning and VQA. They now encompass more complex reasoning tasks as well as expansions from single-image analysis to temporal or cross-view image analysis. Second, the scale of these datasets is increasing, often comprising tens of thousands of samples. However, as these samples are spread across an ever-growing variety of tasks, only a few hundred samples are typically available for each specific task. Third, question formats in the datasets can be categorized into three types: single choice, multiple choice, and open ended. For single-choice and multiple-choice questions, each question is accompanied by a set of candidate answers, facilitating the objective assessment of model performance. In contrast, open-ended questions allow models to generate answers freely, closely reflecting real-world scenarios where users may not know the answer in advance. Next, we present a detailed discussion of these datasets.

1) *RSIEval and VRSBench*: RSIEval [17] and VRSBench [194] both focus on common tasks. RSIEval, which relies entirely on human annotation, is limited in scale, with only 100 samples for image captioning and 936 samples for VQA. Such a small dataset may be insufficient for evaluating a model’s practicality and robustness. In contrast, VRSBench leverages a semiautomatic creation pipeline, substantially increasing the dataset size. It contains 29,614, 52,472, and 123,221 samples for image captioning, VG, and VQA, respectively, with LLaVA-1.5 [131], GeoChat [18], and GPT-4V achieving the best performance in each task.

- 2) *FIT-RSRC and LHRS-Bench*: FIT-RSRC [25] and LHRS-Bench [24] challenge a model’s capability to understand relationships between objects in images. Specifically, FIT-RSRC examines the comprehension of semantic relationships among objects, using terms like “run along” and “around” to describe these relationships. It features four types of questions: querying the relationship between two objects, the existence of a specific relationship, the subject of a relationship, and the object of a relationship. Furthermore, each type of question contains unanswerable variants to assess the model’s robustness and veracity. LHRS-Bot, on the other hand, targets spatial relationships, describing them using terms like “top,” “middle,” and “bottom.” It also includes tasks such as object counting, object attribute recognition, image property recognition, and visual reasoning. However, with only 690 samples in total and some tasks containing merely a few dozen samples, its scale is highly inadequate. On these datasets, SkySenseGPT [25] and LHRS-Bot [24] demonstrate the best performance, respectively.
- 3) *COREval*: COREval [258] expands the evaluation of model capabilities from perception to reasoning. In terms of perception, it encompasses image-level comprehension, single-instance identification, and cross-instance discernment. For reasoning, it emphasizes inferring attributes of scenes or instances, such as the CO₂ emissions or population density of a scene, the height of a building, or the imaging season of an image. These two evaluations are divided into six subdimensions and 22 specific tasks, with a total of 6,263 samples. Recognizing the significant regional intraclass variations in remote sensing images, COREval incorporates images sourced from multiple satellites (*Landsat-8*, *SDGSAT-1*, and *Sentinel-1/2*) and geographic databases (Google Earth, Natural Earth, and SWISSIMAGE), offering geographic coverage of 50 cities spanning six continents. On this dataset, 13 open source vision-language models from both general and remote sensing domains are evaluated. The experimental results reveal that while existing models perform well in image-level comprehension, they struggle with fine-grained instance perception and complex reasoning tasks.
- 4) *VLEO-Bench and GEOBench-VLM*: VLEO-Bench [261] and GEOBench-VLM [259], from the perspective of remote sensing applications (e.g., urban monitoring and disaster management), integrate remote sensing image analysis tasks such as temporal analysis of changes, counting objects, and understanding relationships between objects. In VLEO-Bench, three types of tasks are included: scene understanding, which tests a model’s ability to combine high-level image semantics with knowledge expressed in language; object localization and counting, which evaluate fine-grained perception; and change detection, which assesses a model’s capability to identify differences between multiple images. Experimental results on VLEO-Bench show that

TABLE 14. SUMMARY OF BENCHMARK DATASETS FOR VISION-LANGUAGE MODELS IN REMOTE SENSING.

DATASET	DATA SOURCE			TASK	NO. OF SAMPLE	QUESTION FORMAT	PUBLIC
FIT-RSRC [25] ^a	—			Relation comprehension	—	Single choice	✓
RSIEval [17] ^b	DOTA-v1.5 [198]			Image captioning	100	Open ended	✓
				VQA	936	Open ended	
VRSBench [194] ^c	DIOR [199]	DOTA v2 [198]		Image captioning	29,614	Open ended	✓
				VG	52,472		
				VQA	123,221		
LHRS-Bench [24] ^{†d}	Google Earth			Object counting	690	Single choice	✓
				Visual reasoning			
				Object attribute recognition			
				Image property recognition			
				ORR			
COREval [258] [†]	Google Earth	Landsat-8	Natural Earth	Image-level comprehension	3,257	Single choice	✗
	SDGSAT-1	Sentinel-1	Sentinel-2	Single choice	1,244	Single choice	
	SWISSIMAGE			Cross-instance discernment	562	Open ended	
				Attribute reasoning	300	Single choice	
				Assessment reasoning	400	Single choice	
				Common sense reasoning	500	Single choice	
VLEO-Bench [261] ^e	Aerial Landmark [261]	Animal Detection [264]	BigEarthNet [68]	Location recognition	602	Single choice	✓
	COWC [262]	DIOR-RSVG [144]	fMoW-WILDS [265]	Image captioning	1,009	Open ended	
	NeonTreeEvaluation [263]	PatternNet [74]	RSICD [78]	LULC classification	3,000	Single choice	
	xBD [256]			Multilabel LULC classification	1,000	Multiple choice	
				VG	—	Open ended	
				Object counting	2,239	Open ended	
				Change detection	—	Open ended	
GEOBench-VLM [259] ^{†f}	AiRound [266]	COWC [262]	DeepGlobe [252]	Scene understanding	10,000	Single choice	✓
	Deforestation [267]	DIOR [199]	DOTA [198]	Object classification		Single choice	
	FAIR1M [225]	FGSCR-42 [239]	FireRisk [274]	Object localization and detection		Single choice	
	fMoW [60]	Forest Damage [271]	FPCD [275]	Event detection		Single choice	
	GeoNRW [268]	GVLN [272]	MtSCCD [276]	Caption generation		Open ended	
	NASA Marine Debris [269]	NWPU-RESICS45 [56]	PASTIS [277]	Semantic segmentation		Open ended	
	PatternNet [74]	QuakeSet [273]	RarePlanes [278]	Temporal understanding		Single choice	
	So2Sat [270]	xBD [256]		Nonoptical imagery			
UrBench [196] [†]	Cityscapes [279]	Google Earth	MTSD [281]	Geo-localization	4,562	Single choice	✗
	IM2GPS [280]	Google Street View	VIGOR [282]	Scene reasoning	1,872	Single choice	
				Scene understanding	2,606	Single choice	
				Object understanding	2,968	Open ended	
						Single choice	
GeoText-1652 [283] ^g	University-1652 [284]			Image-text retrieval	1,652	N/A	✓
DIOR-RSVG [144] ^h	DIOR [199]			VG	38,320	N/A	✓
RemoteCount [20]	DOTA [198]			Object counting	947	N/A	✗
SATIN [257] ⁱ	AID [70]	AWTP [288]	BC Scenes [292]	Land cover classification	201,000	N/A	✓
	BCS Scenes [285]	Canadian Crop-land [289]	CLRS [145]	Land use classification	260,205		
	EuroSAT [71]	GID [253]	Million-AID [67]	Hierarchical land use classification	34,000		
	MLRSNet [72]	MultiScene [290]	NaSC-TG2 [146]	Complex scene classification	135,261		
	NWPU-RESISC45 [56]	Optimal-31 [73]	PatternNet [74]	Rare scene classification	105,840		
	Post Hurricane [286]	RSC11 [76]	RSD46-WHU [204]	False color scene classification	39,326		
	RSI-CB256 [77]	RSSCN7 [75]	SAT-4 [287]				
	SAT-6 [287]	SIRI-WHU [79]	SISI [293]				
	UCM [80]	USTC-SmokeRS [291]	WHU-RS19 [142]				

^a<https://huggingface.co/datasets/II-13/FIT-RS/tree/main/FIT-RSRC>; ^b<https://github.com/Lavender105/RSGPT>; ^c<https://huggingface.co/datasets/xiang709/VRSBench>; ^dhttps://huggingface.co/datasets/PumpkinCat/LHRS_Data/tree/main/LHRS-Bench; ^e<https://huggingface.co/collections/mit-ei/vleo-benchmark-datasets-65b789b0466555489c0d70>; ^f<https://huggingface.co/datasets/aialliance/GEOBench-VLM>; ^g<https://multimodalgeo.github.io/GeoText/>; ^h<https://github.com/zhanyang-nwpu/rsvg-pytorch>; ⁱ<https://huggingface.co/datasets/jonathan-roberts1/SATIN>.

[†]indicates that the dataset's task follows a hierarchical taxonomy. In this table, only the broad dimensions of task design are displayed rather than specific tasks. N/A: not applicable.

although state-of-the-art models like GPT-4V achieve strong performance in scene understanding, their poor

spatial reasoning limits their effectiveness in object localization, object counting, and change detection

tasks. GEOBench-VLM, on the other hand, includes more than 10,000 questions across 31 tasks, categorized into eight broad categories. Compared to VLEO-Bench, GEOBench-VLM offers a more diverse range of tasks, including unique ones like referring expression segmentation and nonoptical imagery analysis. Additionally, unlike VLEO-Bench, which confines change detection to counting damaged buildings and estimating damage severity, GEOBench-VLM's temporal understanding covers a wider range of tasks. These include detecting the presence of changes, reasoning about change causes, assessing disaster impacts, and classifying crop types based on long-term time-series images. Detailed experiments with 10 vision-language models show that none excels across all tasks in GEOBench-VLM. Significant effort is still required to develop remote sensing-specific models capable of addressing challenging tasks like referring expression segmentation.

- 5) *UrBench*: UrBench [196] concentrates on exploring the potential applications of vision-language models in urban scenarios. It comprises 11,600 questions across 14 tasks, spanning four dimensions: geo-localization, scene reasoning, scene understanding, and object understanding. Unlike the previously mentioned datasets that include only single-view questions in satellite or aerial images, UrBench also incorporates cross-view questions in which each question pairs images of the same scenario captured from satellite and street views. These questions involve tasks such as image retrieval, orientation identification, camera localization, road understanding, object attribute recognition, and object matching. Meanwhile, they encompass region-level questions, which examine a model's ability to understand urban scenarios at a region level, and role-level questions, which evaluate its potential to assist humans in daily life. Experiments with 21 general-purpose models on UrBench reveal that current models still lag significantly behind human experts in urban environments. They struggle to understand multi-view image relations, and their performance varies inconsistently across different views.

GENERAL-PURPOSE DATASETS

General-purpose datasets are designed to evaluate various types of vision-language models. These datasets typically target specific remote sensing multimodal tasks, with samples consisting of images and task ground truth presented in text form. As summarized in [2], commonly used general-purpose datasets for vision-language model evaluation have been well documented. This section, therefore, focuses on the latest research advancements from the past two years, as outlined in Table 14. GeoText-1652 [283] is a language-guided geo-localization benchmark built on the University-1652 image dataset [284]. Each image includes detailed image-level descriptions as well as region-level brief descriptions with corresponding bounding boxes. DIOR-RSVG [144] is designed for VG toward remote sensing. Through an

automatic expression generation method, it contains 38,320 image-expression-box triplets. The large scale of this dataset makes it a popular choice for evaluating vision-language models. RemoteCount [20] evaluates object-counting abilities. Each image is paired with a human-annotated caption, such as "a photo of nine tennis courts." As a result, this dataset is small in scale, containing only 947 samples. To meet the input requirements of contrastive learning-based foundation models, the original caption is augmented with nine additional captions by replacing the number in the caption with all numbers from one to 10.

SATIN [257] sources images from 27 remote sensing datasets and is designed to evaluate the classification capabilities of models for satellite images. This dataset covers six classification tasks: land cover, land use, hierarchical land use, complex scenes, rare scenes, and false color scenes. Among these, the hierarchical land use task tests the ability to classify land use across varying levels of granularity, while the complex scene task leverages the large view fields of remote sensing images to assess the capability of identifying multiple land use types within a single image. Experiments with 40 vision-language models on SATIN in a zero-shot setting show that this dataset poses a significant challenge, with even models trained on billions of natural images achieving an accuracy of just over 50%.

CONCLUSION AND FUTURE DIRECTIONS

From the perspectives of models and datasets, we have covered the advancements in VLM for remote sensing, knowing how remote sensing images and natural language can be effectively bridged, which remote sensing tasks existing vision-language models can address, and which datasets are suitable for developing and testing vision-language models. Naturally, this raises two important questions.

- 1) Are existing vision-language models adequate for practical applications?
- 2) If not, which directions are worth pursuing to advance this field further?

The answer to the first question is, unsurprisingly, no. VLM remains a highly challenging task and is far from meeting practical needs. In this section, we aim to share insights on future research directions from two perspectives: models and datasets.

EFFECTIVE REPRESENTATION AND ALIGNMENT FOR CROSS-MODAL DATA

A growing trend seeks to advance vision-language models to accommodate a wide range of remote sensing images, including optical, SAR, and IR, thereby enabling the acquisition of more comprehensive information about Earth's surface [122], [129]. However, in applications such as disaster risk assessment, these models may need to integrate additional information sources beyond remote sensing images, such as geospatial vector data and social media, to perform complex reasoning [294], [295]. Geospatial vector data present complex data structures in the form of points, polylines, polygons, and networks.

Meanwhile, social media encompasses texts in various languages (e.g., Chinese, English, and French), diverse types of images (e.g., photographs and emojis), videos, and more. This complexity and diversity pose challenges for models in comprehending the embedded information and associating it with remote sensing data. Consequently, there is a pressing need for effective representation and alignment across a broader scope of cross-modal data.

REQUIREMENTS ARBITRARILY DESCRIBED IN NATURAL LANGUAGE

Existing instructions for prompting vision-language models are usually definite, such as using task identifiers to specify particular remote sensing tasks [18], [43] or providing candidate answers [24]. However, in more realistic and practical scenarios, requirements described in natural language tend to be vague and complex, involving the sequential execution of multiple tasks. For example, given an instruction like “Please assess the water quality in the indicated area of the image,” the model must be capable of decomposing water quality assessment into two subtasks—water detection and quantitative retrieval—and then accomplish each task in sequence. This creates the need to advance further models’ language understanding capability to adapt to users’ arbitrary or flexible demands.

ENHANCING THE RELIABILITY OF MODEL ANSWERS VIA EXPERT EXPLANATIONS

Existing vision-language models take language instructions and visual representations as input, producing image analysis results in the form of natural language. However, these models typically do not provide expert explanations for their answers, leading users to doubt the reliability of the outputs, especially in tasks requiring complex reasoning rather than simple recognition. For instance, in precision agriculture, where the model may be used to guide pesticide spraying schedules, the lack of explanations regarding crop diseases and pest conditions makes it hard to convince farmers to follow the recommendations. Consequently, expert explanations accompanying image analysis results are necessary as they not only enhance the reliability and interpretability of the model’s answers but also offer insight into the decision-making process, thereby fostering user trust. A recent initiative [133] has begun exploring the feasibility of developing the vision-language model capable of performing fine-grained ship classification while providing reasoning behind its classification. This is accomplished by integrating domain knowledge into the construction of the instruction-following dataset.

CONTINUALLY ADAPTING VISION-LANGUAGE MODELS

It is well known that new remote sensing images are collected daily from around the world, and human demands may change accordingly. This necessitates continually adapting vision-language models to these dynamic changes rather than relying solely on one-time pretraining and fine-tuning. Combining newly collected data with previously

gathered data for continual learning is a straightforward and effective approach. However, the growing volume of data poses significant challenges in terms of computational and storage costs. Training the model exclusively on new data risks catastrophic forgetting [296], [297]. Therefore, it is crucial for the research community to explore effective learning strategies that enable vision-language models to learn new knowledge from new data while maintaining old knowledge.

MORE DIVERSE AND RICH REMOTE SENSING MULTIMODAL DATASET

The original CLIP model was trained on 400 million image-text pairs, whereas the largest remote sensing pre-training dataset (Git-10M [158]) contains only 10 million pairs, predominantly limited to optical images. This data scarcity restricts models’ capacity to capture a broad range of visual concepts, a challenge that is particularly pressing in remote sensing. Variations in imaging conditions, sensor parameters, and geographic locations result in highly varied visual characteristics of objects in remote sensing images [67]. Therefore, substantial efforts are required to create image-text datasets that encompass diverse ground objects and a rich variety of remote sensing image modalities.

CHALLENGING AND APPLICATION-SPECIFIC BENCHMARKS

Most benchmark datasets [17], [194] for model evaluation are limited to a narrow range of remote sensing tasks (e.g., VQA and image captioning), whereas contemporary vision-language models [25], [42], [134] in remote sensing demonstrate versatile capabilities across diverse image analysis tasks. Furthermore, while some benchmarks include multiple tasks [258], [259], each task typically has a limited number of samples. Such datasets are insufficient for thoroughly testing and comparing the performance of different vision-language models. Beyond general-purpose benchmarks, exploring application-specific benchmarks is also a promising research direction as some efforts have focused on developing versatile vision-language models tailored to specific real-world applications [129], [133].

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REFERENCES

- [1] L. Bashmal, Y. Bazi, F. Melgani, M. M. Al Rahhal, and M. A. Al Zuair, "Language integration in remote sensing: Tasks, datasets, and future directions," *IEEE Geosci. Remote Sens. Mag.*, vol. 11, no. 4, pp. 63–93, Dec. 2023, doi: [10.1109/MGRS.2023.3316438](https://doi.org/10.1109/MGRS.2023.3316438).
- [2] X. Li, C. Wen, Y. Hu, Z. Yuan, and X. X. Zhu, "Vision-language models in remote sensing: Current progress and future trends," *IEEE Geosci. Remote Sens. Mag.*, vol. 12, no. 2, pp. 32–66, Jun. 2024, doi: [10.1109/MGRS.2024.3383473](https://doi.org/10.1109/MGRS.2024.3383473).
- [3] B. Qu, X. Li, D. Tao, and X. Lu, "Deep semantic understanding of high resolution remote sensing image," in *Proc. Int. Conf. Comput. Inf. Telecom. Syst. (CITS)*, 2016, pp. 1–5, doi: [10.1109/CITS.2016.7546397](https://doi.org/10.1109/CITS.2016.7546397).
- [4] S. Lobry, D. Marcos, J. Murray, and D. Tuia, "RSVQA: Visual question answering for remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8555–8566, Dec. 2020, doi: [10.1109/TGRS.2020.2988782](https://doi.org/10.1109/TGRS.2020.2988782).
- [5] T. Abdullah, Y. Bazi, M. M. Al Rahhal, M. L. Mekhalif, L. Rangarajan, and M. Zuair, "TextRS: Deep bidirectional triplet network for matching text to remote sensing images," *Remote Sens.*, vol. 12, no. 3, p. 405, 2020, doi: [10.3390/rs12030405](https://doi.org/10.3390/rs12030405).
- [6] M. B. Bejiga, F. Melgani, and A. Vascotto, "Retro-remote sensing: Generating images from ancient texts," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 3, pp. 950–960, Mar. 2019, doi: [10.1109/JSTARS.2019.2895693](https://doi.org/10.1109/JSTARS.2019.2895693).
- [7] Z. Zhang, W. Zhang, W. Diao, M. Yan, X. Gao, and X. Sun, "VAA: Visual aligning attention model for remote sensing image captioning," *IEEE Access*, vol. 7, pp. 137,355–137,364, 2019, doi: [10.1109/ACCESS.2019.2942154](https://doi.org/10.1109/ACCESS.2019.2942154).
- [8] Y. Li, S. Fang, L. Jiao, R. Liu, and R. Shang, "A multi-level attention model for remote sensing image captions," *Remote Sens.*, vol. 12, no. 6, p. 939, 2020, doi: [10.3390/rs12060939](https://doi.org/10.3390/rs12060939).
- [9] R. Zhao, Z. Shi, and Z. Zou, "High-resolution remote sensing image captioning based on structured attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, doi: [10.1109/TGRS.2021.3070383](https://doi.org/10.1109/TGRS.2021.3070383).
- [10] U. Zia, M. M. Riaz, and A. Ghafoor, "Transforming remote sensing images to textual descriptions," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 108, Apr. 2022, Art. no. 102741, doi: [10.1016/j.jag.2022.102741](https://doi.org/10.1016/j.jag.2022.102741).
- [11] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.
- [12] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 770–778, doi: [10.1109/CVPR.2016.90](https://doi.org/10.1109/CVPR.2016.90).
- [13] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997, doi: [10.1162/neco.1997.9.8.1735](https://doi.org/10.1162/neco.1997.9.8.1735).
- [14] A. Vaswani et al., "Attention is all you need," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst.*, 2017, vol. 30, pp. 6000–6010.
- [15] Z. Zhang, T. Zhao, Y. Guo, and J. Yin, "RS5M and geoRSClip: A large scale vision-language dataset and a large vision-language model for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–23, 2024, doi: [10.1109/TGRS.2024.3449154](https://doi.org/10.1109/TGRS.2024.3449154).
- [16] Z. Wang, R. Prabha, T. Huang, J. Wu, and R. Rajagopal, "SkyScript: A large and semantically diverse vision-language dataset for remote sensing," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, pp. 5805–5813.
- [17] Y. Hu, J. Yuan, C. Wen, X. Lu, Y. Liu, and X. Li, "RSGPT: A remote sensing vision language model and benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 224, pp. 272–286, Jun. 2025, doi: [10.1016/j.isprsjprs.2025.03.028](https://doi.org/10.1016/j.isprsjprs.2025.03.028).
- [18] K. Kuckreja, M. S. Danish, M. Naseer, A. Das, S. Khan, and F. S. Khan, "GeoChat: Grounded large vision-language model for remote sensing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 27,831–27,840, doi: [10.1109/CVPR52733.2024.02629](https://doi.org/10.1109/CVPR52733.2024.02629).
- [19] R. Hadsell, S. Chopra, and Y. LeCun, "Dimensionality reduction by learning an invariant mapping," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2006, vol. 2, pp. 1735–1742, doi: [10.1109/CVPR.2006.100](https://doi.org/10.1109/CVPR.2006.100).
- [20] F. Liu et al., "RemoteCLIP: A vision language foundation model for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–16, 2024, doi: [10.1109/TGRS.2024.3390838](https://doi.org/10.1109/TGRS.2024.3390838).
- [21] S. Khanna et al., "DiffusionSat: A generative foundation model for satellite imagery," in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2023, pp. 1–15.
- [22] D. Tang, X. Cao, X. Hou, Z. Jiang, J. Liu, and D. Meng, "CRS-Diff: Controllable remote sensing image generation with diffusion model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–14, 2024, doi: [10.1109/TGRS.2024.3453414](https://doi.org/10.1109/TGRS.2024.3453414).

- [23] H. Liu, C. Li, Q. Wu, and Y. J. Lee, "Visual instruction tuning," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2023, vol. 36, pp. 34,892–34,916.
- [24] D. Muhtar, Z. Li, F. Gu, X. Zhang, and P. Xiao, "LHRS-Bot: Empowering remote sensing with VGI-enhanced large multimodal language model," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2024, pp. 440–457.
- [25] J. Luo et al., "SkySenseGPT: A fine-grained instruction tuning dataset and model for remote sensing vision-language understanding," 2024, *arXiv:2406.10100*.
- [26] Y. Zhou et al., "Towards vision-language geo-foundation model: A survey," 2024, *arXiv:2406.09385*.
- [27] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. 38th Int. Conf. Mach. Learn. (ICML)*, 2021, pp. 8748–8763.
- [28] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 10,684–10,695, doi: [10.1109/CVPR52688.2022.01042](https://doi.org/10.1109/CVPR52688.2022.01042).
- [29] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," 2018, *arXiv:1807.03748*.
- [30] Y. Li, Y. Pan, T. Yao, and T. Mei, "Comprehending and ordering semantics for image captioning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 17,990–17,999, doi: [10.1109/CVPR52688.2022.01746](https://doi.org/10.1109/CVPR52688.2022.01746).
- [31] H. Wang, Y. Li, H. Yao, and X. Li, "CLIPN for zero-shot OOD detection: Teaching CLIP to say no," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 1802–1812, doi: [10.1109/ICCV51070.2023.00173](https://doi.org/10.1109/ICCV51070.2023.00173).
- [32] Z. Zhou, Y. Lei, B. Zhang, L. Liu, and Y. Liu, "ZegCLIP: Towards adapting clip for zero-shot semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 11,175–11,185, doi: [10.1109/CVPR52729.2023.01075](https://doi.org/10.1109/CVPR52729.2023.01075).
- [33] U. Mall, C. P. Phoo, M. K. Liu, C. Vondrick, B. Hariharan, and K. Bala, "Remote sensing vision-language foundation models without annotations via ground remote alignment," in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2024, pp. 1–13.
- [34] Z. Song et al., "Set-CLIP: Exploring aligned semantic from low-alignment multimodal data through a distribution view," 2024, *arXiv:2406.05766*.
- [35] X. Li, C. Wen, Y. Hu, and N. Zhou, "RS-CLIP: Zero shot remote sensing scene classification via contrastive vision-language supervision," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 124, Nov. 2023, Art. no. 103497, doi: [10.1016/j.jag.2023.103497](https://doi.org/10.1016/j.jag.2023.103497).
- [36] S. Wang et al., "Multi-view feature fusion and visual prompt for remote sensing image captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–17, 2024, doi: [10.1109/TGRS.2024.3426359](https://doi.org/10.1109/TGRS.2024.3426359).
- [37] M. M. Al Rahhal, Y. Bazi, H. Elgibreen, and M. Zuair, "Vision-language models for zero-shot classification of remote sensing images," *Appl. Sci.*, vol. 13, no. 22, p. 12462, 2023.
- [38] W. Xu, J. Wang, Z. Wei, M. Peng, and Y. Wu, "Deep semantic-visual alignment for zero-shot remote sensing image scene classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 198, pp. 140–152, Apr. 2023, doi: [10.1016/j.isprsjprs.2023.02.012](https://doi.org/10.1016/j.isprsjprs.2023.02.012).
- [39] A. Zavras, D. Michail, B. Demir, and I. Papoutsis, "Mind the modality gap: Towards a remote sensing vision-language model via cross-modal alignment," 2024, *arXiv:2402.09816*.
- [40] Y. Zhao, M. Zhang, B. Yang, Z. Zhang, J. Kang, and J. Gong, "LuoJiaHOG: A hierarchy oriented geo-aware image caption dataset for remote sensing image-text retrieval," *ISPRS J. Photogramm. Remote Sens.*, vol. 222, pp. 130–151, Apr. 2025, doi: [10.1016/j.isprsjprs.2025.02.009](https://doi.org/10.1016/j.isprsjprs.2025.02.009).
- [41] D. Zhu, J. Chen, X. Shen, X. Li, and M. Elhoseiny, "MiniGPT-4: Enhancing vision-language understanding with advanced large language models," in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2024.
- [42] C. Pang et al., "VHM: Versatile and honest vision language model for remote sensing image analysis," in *Proc. AAAI Conf. Artif. Intell.*, 2025, vol. 39, pp. 6381–6388.
- [43] Y. Zhan, Z. Xiong, and Y. Yuan, "SkyEyeGPT: Unifying remote sensing vision-language tasks via instruction tuning with large language model," *ISPRS J. Photogramm. Remote Sens.*, vol. 221, pp. 64–77, Mar. 2025, doi: [10.1016/j.isprsjprs.2025.01.020](https://doi.org/10.1016/j.isprsjprs.2025.01.020).
- [44] W. Zhang, M. Cai, T. Zhang, Y. Zhuang, J. Li, and X. Mao, "EarthMarker: A visual prompting multimodal large language model for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–19, 2024, doi: [10.1109/TGRS.2024.3523505](https://doi.org/10.1109/TGRS.2024.3523505).
- [45] D. P. Kingma and M. Welling, "Auto-encoding variational bayes," 2013, *arXiv:1312.6114*.
- [46] Y. Liu, J. Yue, S. Xia, P. Ghamisi, W. Xie, and L. Fang, "Diffusion models meet remote sensing: Principles, methods, and perspectives," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–22, 2024, doi: [10.1109/TGRS.2024.3464685](https://doi.org/10.1109/TGRS.2024.3464685).
- [47] A. Arrabi, X. Zhang, W. Sultan, C. Chen, and S. Wshah, "Cross-view meets diffusion: Aerial image synthesis with geometry and text guidance," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, 2025, pp. 5356–5366, doi: [10.1109/WACV61041.2025.00523](https://doi.org/10.1109/WACV61041.2025.00523).
- [48] L. Pang et al., "HSIGene: A foundation model for hyperspectral image generation," 2024, *arXiv:2409.12470*.
- [49] Z. Yu, C. Liu, L. Liu, Z. Shi, and Z. Zou, "MetaEarth: A generative foundation model for global-scale remote sensing image generation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 3, pp. 1764–1781, 2025, doi: [10.1109/TPAMI.2024.3507010](https://doi.org/10.1109/TPAMI.2024.3507010).
- [50] Q. Cheng, Y. Xu, and Z. Huang, "VCC-DiffNet: Visual conditional control diffusion network for remote sensing image captioning," *Remote Sens.*, vol. 16, no. 16, p. 2961, 2024, doi: [10.3390/rs16162961](https://doi.org/10.3390/rs16162961).
- [51] X. Yu, Y. Li, J. Ma, C. Li, and H. Wu, "Diffusion-RSCC: Diffusion probabilistic model for change captioning in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–13, 2025, doi: [10.1109/TGRS.2025.3554360](https://doi.org/10.1109/TGRS.2025.3554360).
- [52] Y. Xing, L. Qu, S. Zhang, J. Feng, X. Zhang, and Y. Zhang, "Empower generalizability for pansharpening through text-modulated diffusion model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–12, 2024, doi: [10.1109/TGRS.2024.3434431](https://doi.org/10.1109/TGRS.2024.3434431).
- [53] J. Wang et al., "Leveraging visual language model and generative diffusion model for zero-shot sar target recognition," *Remote Sens.*, vol. 16, no. 16, p. 2927, 2024, doi: [10.3390/rs16162927](https://doi.org/10.3390/rs16162927).

- [54] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2021, pp. 1–12.
- [55] S. Mo, M. Kim, K. Lee, and J. Shin, "S-CLIP: Semi-supervised vision-language learning using few specialist captions," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2023, vol. 36, pp. 61,187–61,212.
- [56] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017, doi: [10.1109/JPROC.2017.2675998](https://doi.org/10.1109/JPROC.2017.2675998).
- [57] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2020, pp. 1597–1607.
- [58] G. Mai, N. Lao, Y. He, J. Song, and S. Ermon, "Csp: Self-supervised contrastive spatial pre-training for geospatial-visual representations," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2023, pp. 23,498–23,515.
- [59] G. Mai, K. Janowicz, B. Yan, R. Zhu, L. Cai, and N. Lao, "Multi-scale representation learning for spatial feature distributions using grid cells," in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2020, pp. 1–12.
- [60] G. Christie, N. Fendley, J. Wilson, and R. Mukherjee, "Functional map of the world," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2018, pp. 6172–6180, doi: [10.1109/CVPR.2018.00646](https://doi.org/10.1109/CVPR.2018.00646).
- [61] V. V. Cepeda, G. K. Nayak, and M. Shah, "GeoCLIP: Clip-inspired alignment between locations and images for effective worldwide geo-localization," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2023, vol. 36, pp. 8690–8701.
- [62] M. Tancik et al., "Fourier features let networks learn high frequency functions in low dimensional domains," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2020, vol. 33, pp. 7537–7547.
- [63] M. Larson, M. Soleymani, G. Gravier, B. Ionescu, and G. J. Jones, "The benchmarking initiative for multimedia evaluation: MediaEval 2016," *IEEE Multimedia Mag.*, vol. 24, no. 1, pp. 93–96, Jan./Mar. 2017, doi: [10.1109/MMUL.2017.9](https://doi.org/10.1109/MMUL.2017.9).
- [64] K. Klemmer, E. Rolf, C. Robinson, L. Mackey, and M. Rußwurm, "SatCLIP: Global, general-purpose location embeddings with satellite imagery," in *Proc. AAAI Conf. Artif. Intell.*, vol. 39, pp. 4347–4355, 2025.
- [65] M. Rußwurm, K. Klemmer, E. Rolf, R. Zbinden, and D. Tuia, "Geographic location encoding with spherical harmonics and sinusoidal representation networks," in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2024, pp. 1–12.
- [66] J. Pan, M. Ma, Q. Ma, C. Bai, and S. Chen, "PIR: Remote sensing image-text retrieval with prior instruction representation learning," 2024, *arXiv:2405.10160*.
- [67] Y. Long et al., "On creating benchmark dataset for aerial image interpretation: Reviews, guidances, and million-aid," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4205–4230, 2021, doi: [10.1109/JSTARS.2021.3070368](https://doi.org/10.1109/JSTARS.2021.3070368).
- [68] G. Sumbul, M. Charfuelan, B. Demir, and V. Markl, "Bigearthnet: A large-scale benchmark archive for remote sensing image understanding," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, 2019, pp. 5901–5904, doi: [10.1109/IGARSS.2019.8900532](https://doi.org/10.1109/IGARSS.2019.8900532).
- [69] J. Li, D. Li, S. Savarese, and S. Hoi, "BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2023, pp. 19,730–19,742.
- [70] G.-S. Xia et al., "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017, doi: [10.1109/TGRS.2017.2685945](https://doi.org/10.1109/TGRS.2017.2685945).
- [71] P. Helber, B. Bischke, A. Dengel, and D. Borth, "EuroSAT: A novel dataset and deep learning benchmark for land use and land cover classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 7, pp. 2217–2226, Jul. 2019, doi: [10.1109/JSTARS.2019.2918242](https://doi.org/10.1109/JSTARS.2019.2918242).
- [72] X. Qi et al., "MLRSNet: A multi-label high spatial resolution remote sensing dataset for semantic scene understanding," *ISPRS J. Photogramm. Remote Sens.*, vol. 169, pp. 337–350, Nov. 2020, doi: [10.1016/j.isprsjprs.2020.09.020](https://doi.org/10.1016/j.isprsjprs.2020.09.020).
- [73] Q. Wang, S. Liu, J. Chanussot, and X. Li, "Scene classification with recurrent attention of VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1155–1167, 2018, doi: [10.1109/TGRS.2018.2864987](https://doi.org/10.1109/TGRS.2018.2864987).
- [74] W. Zhou, S. Newsam, C. Li, and Z. Shao, "PatternNet: A benchmark dataset for performance evaluation of remote sensing image retrieval," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 197–209, Nov. 2018, doi: [10.1016/j.isprsjprs.2018.01.004](https://doi.org/10.1016/j.isprsjprs.2018.01.004).
- [75] Q. Zou, L. Ni, T. Zhang, and Q. Wang, "Deep learning based feature selection for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 11, pp. 2321–2325, Nov. 2015, doi: [10.1109/LGRS.2015.2475299](https://doi.org/10.1109/LGRS.2015.2475299).
- [76] L. Zhao, P. Tang, and L. Huo, "Feature significance-based multibag-of-visual-words model for remote sensing image scene classification," *J. Appl. Remote Sens.*, vol. 10, no. 3, 2016, Art. no. 035004, doi: [10.1117/1.JRS.10.035004](https://doi.org/10.1117/1.JRS.10.035004).
- [77] H. Li et al., "RSI-CB: A large-scale remote sensing image classification benchmark using crowdsourced data," *Sensors*, vol. 20, no. 6, p. 1594, 2020, doi: [10.3390/s20061594](https://doi.org/10.3390/s20061594).
- [78] X. Lu, B. Wang, X. Zheng, and X. Li, "Exploring models and data for remote sensing image caption generation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2183–2195, Apr. 2018, doi: [10.1109/TGRS.2017.2776321](https://doi.org/10.1109/TGRS.2017.2776321).
- [79] B. Zhao, Y. Zhong, G.-S. Xia, and L. Zhang, "Dirichlet-derived multiple topic scene classification model for high spatial resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 2108–2123, Apr. 2016, doi: [10.1109/TGRS.2015.2496185](https://doi.org/10.1109/TGRS.2015.2496185).
- [80] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. ACM SIGSPATIAL Int. Conf. Adv. Geogr. Inf. Syst. (SIGSPATIAL)*, 2010, pp. 270–279.
- [81] G.-S. Xia, W. Yang, J. Delon, Y. Gousseau, H. Sun, and H. Maitre, "Structural high-resolution satellite image indexing," in *Proc. ISPRS TC VII Symp.-100 Years ISPRS*, 2010, vol. 38, pp. 298–303.
- [82] Z. Yuan et al., "Exploring a fine-grained multiscale method for cross-modal remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–19, 2022, doi: [10.1109/TGRS.2021.3078451](https://doi.org/10.1109/TGRS.2021.3078451).

- [83] T. Wei, W. Yuan, J. Luo, W. Zhang, and L. Lu, "VLCA: Vision-language aligning model with cross-modal attention for bilingual remote sensing image captioning," *J. Syst. Eng. Electron.*, vol. 34, no. 1, pp. 9–18, Feb. 2023, doi: [10.23919/JSEE.2023.000035](https://doi.org/10.23919/JSEE.2023.000035).
- [84] M.-H. Guo, Z.-N. Liu, T.-J. Mu, and S.-M. Hu, "Beyond self-attention: External attention using two linear layers for visual tasks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 5, pp. 5436–5447, 2022, doi: [10.1109/TPAMI.2022.3211006](https://doi.org/10.1109/TPAMI.2022.3211006).
- [85] C. Yang, Z. Li, and L. Zhang, "Bootstrapping interactive image-text alignment for remote sensing image captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–12, 2024, doi: [10.1109/TGRS.2024.3359316](https://doi.org/10.1109/TGRS.2024.3359316).
- [86] C. Yang, Z. Li, and L. Zhang, "MGIMM: Multi-granularity instruction multimodal model for attribute-guided remote sensing image detailed description," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–13, 2024, doi: [10.1109/TGRS.2024.3497976](https://doi.org/10.1109/TGRS.2024.3497976).
- [87] R. Ricci, Y. Bazi, and F. Melgani, "Machine-to-machine visual dialoguing with ChatGPT for enriched textual image description," *Remote Sens.*, vol. 16, no. 3, p. 441, 2024, doi: [10.3390/rs16030441](https://doi.org/10.3390/rs16030441).
- [88] C. Liu, R. Zhao, J. Chen, Z. Qi, Z. Zou, and Z. Shi, "A decoupling paradigm with prompt learning for remote sensing image change captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–18, 2023, doi: [10.1109/TGRS.2023.3321752](https://doi.org/10.1109/TGRS.2023.3321752).
- [89] L. Mi, X. Dai, J. Castillo-Navarro, and D. Tuia, "Knowledge-aware text-image retrieval for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–13, 2024, doi: [10.1109/TGRS.2024.3486977](https://doi.org/10.1109/TGRS.2024.3486977).
- [90] J. Li, D. Li, C. Xiong, and S. Hoi, "Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2022, pp. 12,888–12,900.
- [91] B. Psomas et al., "Composed image retrieval for remote sensing," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, 2024, pp. 8526–8534.
- [92] L. Lan, F. Wang, X. Zheng, Z. Wang, and X. Liu, "Efficient prompt tuning of large vision-language model for fine-grained ship classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–10, 2025, doi: [10.1109/TGRS.2024.3509721](https://doi.org/10.1109/TGRS.2024.3509721).
- [93] Y. Tay, M. C. Phan, L. A. Tuan, and S. C. Hui, "Learning to rank question answer pairs with holographic dual LSTM architecture," in *Proc. Int. ACM SIGIR Conf. Res. Dev. Inf. Retr. (SIGIR)*, 2017, pp. 695–704.
- [94] J. Zhang et al., "Text2Seg: Zero-shot remote sensing image semantic segmentation via text-guided visual foundation models," in *Proc. ACM SIGSPATIAL Int. Workshop AI Geographic Knowl. Discov.*, 2024, pp. 63–66.
- [95] Y. Lin, K. Suzuki, and S. Sogo, "Practical techniques for vision-language segmentation model in remote sensing," *Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci.*, vol. XLVIII-2-2024, pp. 203–210, Jun. 2024, doi: [10.5194/isprs-archives-XLVIII-2-2024-203-2024](https://doi.org/10.5194/isprs-archives-XLVIII-2-2024-203-2024).
- [96] T. Lüddecke and A. Ecker, "Image segmentation using text and image prompts," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 7086–7096.
- [97] K. Li, R. Liu, X. Cao, D. Meng, and Z. Wang, "SegEarth-OV: Towards-free open-vocabulary segmentation for remote sensing images," 2024, *arXiv:2410.01768*.
- [98] X. Tan, G. Chen, T. Wang, J. Wang, and X. Zhang, "Segment change model (SCM) for unsupervised change detection in VHR remote sensing images: A case study of buildings," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, 2024, pp. 8577–8580, doi: [10.1109/IGARSS53475.2024.10642429](https://doi.org/10.1109/IGARSS53475.2024.10642429).
- [99] S. Dong, L. Wang, B. Du, and X. Meng, "ChangeCLIP: Remote sensing change detection with multimodal vision-language representation learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 208, pp. 53–69, Feb. 2024, doi: [10.1016/j.isprsjprs.2024.01.004](https://doi.org/10.1016/j.isprsjprs.2024.01.004).
- [100] K. Li, X. Cao, and D. Meng, "A new learning paradigm for foundation model-based remote-sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–12, 2024, doi: [10.1109/TGRS.2024.3365825](https://doi.org/10.1109/TGRS.2024.3365825).
- [101] Y. Bazi, M. M. A. Rahhal, M. L. Mekhalfi, M. A. A. Zuair, and F. Melgani, "Bi-modal transformer-based approach for visual question answering in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022, doi: [10.1109/TGRS.2022.3192460](https://doi.org/10.1109/TGRS.2022.3192460).
- [102] M. Czerkawski, R. Atkinson, and C. Tachtatzis, "Detecting cloud presence in satellite images using the RGB-based clip vision-language model," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, 2023, pp. 5170–5173, doi: [10.1109/IGARSS52108.2023.10282016](https://doi.org/10.1109/IGARSS52108.2023.10282016).
- [103] H. Tang, W. Zhao, G. Hu, Y. Xiao, Y. Li, and H. Wang, "Text-guided diverse image synthesis for long-tailed remote sensing object classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–13, 2024, doi: [10.1109/TGRS.2024.3422095](https://doi.org/10.1109/TGRS.2024.3422095).
- [104] S. Xu, C. Zhang, L. Fan, G. Meng, S. Xiang, and J. Ye, "AddressCLIP: Empowering vision-language models for city-wide image address localization," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2024, pp. 76–92.
- [105] A. Radford et al., "Language models are unsupervised multi-task learners," *OpenAI Blog*, vol. 1, no. 8, p. 9, 2019.
- [106] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguist.: Hum. Lang. Technol. (NAACL-HLT)*, 2019, vol. 1, pp. 4171–4186.
- [107] S. Zhang et al., "OPT: Open pre-trained transformer language models," 2022, *arXiv:2205.01068*.
- [108] T. Brown et al., "Language models are few-shot learners," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, vol. 33, pp. 1877–1901.
- [109] P. Gao et al., "CLIP-adapter: Better vision-language models with feature adapters," *Int. J. Comput. Vis.*, vol. 132, no. 2, pp. 581–595, 2024, doi: [10.1007/s11263-023-01891-x](https://doi.org/10.1007/s11263-023-01891-x).
- [110] Y. Li, D. Kong, Y. Zhang, Y. Tan, and L. Chen, "Robust deep alignment network with remote sensing knowledge graph for zero-shot and generalized zero-shot remote sensing image scene classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 179, pp. 145–158, Sep. 2021, doi: [10.1016/j.isprsjprs.2021.08.001](https://doi.org/10.1016/j.isprsjprs.2021.08.001).
- [111] R. Speer, J. Chin, and C. Havasi, "ConceptNet 5.5: An open multilingual graph of general knowledge," in *Proc. AAAI Conf. Artif. Intell.*, 2017, vol. 31, pp. 4444–4451.
- [112] A. Kirillov et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 3992–4003, doi: [10.1109/ICCV51070.2023.00371](https://doi.org/10.1109/ICCV51070.2023.00371).

- [113] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, doi: [10.1109/TGRS.2021.3095166](https://doi.org/10.1109/TGRS.2021.3095166).
- [114] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, 2022, pp. 207–210, doi: [10.1109/IGARSS46834.2022.9883686](https://doi.org/10.1109/IGARSS46834.2022.9883686).
- [115] K. Zhou, J. Yang, C. C. Loy, and Z. Liu, "Learning to prompt for vision-language models," *Int. J. Comput. Vis.*, vol. 130, no. 9, pp. 2337–2348, 2022, doi: [10.1007/s11263-022-01653-1](https://doi.org/10.1007/s11263-022-01653-1).
- [116] Y. Fang et al., "EVA: Exploring the limits of masked visual representation learning at scale," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 19,358–19,369, doi: [10.1109/CVPR52729.2023.01855](https://doi.org/10.1109/CVPR52729.2023.01855).
- [117] W. Dai et al., "InstructBLIP: Towards general-purpose vision-language models with instruction tuning," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2023, pp. 49,250–49,267.
- [118] W.-L. Chiang et al., "Vicuna: An open-source Chatbot impressing GPT-4 with 90%* Chatgpt quality," *LMSYS Org*, Mar. 2023. [Online]. Available: <https://lmsys.org/blog/2023-03-30-vicuna/>
- [119] H. Touvron et al., "Llama 2: Open foundation and fine-tuned chat models," 2023, *arXiv:2307.09288*.
- [120] J. Chen et al., "MiniGPT-v2: Large language model as a unified interface for vision-language multi-task learning," 2023, *arXiv:2310.09478*.
- [121] E. J. Hu et al., "LoRA: Low-rank adaptation of large language models," in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2022, pp. 1–13.
- [122] W. Zhang, M. Cai, T. Zhang, Y. Zhuang, and X. Mao, "EarthGPT: A universal multimodal large language model for multisensor image comprehension in remote sensing domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–20, 2024, doi: [10.1109/TGRS.2024.3409624](https://doi.org/10.1109/TGRS.2024.3409624).
- [123] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," *Trans. Mach. Learn. Res.*, pp. 1–31, 2024.
- [124] C. Schuhmann et al., "LAION-400m: Open dataset of CLIP-filtered 400 million image-text pairs," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2021, pp. 1–5.
- [125] X. Chen et al., "Microsoft COCO captions: Data collection and evaluation server," 2015, *arXiv:1504.00325*.
- [126] P. Gao et al., "Llama-adapter v2: Parameter-efficient visual instruction model," 2023, *arXiv:2304.15010*.
- [127] S. Kazemzadeh, V. Ordonez, M. Matten, and T. Berg, "ReferIt-Game: Referring to objects in photographs of natural scenes," in *Proc. Conf. Empirical Methods Nat. Lang. Process. (EMNLP)*, 2014, pp. 787–798.
- [128] L. Yu, P. Poirson, S. Yang, A. C. Berg, and T. L. Berg, "Modeling context in referring expressions," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2016, pp. 69–85.
- [129] W. Zhang, M. Cai, T. Zhang, G. Lei, Y. Zhuang, and X. Mao, "Popeye: A unified visual-language model for multi-source ship detection from remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 20,050–20,063, 2024, doi: [10.1109/JSTARS.2024.3488034](https://doi.org/10.1109/JSTARS.2024.3488034).
- [130] H. Touvron et al., "LLaMA: Open and efficient foundation language models," 2023, *arXiv:2302.13971*.
- [131] H. Liu, C. Li, Y. Li, and Y. J. Lee, "Improved baselines with visual instruction tuning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 26,286–26,296, doi: [10.1109/CVPR52733.2024.02484](https://doi.org/10.1109/CVPR52733.2024.02484).
- [132] Y. Bazi, L. Bashmal, M. M. Al Rahhal, R. Ricci, and F. Melgani, "RS-LLaVA: A large vision-language model for joint captioning and question answering in remote sensing imagery," *Remote Sens.*, vol. 16, no. 9, pp. 1477, 2024, doi: [10.3390/rs16091477](https://doi.org/10.3390/rs16091477).
- [133] M. Guo, M. Wu, Y. Shen, H. Li, and C. Tao, "IfShip: A large vision-language model for interpretable fine-grained ship classification via domain knowledge-enhanced instruction tuning," 2024, *arXiv:2408.06631*.
- [134] J. A. Irvin et al., "TEOChat: A large vision-language assistant for temporal earth observation data," in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2025, pp. 1–16.
- [135] B. Lin et al., "Video-llava: Learning united visual representation by alignment before projection," in *Proc. Conf. Empirical Methods Nat. Lang. Process. (EMNLP)*, 2024, pp. 5971–5984.
- [136] J. Ainslie, J. Lee-Thorp, M. De Jong, Y. Zemlyanskiy, F. Lebrón, and S. Sanghai, "GQA: Training generalized multi-query transformer models from multi-head checkpoints," in *Proc. Conf. Empirical Methods Nat. Lang. Process. (EMNLP)*, 2023, pp. 4895–4901.
- [137] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, "BLEU: A method for automatic evaluation of machine translation," in *Proc. Annu. Meet. Assoc. Comput. Linguist. (ACL)*, 2002, pp. 311–318.
- [138] S. Banerjee and A. Lavie, "METEOR: An automatic metric for MT evaluation with improved correlation with human judgments," in *Proc. ACL Workshop Intrinsic Extrinsic Eval. Measures Mach. Transl. Summarization*, 2005, pp. 65–72.
- [139] C.-Y. Lin, "ROUGE: A package for automatic evaluation of summaries," in *Proc. Text Summarization Branches Out*, 2004, pp. 74–81.
- [140] R. Vedantam, C. L. Zitnick, and D. Parikh, "CIDEr: Consensus-based image description evaluation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2015, pp. 4566–4575, doi: [10.1109/CVPR.2015.7299087](https://doi.org/10.1109/CVPR.2015.7299087).
- [141] Q. Cheng, H. Huang, Y. Xu, Y. Zhou, H. Li, and Z. Wang, "NW-PU-captions dataset and MLCA-net for remote sensing image captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–19, 2022, doi: [10.1109/TGRS.2022.3201474](https://doi.org/10.1109/TGRS.2022.3201474).
- [142] D. Dai and W. Yang, "Satellite image classification via two-layer sparse coding with biased image representation," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 1, pp. 173–176, Jan. 2011, doi: [10.1109/LGRS.2010.2055033](https://doi.org/10.1109/LGRS.2010.2055033).
- [143] Y. Sun, S. Feng, X. Li, Y. Ye, J. Kang, and X. Huang, "Visual grounding in remote sensing images," in *Proc. ACM Int. Conf. Multimedia*, 2022, pp. 404–412.
- [144] Y. Zhan, Z. Xiong, and Y. Yuan, "RSVG: Exploring data and models for visual grounding on remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–13, 2023, doi: [10.1109/TGRS.2023.3250471](https://doi.org/10.1109/TGRS.2023.3250471).
- [145] H. Li et al., "CLRS: Continual learning benchmark for remote sensing image scene classification," *Sensors*, vol. 20, no. 4, p. 1226, 2020, doi: [10.3390/s20041226](https://doi.org/10.3390/s20041226).

- [146] Z. Zhou et al., "NaSC-TG2: Natural scene classification with tianhong-2 remotely sensed imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3228–3242, 2021, doi: [10.1109/JSTARS.2021.3063096](https://doi.org/10.1109/JSTARS.2021.3063096).
- [147] B. Zhu et al., "METER-ML: A multi-sensor earth observation benchmark for automated methane source mapping," 2022, *arXiv:2207.11166*.
- [148] X. Huang, Y. Cao, and J. Li, "An automatic change detection method for monitoring newly constructed building areas using time-series multi-view high-resolution optical satellite images," *Remote Sens. Environ.*, vol. 244, Jul. 2020, Art. no. 111802, doi: [10.1016/j.rse.2020.111802](https://doi.org/10.1016/j.rse.2020.111802).
- [149] S. Holail et al., "Time-series satellite remote sensing reveals gradually increasing war damage in the gaza strip," *Natl. Sci. Rev.*, vol. 11, no. 9, 2024, Art. no. nwae304, doi: [10.1093/nsr/nwae304](https://doi.org/10.1093/nsr/nwae304).
- [150] Z. Lin, F. Zhu, Y. Kong, Q. Wang, and J. Wang, "SRSG and S2SG: A model and a dataset for scene graph generation of remote sensing images from segmentation results," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022, doi: [10.1109/TGRS.2022.3185678](https://doi.org/10.1109/TGRS.2022.3185678).
- [151] Y. Yang, X. Tang, Y.-M. Cheung, X. Zhang, and L. Jiao, "SAGN: Semantic-aware graph network for remote sensing scene classification," *IEEE Trans. Image Process.*, vol. 32, pp. 1011–1025, 2023, doi: [10.1109/TIP.2023.3238310](https://doi.org/10.1109/TIP.2023.3238310).
- [152] F. Bastani, P. Wolters, R. Gupta, J. Ferdinando, and A. Kembhavi, "SatlasPretrain: A large-scale dataset for remote sensing image understanding," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 16,772–16,782.
- [153] A. Van Etten, D. Lindenbaum, and T. M. Bacastow, "SpaceNet: A remote sensing dataset and challenge series," 2018, *arXiv:1807.01232*.
- [154] L. Zhang, A. Rao, and M. Agrawala, "Adding conditional control to text-to-image diffusion models," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 3813–3824, doi: [10.1109/ICCV51070.2023.00355](https://doi.org/10.1109/ICCV51070.2023.00355).
- [155] S. Sastry, S. Khanal, A. Dhakal, and N. Jacobs, "GeoSynth: Contextually-aware high-resolution satellite image synthesis," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR) Workshops*, 2024, pp. 460–470, doi: [10.1109/CVPRW63382.2024.00051](https://doi.org/10.1109/CVPRW63382.2024.00051).
- [156] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2020, vol. 33, pp. 6840–6851.
- [157] X. Wang et al., "Esrgan: Enhanced super-resolution generative adversarial networks," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 63–79.
- [158] C. Liu, K. Chen, R. Zhao, Z. Zou, and Z. Shi, "Text2Earth: Unlocking text-driven remote sensing image generation with a global-scale dataset and a foundation model," *IEEE Geosci. Remote Sens. Mag.*, early access, May 6, 2025, doi: [10.1109/MGRS.2025.3560455](https://doi.org/10.1109/MGRS.2025.3560455).
- [159] C. Yi et al., "Aerial hyperspectral remote sensing classification dataset of Xiongan new area (Matiwan village)," *J. Remote Sens.*, vol. 24, no. 11, pp. 1299–1306, 2020.
- [160] N. Yokoya and A. Iwasaki, "Airborne hyperspectral data over Chikusei," Univ. of Tokyo, Tokyo, Japan, Tech. Rep. SAL-2016-05-27, May 2016.
- [161] "Houston hyperspectral dataset," I. G. D. F. Contest, 2013. [Online]. Available: https://hyperspectral.ee.uh.edu/?page_id=459
- [162] "Houston hyperspectral dataset," I. G. D. F. Contest, 2018. [Online]. Available: https://hyperspectral.ee.uh.edu/?page_id=1075
- [163] X. Li et al., "A multiscale dataset for understanding complex eco-hydrological processes in a heterogeneous oasis system," *Sci. Data*, vol. 4, Jun. 2017, Art. no. 170083, doi: [10.1038/sdata.2017.83](https://doi.org/10.1038/sdata.2017.83).
- [164] A. Q. Nichol and P. Dhariwal, "Improved denoising diffusion probabilistic models," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2021, pp. 8162–8171.
- [165] A. Sebaq and M. ElHelw, "RsDiff: Remote sensing image generation from text using diffusion model," *Neural Comput. Appl.*, vol. 36, no. 36, pp. 23,103–23,111, 2024, doi: [10.1007/s00521-024-10363-3](https://doi.org/10.1007/s00521-024-10363-3).
- [166] C. Saharia et al., "Photorealistic text-to-image diffusion models with deep language understanding," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2022, vol. 35, pp. 36,479–36,494.
- [167] C. Raffel et al., "Exploring the limits of transfer learning with a unified text-to-text transformer," *J. Mach. Learn. Res.*, vol. 21, no. 1, pp. 1–67, 2020.
- [168] S. Xie and Z. Tu, "Holistically-nested edge detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2015, pp. 1395–1403, doi: [10.1109/ICCV.2015.164](https://doi.org/10.1109/ICCV.2015.164).
- [169] Y. Xu, W. Xu, D. Cheung, and Z. Tu, "Line segment detection using transformers without edges," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2021, pp. 4255–4264, doi: [10.1109/CVPR46437.2021.00424](https://doi.org/10.1109/CVPR46437.2021.00424).
- [170] E. Simo-Serra, S. Iizuka, K. Sasaki, and H. Ishikawa, "Learning to simplify: Fully convolutional networks for rough sketch cleanup," *ACM Trans. Graph.*, vol. 35, no. 4, pp. 1–11, 2016, doi: [10.1145/2897824.2925972](https://doi.org/10.1145/2897824.2925972).
- [171] S. Zhao et al., "Uni-ControlNet: All-in-one control to text-to-image diffusion models," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2023, vol. 36, pp. 11,127–11,150.
- [172] Y. Dai, F. Gieseke, S. Oehmcke, Y. Wu, and K. Barnard, "Attentional feature fusion," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, 2021, pp. 3559–3568, doi: [10.1109/WACV48630.2021.00360](https://doi.org/10.1109/WACV48630.2021.00360).
- [173] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, "GANs trained by a two time-scale update rule converge to a local Nash equilibrium," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017, vol. 30, pp. 6629–6640.
- [174] T. Salimans, I. Goodfellow, W. Zaremba, V. Cheung, A. Radford, and X. Chen, "Improved techniques for training GANs," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2016, vol. 29, pp. 2234–2242.
- [175] L. Liu, B. Chen, H. Chen, Z. Zou, and Z. Shi, "Diverse hyperspectral remote sensing image synthesis with diffusion models," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–16, 2023, doi: [10.1109/TGRS.2023.3335975](https://doi.org/10.1109/TGRS.2023.3335975).
- [176] C. Zhao, Y. Ogawa, S. Chen, Z. Yang, and Y. Sekimoto, "Label freedom: Stable diffusion for remote sensing image semantic segmentation data generation," in *Proc. IEEE Int. Conf. Big Data (BigData)*, 2023, pp. 1022–1030, doi: [10.1109/BigData59044.2023.10386381](https://doi.org/10.1109/BigData59044.2023.10386381).
- [177] Z. Yuan et al., "Efficient and controllable remote sensing fake sample generation based on diffusion model," *IEEE Trans.*

- Geosci. Remote Sens.*, vol. 61, pp. 1–12, 2023, doi: [10.1109/TGRS.2023.3268331](https://doi.org/10.1109/TGRS.2023.3268331).
- [178] Y. Xiao, Q. Yuan, K. Jiang, J. He, X. Jin, and L. Zhang, "EDiffSR: An efficient diffusion probabilistic model for remote sensing image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–14, 2024, doi: [10.1109/TGRS.2023.3341437](https://doi.org/10.1109/TGRS.2023.3341437).
- [179] F. Meng et al., "A conditional diffusion model with fast sampling strategy for remote sensing image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–16, 2024, doi: [10.1109/TGRS.2024.3458009](https://doi.org/10.1109/TGRS.2024.3458009).
- [180] X. Zhao and K. Jia, "Cloud removal in remote sensing using sequential-based diffusion models," *Remote Sens.*, vol. 15, no. 11, p. 2861, 2023, doi: [10.3390/rs15112861](https://doi.org/10.3390/rs15112861).
- [181] R. Jing, F. Duan, F. Lu, M. Zhang, and W. Zhao, "Denoising diffusion probabilistic feature-based network for cloud removal in sentinel-2 imagery," *Remote Sens.*, vol. 15, no. 9, p. 2217, 2023, doi: [10.3390/rs15092217](https://doi.org/10.3390/rs15092217).
- [182] J. Jia, G. Lee, Z. Wang, L. Zhi, and Y. He, "Siamese meets diffusion network: SMDNet for enhanced change detection in high-resolution RS imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 8189–8202, 2024, doi: [10.1109/JSTARS.2024.3384545](https://doi.org/10.1109/JSTARS.2024.3384545).
- [183] Y. Wen, Z. Zhang, Q. Cao, and G. Niu, "TransC-GD-CD: Transformer-based conditional generative diffusion change detection model," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 7144–7158, 2024, doi: [10.1109/JSTARS.2024.3373201](https://doi.org/10.1109/JSTARS.2024.3373201).
- [184] N. Chen, J. Yue, L. Fang, and S. Xia, "SpectralDiff: A generative framework for hyperspectral image classification with diffusion models," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–16, 2023, doi: [10.1109/TGRS.2023.3310023](https://doi.org/10.1109/TGRS.2023.3310023).
- [185] J. Austin, D. D. Johnson, J. Ho, D. Tarlow, and R. Van Den Berg, "Structured denoising diffusion models in discrete state-spaces," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2021, vol. 34, pp. 17,981–17,993.
- [186] Y. Yang, T. Liu, Y. Pu, L. Liu, Q. Zhao, and Q. Wan, "Remote sensing image change captioning using multi-attentive network with diffusion model," *Remote Sens.*, vol. 16, no. 21, p. 4083, 2024, doi: [10.3390/rs16214083](https://doi.org/10.3390/rs16214083).
- [187] C. Saharia, J. Ho, W. Chan, T. Salimans, D. J. Fleet, and M. Norouzi, "Image super-resolution via iterative refinement," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 4713–4726, Apr. 2023, doi: [10.1109/TPAMI.2022.3204461](https://doi.org/10.1109/TPAMI.2022.3204461).
- [188] W. G. C. Bandara, N. G. Nair, and V. M. Patel, "DDPM-CD: Denoising diffusion probabilistic models as feature extractors for change detection," 2022, *arXiv:2206.11892*.
- [189] M. Czerwinski and C. Tachatzis, "Exploring the capability of text-to-image diffusion models with structural edge guidance for multispectral satellite image inpainting," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024, doi: [10.1109/LGRS.2024.3370212](https://doi.org/10.1109/LGRS.2024.3370212).
- [190] Z. Wang, Z. Hao, Y. Zhang, Y. Feng, and Y. Guo, "Up-Diff: Latent diffusion model for remote sensing urban prediction," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025, doi: [10.1109/LGRS.2024.3520133](https://doi.org/10.1109/LGRS.2024.3520133).
- [191] Q. Meng, W. Shi, S. Li, and L. Zhang, "PanDiff: A novel pan-sharpening method based on denoising diffusion probabilistic model," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–17, 2023, doi: [10.1109/TGRS.2023.3279864](https://doi.org/10.1109/TGRS.2023.3279864).
- [192] D. Tochilkin et al., "TripoSR: Fast 3D object reconstruction from a single image," 2024, *arXiv:2403.02151*.
- [193] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 11,966–11,976, doi: [10.1109/CVPR52688.2022.01167](https://doi.org/10.1109/CVPR52688.2022.01167).
- [194] X. Li, J. Ding, and M. Elhoseiny, "VRSBench: A versatile vision-language benchmark dataset for remote sensing image understanding," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS) Datasets Benchmarks Track*, 2024, pp. 1–13.
- [195] Z. Yuan, Z. Xiong, L. Mou, and X. X. Zhu, "ChatEarthNet: A global-scale image-text dataset empowering vision-language geo-foundation models," *Earth Syst. Sci. Data*, vol. 17, no. 3, pp. 1245–1263, 2025, doi: [10.5194/essd-17-1245-2025](https://doi.org/10.5194/essd-17-1245-2025).
- [196] B. Zhou et al., "UrBench: A comprehensive benchmark for evaluating large multimodal models in multi-view urban scenarios," in *Proc. AAAI Conf. Artif. Intell.*, 2025, vol. 39, pp. 10,707–10,715.
- [197] J. Ge, Y. Zheng, K. Guo, and J. Liang, "RSTeller: Scaling up visual language modeling in remote sensing with rich linguistic semantics from openly available data and large language models," 2024, *arXiv:2408.14744*.
- [198] J. Ding et al., "Object detection in aerial images: A large-scale benchmark and challenges," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 7778–7796, Nov. 2022, doi: [10.1109/TPAMI.2021.3117983](https://doi.org/10.1109/TPAMI.2021.3117983).
- [199] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020, doi: [10.1016/j.isprsjprs.2019.11.023](https://doi.org/10.1016/j.isprsjprs.2019.11.023).
- [200] "ChatGPT," OpenAI. Accessed: Oct. 30, 2024. [Online]. Available: <https://openai.com/chatgpt/overview/>
- [201] S. Vujasinović, S. Becker, T. Breuer, S. Bullinger, N. Scherer-Negenborn, and M. Arens, "Integration of the 3D environment for UAV onboard visual object tracking," *Appl. Sci.*, vol. 10, no. 21, p. 7622, 2020, doi: [10.3390/app10217622](https://doi.org/10.3390/app10217622).
- [202] S. Waqas Zamir et al., "iSAID: A large-scale dataset for instance segmentation in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR) Workshops*, 2019, pp. 28–37.
- [203] "2D semantic labeling contest - Potsdam," Int. Soc. for Photogrammetry and Remote Sens. (ISPRS), Hannover, Germany, 2012. [Online]. Available: <https://www.isprs.org/resources/datasets/benchmarks/UrbanSemLab/2d-sem-label-potsdam.aspx>
- [204] Y. Long, Y. Gong, Z. Xiao, and Q. Liu, "Accurate object localization in remote sensing images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2486–2498, May 2017, doi: [10.1109/TGRS.2016.2645610](https://doi.org/10.1109/TGRS.2016.2645610).
- [205] "2D semantic labeling - Vaihingen data," Int. Soc. for Photogrammetry and Remote Sens. (ISPRS), Hannover, Germany, 2012. [Online]. Available: <https://www.isprs.org/resources/datasets/benchmarks/UrbanSemLab/2d-sem-label-vaihingen.aspx>
- [206] M.-R. Hsieh, Y.-L. Lin, and W. H. Hsu, "Drone-based object counting by spatially regularized regional proposal network,"

- in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2017, pp. 4165–4173, doi: [10.1109/ICCV.2017.446](https://doi.org/10.1109/ICCV.2017.446).
- [207] Y. Zhang, Y. Yuan, Y. Feng, and X. Lu, “Hierarchical and robust convolutional neural network for very high-resolution remote sensing object detection,” *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5535–5548, Aug. 2019, doi: [10.1109/TGRS.2019.2900302](https://doi.org/10.1109/TGRS.2019.2900302).
- [208] H. Chen and Z. Shi, “A spatial-temporal attention-based method and a new dataset for remote sensing image change detection,” *Remote Sens.*, vol. 12, no. 10, p. 1662, 2020, doi: [10.3390/rs12101662](https://doi.org/10.3390/rs12101662).
- [209] A. Robicquet, A. Sadeghian, A. Alahi, and S. Savarese, “Learning social etiquette: Human trajectory understanding in crowded scenes,” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2016, pp. 549–565.
- [210] P. Zhu et al., “Detection and tracking meet drones challenge,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 7380–7399, Nov. 2022, doi: [10.1109/TPAMI.2021.3119563](https://doi.org/10.1109/TPAMI.2021.3119563).
- [211] Z. Liu, L. Yuan, L. Weng, and Y. Yang, “A high resolution optical satellite image dataset for ship recognition and some new baselines,” in *Proc. Int. Conf. Pattern Recognit. Appl. Methods (ICPRAM)*, 2017, vol. 1, pp. 324–331.
- [212] J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, “LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS) Datasets Benchmarks Track*, 2021, pp. 1–12.
- [213] “CrowdAI mapping challenge: CrowdAI dataset.” Alcrowd. Accessed: Oct. 30, 2024. [Online]. Available: https://www.crowdai.org/challenges/mapping-challenge/dataset_files
- [214] L. Liu and H. Li, “Lending orientation to neural networks for cross-view geo-localization,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 5624–5633, doi: [10.1109/CVPR.2019.00577](https://doi.org/10.1109/CVPR.2019.00577).
- [215] S. Workman, R. Souvenir, and N. Jacobs, “Wide-area image geolocalization with aerial reference imagery,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2015, pp. 3961–3969, doi: [10.1109/ICCV.2015.451](https://doi.org/10.1109/ICCV.2015.451).
- [216] R. Anil et al., “Gemini: A family of highly capable multimodal models,” 2023, *arXiv:2312.11805*.
- [217] Y. Wang, N. A. A. Braham, Z. Xiong, C. Liu, C. M. Albrecht, and X. X. Zhu, “SSL4EO-S12: A large-scale multimodal, multitemporal dataset for self-supervised learning in earth observation [Software and Data Sets],” *IEEE Geosci. Remote Sens. Mag.*, vol. 11, no. 3, pp. 98–106, 2023, doi: [10.1109/MGRS.2023.3281651](https://doi.org/10.1109/MGRS.2023.3281651).
- [218] M. Mendieta, B. Han, X. Shi, Y. Zhu, and C. Chen, “Towards geospatial foundation models via continual pretraining,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 16,760–16,770, doi: [10.1109/ICCV51070.2023.01541](https://doi.org/10.1109/ICCV51070.2023.01541).
- [219] H. Li et al., “RSI-CB: A large scale remote sensing image classification benchmark via crowdsource data,” 2017, *arXiv:1705.10450*.
- [220] A. Q. Jiang et al., “Mixtral of experts,” 2024, *arXiv:2401.04088*.
- [221] P. Sharma, N. Ding, S. Goodman, and R. Soricut, “Conceptual captions: A cleaned, hypernymed, image alt-text dataset for automatic image captioning,” in *Proc. 56th Annu. Meet. Assoc. Comput. Linguist. (ACL)*, 2018, pp. 2556–2565.
- [222] Y. Han, X. Yang, T. Pu, and Z. Peng, “Fine-grained recognition for oriented ship against complex scenes in optical remote sensing images,” *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2021, doi: [10.1109/TGRS.2021.3123666](https://doi.org/10.1109/TGRS.2021.3123666).
- [223] S. Wei, X. Zeng, Q. Qu, M. Wang, H. Su, and J. Shi, “HRSID: A high-resolution SAR images dataset for ship detection and instance segmentation,” *IEEE Access*, vol. 8, pp. 120,234–120,254, 2020, doi: [10.1109/ACCESS.2020.3005861](https://doi.org/10.1109/ACCESS.2020.3005861).
- [224] T. Zhang et al., “SAR ship detection dataset (SSDD): Official release and comprehensive data analysis,” *Remote Sens.*, vol. 13, no. 18, p. 3690, 2021, doi: [10.3390/rs13183690](https://doi.org/10.3390/rs13183690).
- [225] X. Sun et al., “FAIRIM: A benchmark dataset for fine-grained object recognition in high-resolution remote sensing imagery,” *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 116–130, Feb. 2022, doi: [10.1016/j.isprsjprs.2021.12.004](https://doi.org/10.1016/j.isprsjprs.2021.12.004).
- [226] X. Zheng, B. Wang, X. Du, and X. Lu, “Mutual attention inception network for remote sensing visual question answering,” *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2021, doi: [10.1109/TGRS.2021.3079918](https://doi.org/10.1109/TGRS.2021.3079918).
- [227] G. Hoxha and F. Melgani, “A novel SVM-based decoder for remote sensing image captioning,” *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, doi: [10.1109/TGRS.2021.3105004](https://doi.org/10.1109/TGRS.2021.3105004).
- [228] L. Xu et al., “RS-GPT4V: A unified multimodal instruction-following dataset for remote sensing image understanding,” 2024, *arXiv:2406.12479*.
- [229] M. Rahnemoonfar, T. Chowdhury, A. Sarkar, D. Varshney, M. Yari, and R. R. Murphy, “FloodNet: A high resolution aerial imagery dataset for post flood scene understanding,” *IEEE Access*, vol. 9, pp. 89,644–89,654, 2021, doi: [10.1109/ACCESS.2021.3090981](https://doi.org/10.1109/ACCESS.2021.3090981).
- [230] L. Bashmal, Y. Bazi, M. M. Al Rahhal, M. Zuair, and F. Melgani, “CapERA: Captioning events in aerial videos,” *Remote Sens.*, vol. 15, no. 8, p. 2139, 2023, doi: [10.3390/rs15082139](https://doi.org/10.3390/rs15082139).
- [231] S. Tian, A. Ma, Z. Zheng, and Y. Zhong, “Hi-UCD: A large-scale dataset for urban semantic change detection in remote sensing imagery,” 2020, *arXiv:2011.03247*.
- [232] D. Wang et al., “SAMRS: Scaling-up remote sensing segmentation dataset with segment anything model,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS) Datasets Benchmarks Track*, 2023, vol. 36, pp. 8815–8827.
- [233] Y. Wenqi et al., “MAR20: A benchmark for military aircraft recognition in remote sensing images,” *Natl. Remote Sens. Bull.*, vol. 27, no. 12, pp. 2688–2696, 2024.
- [234] S. Ji, S. Wei, and M. Lu, “Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set,” *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019, doi: [10.1109/TGRS.2018.2858817](https://doi.org/10.1109/TGRS.2018.2858817).
- [235] Y. Lyu, G. Vosselman, G.-S. Xia, A. Yilmaz, and M. Y. Yang, “UAVid: A semantic segmentation dataset for UAV imagery,” *ISPRS J. Photogramm. Remote Sens.*, vol. 165, pp. 108–119, Jul. 2020, doi: [10.1016/j.isprsjprs.2020.05.009](https://doi.org/10.1016/j.isprsjprs.2020.05.009).
- [236] H. Zhu, X. Chen, W. Dai, K. Fu, Q. Ye, and J. Jiao, “Orientation robust object detection in aerial images using deep convolutional neural network,” in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, 2015, pp. 3735–3739, doi: [10.1109/ICIP.2015.7351502](https://doi.org/10.1109/ICIP.2015.7351502).

- [237] K. Li, G. Cheng, S. Bu, and X. You, "Rotation-insensitive and context-augmented object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2337–2348, Apr. 2018, doi: [10.1109/TGRS.2017.2778300](https://doi.org/10.1109/TGRS.2017.2778300).
- [238] "Aerial-Mancar," InfiRay, Yantai, China, 2021. [Online]. Available: http://openai.raytrontek.com/apply/Aerial_mancar.html/
- [239] Y. Di, Z. Jiang, and H. Zhang, "A public dataset for fine-grained ship classification in optical remote sensing images," *Remote Sens.*, vol. 13, no. 4, p. 747, 2021, doi: [10.3390/rs13040747](https://doi.org/10.3390/rs13040747).
- [240] "Infrared-security," InfiRay, Yantai, China, 2021. [Online]. Available: http://openai.raytrontek.com/apply/Infrared_security.html/
- [241] "Double-light-vehicle," InfiRay, Yantai, China, 2021. [Online]. Available: http://openai.raytrontek.com/apply/Double_light_vehicle.html/
- [242] "Sea-shipping," InfiRay, Yantai, China, 2021. [Online]. Available: http://openai.raytrontek.com/apply/Double_light_vehicle.html/
- [243] M. Zhang, F. Chen, and B. Li, "Multistep question-driven visual question answering for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–12, 2023, doi: [10.1109/TGRS.2023.3312479](https://doi.org/10.1109/TGRS.2023.3312479).
- [244] Y. Di, Z. Jiang, H. Zhang, and G. Meng, "A public dataset for ship classification in remote sensing images," *Proc. SPIE*, vol. 11155, pp. 515–521, Oct. 2019.
- [245] J. Suo, T. Wang, X. Zhang, H. Chen, W. Zhou, and W. Shi, "HIT-UAV: A high-altitude infrared thermal dataset for unmanned aerial vehicle-based object detection," *Sci. Data*, vol. 10, no. 1, p. 227, 2023, doi: [10.1038/s41597-023-02066-6](https://doi.org/10.1038/s41597-023-02066-6).
- [246] "Oceanic-Ship," Center for Optics Research and Engineering of Shandong University, Qingdao, Shandong, China, 2020. [Online]. Available: <http://www.gxzx.sdu.edu.cn/info/1133/2174.htm/>
- [247] Y. Li et al., "STAR: A first-ever dataset and a large-scale benchmark for scene graph generation in large-size satellite imagery," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 3, pp. 1832–1849, Mar. 2025, doi: [10.1109/TPAMI.2024.3508072](https://doi.org/10.1109/TPAMI.2024.3508072).
- [248] C. Pang, J. Wu, J. Ding, C. Song, and G.-S. Xia, "Detecting building changes with off-nadir aerial images," *Sci. China Inf. Sci.*, vol. 66, no. 4, 2023, Art. no. 140306, doi: [10.1007/s11432-022-3691-4](https://doi.org/10.1007/s11432-022-3691-4).
- [249] R. Xia et al., "Crtranssar: A visual transformer based on contextual joint representation learning for sar ship detection," *Remote Sens.*, vol. 14, no. 6, p. 1488, 2022, doi: [10.3390/rs14061488](https://doi.org/10.3390/rs14061488).
- [250] X.-Y. Tong, G.-S. Xia, and X. X. Zhu, "Enabling country-scale land cover mapping with meter-resolution satellite imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 196, pp. 178–196, 2023.
- [251] C. Wu, L. Zhang, and L. Zhang, "A scene change detection framework for multi-temporal very high resolution remote sensing images," *Signal Process.*, vol. 124, pp. 184–197, Jul. 2016, doi: [10.1016/j.sigpro.2015.09.020](https://doi.org/10.1016/j.sigpro.2015.09.020).
- [252] I. Demir et al., "DeepGlobe 2018: A challenge to parse the earth through satellite images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR) Workshops*, 2018, pp. 172–17209, doi: [10.1109/CVPRW.2018.00031](https://doi.org/10.1109/CVPRW.2018.00031).
- [253] X.-Y. Tong et al., "Land-cover classification with high-resolution remote sensing images using transferable deep models," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111322, doi: [10.1016/j.rse.2019.111322](https://doi.org/10.1016/j.rse.2019.111322).
- [254] S. Verma, A. Panigrahi, and S. Gupta, "QFabric: Multi-task change detection dataset," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR) Workshops*, 2021, pp. 1052–1061, doi: [10.1109/CVPRW53098.2021.00116](https://doi.org/10.1109/CVPRW53098.2021.00116).
- [255] L. Shen et al., "S2Looking: A satellite side-looking dataset for building change detection," *Remote Sens.*, vol. 13, no. 24, p. 5094, 2021, doi: [10.3390/rs13245094](https://doi.org/10.3390/rs13245094).
- [256] R. Gupta et al., "Creating xbd: A dataset for assessing building damage from satellite imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR) Workshops*, 2019, pp. 10–17.
- [257] J. Roberts, K. Han, and S. Albanie, "SATIN: A multi-task meta-dataset for classifying satellite imagery using vision-language models," 2023, *arXiv:2304.11619*.
- [258] X. An, J. Sun, Z. Gui, and W. He, "COREval: A comprehensive and objective benchmark for evaluating the remote sensing capabilities of large vision-language models," 2024, *arXiv:2411.18145*.
- [259] M. S. Danish et al., "GEOBench-VLM: Benchmarking vision-language models for geospatial tasks," 2024, *arXiv:2411.19325*.
- [260] Z. Xiong, F. Zhang, Y. Wang, Y. Shi, and X. X. Zhu, "EarthNets: Empowering artificial intelligence for earth observation," *IEEE Geosci. Remote Sens. Mag.*, early access, Oct. 23, 2024, doi: [10.1109/MGRS.2024.3466998](https://doi.org/10.1109/MGRS.2024.3466998).
- [261] C. Zhang and S. Wang, "Good at captioning bad at counting: Benchmarking GPT-4V on earth observation data," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR) Workshops*, 2024, pp. 7839–7849, doi: [10.1109/CVPRW63382.2024.00780](https://doi.org/10.1109/CVPRW63382.2024.00780).
- [262] T. N. Mundhenk, G. Konjevod, W. A. Sakla, and K. Boakye, "A large contextual dataset for classification, detection and counting of cars with deep learning," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2016, pp. 785–800.
- [263] B. G. Weinstein et al., "A benchmark dataset for canopy crown detection and delineation in co-registered airborne RGB, LiDAR and hyperspectral imagery from the national ecological observation network," *PLoS Comput. Biol.*, vol. 17, no. 7, 2021, Art. no. e1009180.
- [264] J. A. Eikelboom et al., "Improving the precision and accuracy of animal population estimates with aerial image object detection," *Methods Ecol. Evol.*, vol. 10, no. 11, pp. 1875–1887, 2019.
- [265] P. W. Koh et al., "Wilds: A benchmark of in-the-wild distribution shifts," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2021, pp. 5637–5664.
- [266] G. Machado et al., "AiRound and CV-BrCT: Novel multiview datasets for scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 488–503, 2020, doi: [10.1109/JSTARS.2020.3033424](https://doi.org/10.1109/JSTARS.2020.3033424).
- [267] "Deforestation-satellite-imagery dataset," *CSE499 Deforestation Satellite*, 2024. [Online]. Available: <https://universe.roboflow.com/cse499deforestationssatellite/deforestation-satellite-imagery-335n4>
- [268] G. Baier, A. Deschemps, M. Schmitt, and N. Yokoya, "Synthesizing optical and SAR imagery from land cover maps and auxiliary raster data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022, doi: [10.1109/TGRS.2021.3068532](https://doi.org/10.1109/TGRS.2021.3068532).

- [269] A. Shah, L. Thomas, and M. Maskey, "Marine debris dataset for object detection in planetscope imagery," NASA, Washington, DC, USA, 2021. [Online]. Available: <https://cmr.earthdata.nasa.gov/search/concepts/C2781412735-MLHUB.html>
- [270] X. X. Zhu et al., "So2Sat LCZ42: A benchmark data set for the classification of global local climate zones [Software and Data Sets]," *IEEE Geosci. Remote Sens. Mag.*, vol. 8, no. 3, pp. 76–89, Sep. 2020, doi: [10.1109/MGRS.2020.2964708](https://doi.org/10.1109/MGRS.2020.2964708).
- [271] "Forest damages – Larch casebearer," The Swedish Forest Agency, Jönköping, Sweden, 2021. [Online]. Available: <https://lila.science/datasets/forest-damages-larch-casebearer/>
- [272] X. Zhang, W. Yu, M.-O. Pun, and W. Shi, "Cross-domain landslide mapping from large-scale remote sensing images using prototype-guided domain-aware progressive representation learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 197, pp. 1–17, Mar. 2023, doi: [10.1016/j.isprsjprs.2023.01.018](https://doi.org/10.1016/j.isprsjprs.2023.01.018).
- [273] D. R. Cambrin and P. Garza, "Quakeset: A dataset and low-resource models to monitor earthquakes through sentinel-1," in *Proc. Int. ISCRAM Conf.*, 2024, pp. 1–12.
- [274] S. Shen, S. Seneviratne, X. Wanyan, and M. Kirley, "FireRisk: A remote sensing dataset for fire risk assessment with benchmarks using supervised and self-supervised learning," in *Proc. Int. Conf. Digit. Image Comput.: Tech. Appl. (DICTA)*, 2023, pp. 189–196, doi: [10.1109/DICTA60407.2023.00034](https://doi.org/10.1109/DICTA60407.2023.00034).
- [275] C. Tundia, R. Kumar, O. Damani, and G. Sivakumar, "FPCD: An open aerial VHR dataset for farm pond change detection," 2023, *arXiv:2302.14554*.
- [276] J. Liu, W. Zhou, H. Guan, and W. Zhao, "Similarity learning for land use scene-level change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 6501–6513, 2024, doi: [10.1109/JSTARS.2024.3373401](https://doi.org/10.1109/JSTARS.2024.3373401).
- [277] V. S. F. Garnot and L. Landrieu, "Panoptic segmentation of satellite image time series with convolutional temporal attention networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 4852–4861.
- [278] J. Shermeyer, T. Hossler, A. Van Etten, D. Hogan, R. Lewis, and D. Kim, "RarePlanes: Synthetic data takes flight," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, 2020, pp. 207–217, doi: [10.1109/WACV48630.2021.00025](https://doi.org/10.1109/WACV48630.2021.00025).
- [279] M. Cordts et al., "The cityscapes dataset for semantic urban scene understanding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 3213–3223, doi: [10.1109/CVPR.2016.350](https://doi.org/10.1109/CVPR.2016.350).
- [280] J. Hays and A. A. Efros, "IM2GPS: Estimating geographic information from a single image," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2008, pp. 1–8, doi: [10.1109/CVPR.2008.4587784](https://doi.org/10.1109/CVPR.2008.4587784).
- [281] C. Ertler, J. Mislej, T. Ollmann, L. Porzi, G. Neuhold, and Y. Kuang, "The mapillary traffic sign dataset for detection and classification on a global scale," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 68–84.
- [282] S. Zhu, T. Yang, and C. Chen, "VIGOR: Cross-view image geo-localization beyond one-to-one retrieval," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2021, pp. 3640–3649.
- [283] M. Chu, Z. Zheng, W. Ji, T. Wang, and T.-S. Chua, "Towards natural language-guided drones: Geotext-1652 benchmark with spatial relation matching," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2025, pp. 213–231, doi: [10.1109/CVPR46437.2021.00364](https://doi.org/10.1109/CVPR46437.2021.00364).
- [284] Z. Zheng, Y. Wei, and Y. Yang, "University-1652: A multi-view multi-source benchmark for drone-based geo-localization," in *Proc. ACM Int. Conf. Multimedia*, 2020, pp. 1395–1403.
- [285] K. Nogueira, J. A. Dos Santos, T. Fornazari, T. S. F. Silva, and L. P. Morellato, and R. D. S. Torres, "Towards vegetation species discrimination by using data-driven descriptors," in *Proc. IAPR Workshop Pattern Recognit. Remote Sens. (PRRS)*, 2016, pp. 1–6, doi: [10.1109/PRRS.2016.7867024](https://doi.org/10.1109/PRRS.2016.7867024).
- [286] Q. D. Cao and Y. Choe, "Building damage annotation on post-hurricane satellite imagery based on convolutional neural networks," 2018, *arXiv:1807.01688*.
- [287] S. Basu, S. Ganguly, S. Mukhopadhyay, R. DiBiano, M. Karki, and R. Nemani, "DeepSat: A learning framework for satellite imagery," in *Proc. ACM SIGSPATIAL Int. Conf. Adv. Geogr. Inf. Syst. (SIGSPATIAL)*, 2015, pp. 1–10.
- [288] "Airbus wind turbine patches." Kaggle. Accessed: Oct. 30, 2024. [Online]. Available: <https://www.kaggle.com/datasets/airbusgeo/airbus-wind-turbines-patches>
- [289] A. A. B. Jacques, A. B. Diallo, and E. Lord, "Towards the creation of a canadian land-use dataset for agricultural land classification," in *Proc. Can. Symp. Remote Sens.: Understanding Our World: Remote Sens. Sustain. Future*, 2021, vol. 4, p. 6.
- [290] Y. Hua, L. Mou, P. Jin, and X. X. Zhu, "MultiScene: A large-scale dataset and benchmark for multiscene recognition in single aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2021, doi: [10.1109/TGRS.2021.3110314](https://doi.org/10.1109/TGRS.2021.3110314).
- [291] R. Ba, C. Chen, J. Yuan, W. Song, and S. Lo, "SmokeNet: Satellite smoke scene detection using convolutional neural network with spatial and channel-wise attention," *Remote Sens.*, vol. 11, no. 14, p. 1702, 2019, doi: [10.3390/rs11141702](https://doi.org/10.3390/rs11141702).
- [292] O. A. Penatti, K. Nogueira, and J. A. Dos Santos, "Do deep features generalize from everyday objects to remote sensing and aerial scenes domains?" in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR) Workshops*, 2015, pp. 44–51, doi: [10.1109/CVPRW.2015.7301382](https://doi.org/10.1109/CVPRW.2015.7301382).
- [293] R. Hammell. "Ships in satellite imagery." Kaggle. Accessed: Oct. 30, 2024. [Online]. Available: <https://www.kaggle.com/datasets/rhammell/ships-in-satellite-imagery>
- [294] J. F. Rosser, D. G. Leibovici, and M. J. Jackson, "Rapid flood inundation mapping using social media, remote sensing and topographic data," *Nat. Hazards*, vol. 87, pp. 103–120, Jan. 2017.
- [295] J. Li et al., "Social Media: New perspectives to improve remote sensing for emergency response," *Proc. IEEE*, vol. 105, no. 10, pp. 1900–1912, Oct. 2017, doi: [10.1109/JPROC.2017.2684460](https://doi.org/10.1109/JPROC.2017.2684460).
- [296] Y. Zhai et al., "Investigating the catastrophic forgetting in multimodal large language model fine-tuning," in *Proc. Conf. Parsimony Learn.*, 2024, vol. 234, pp. 202–227.
- [297] D. Zhu et al., "Model tailor: Mitigating catastrophic forgetting in multi-modal large language models," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2024, pp. 62,581–62,598.

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